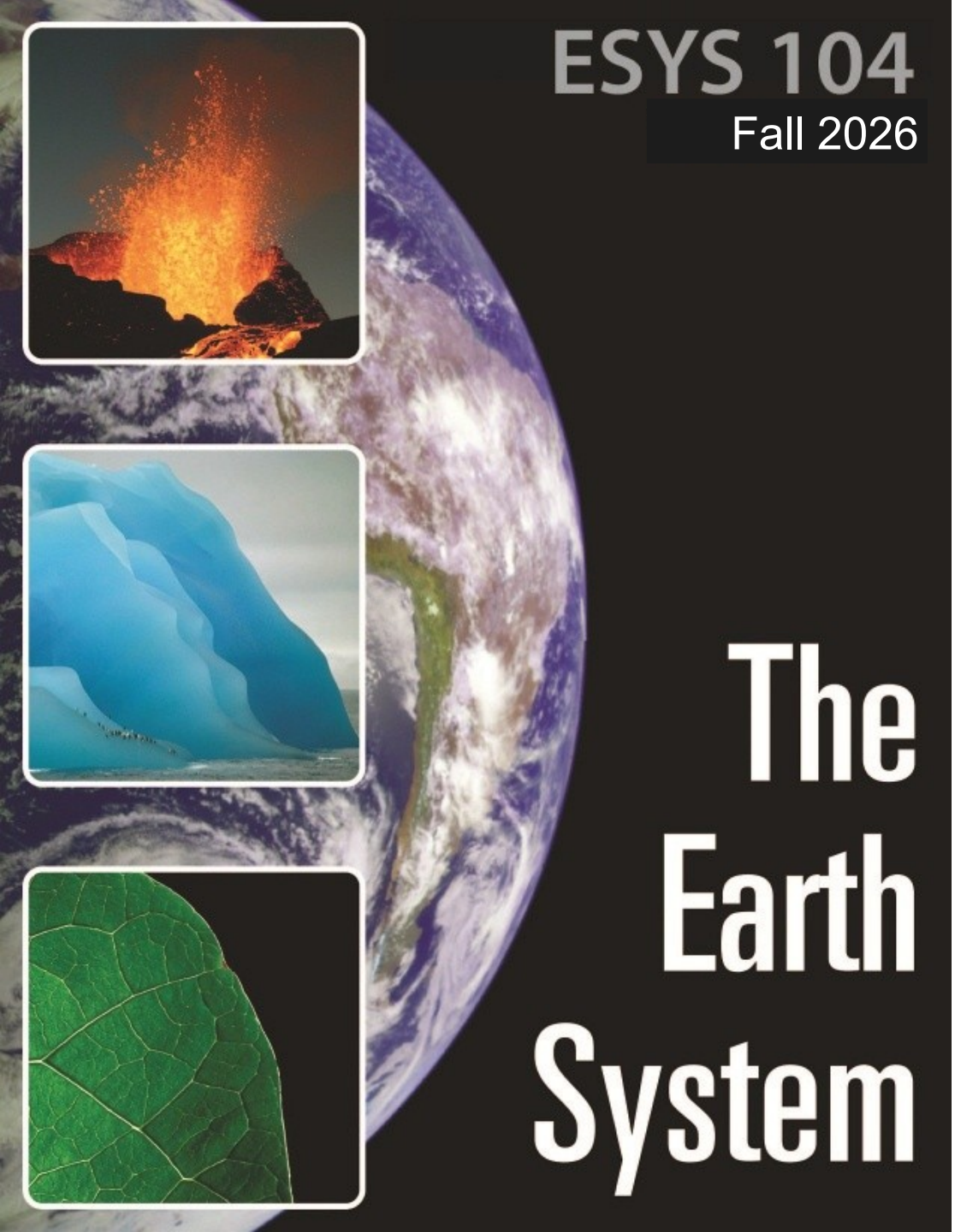
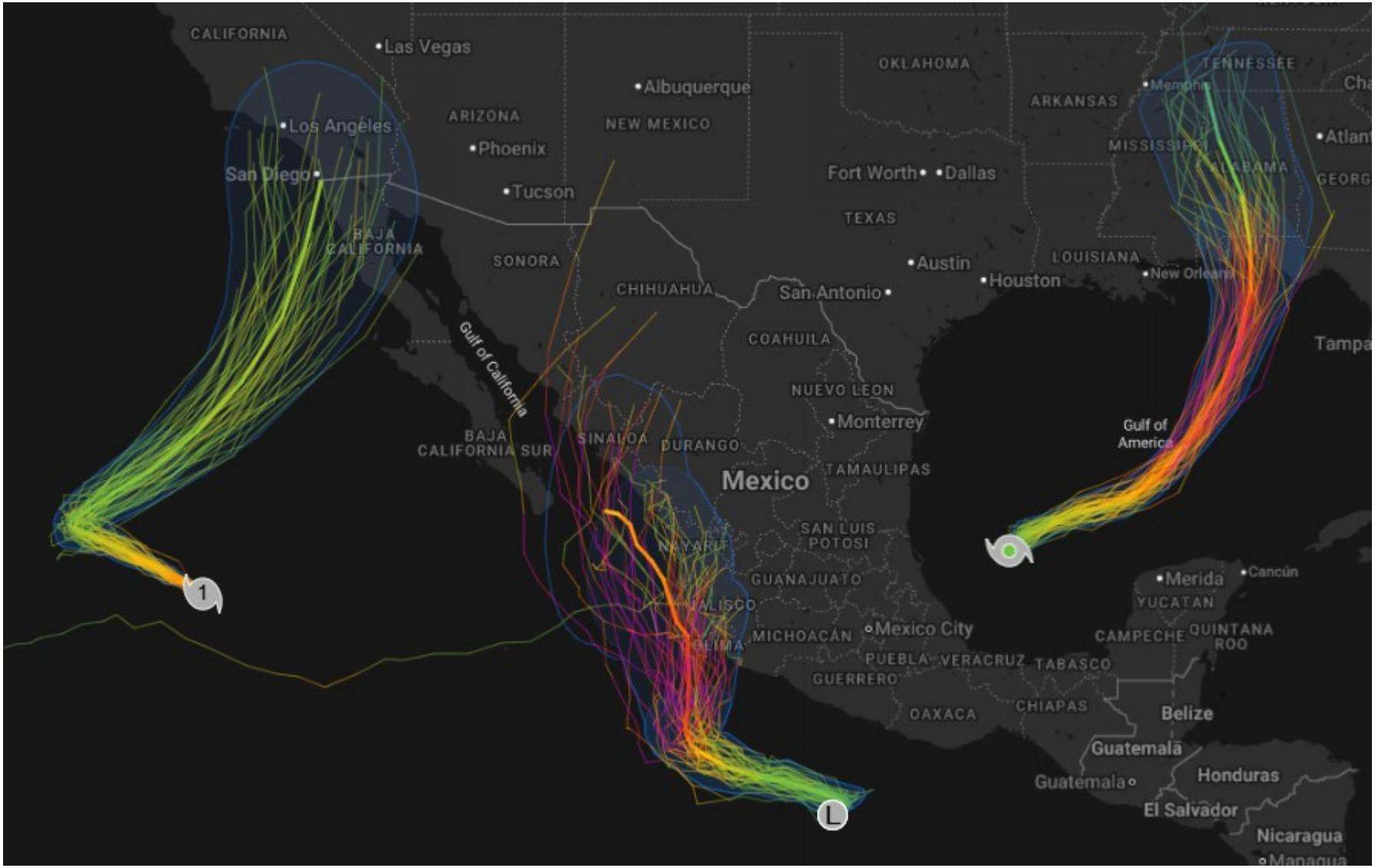


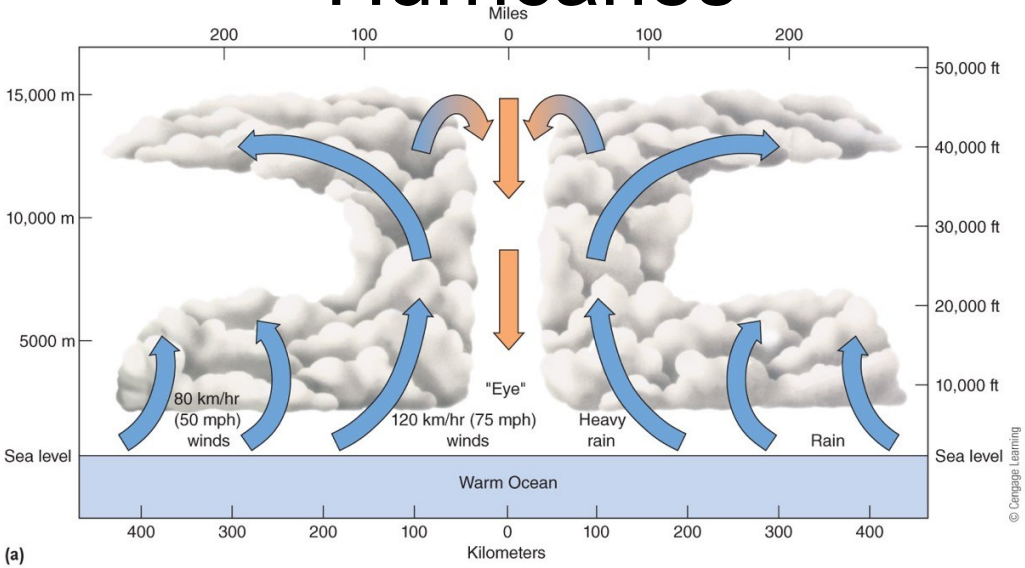

Ecosystems
Feedbacks
Tipping Points

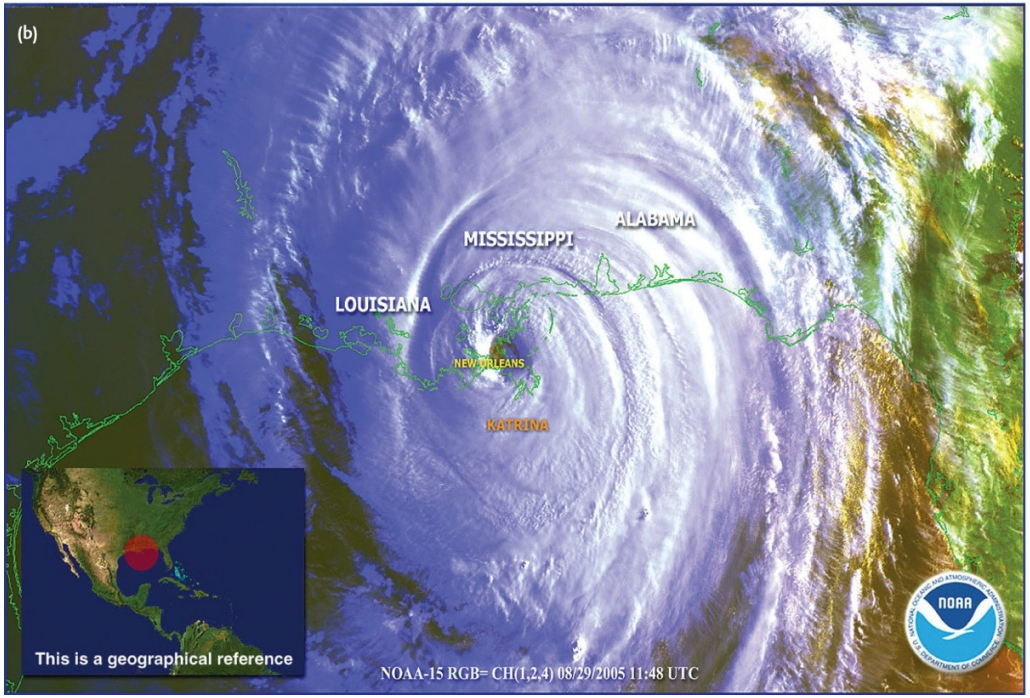
Think of Positive and Negative Feedbacks (post in Assignments)

# Hurricanes!

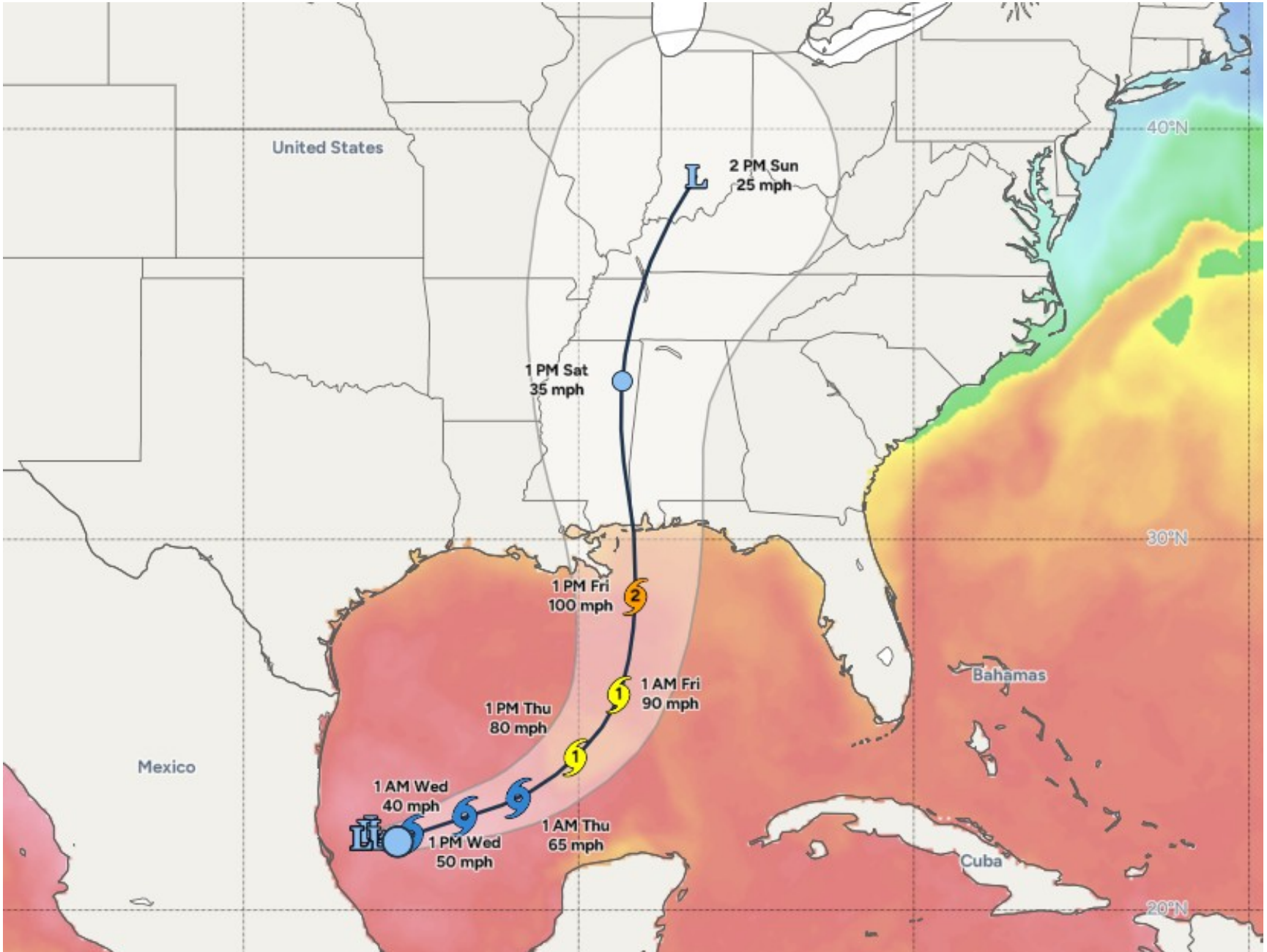

CALIFORNIA
Las Vegas
Albuquerque
OKLAHOMA
TENNESSEE
ARKANSAS
Los Angeles
ARIZONA
NEW MEXICO
Phoenix
San Diego
Tucson
Fort Worth
Dallas
TEXAS
BAJA CALIFORNIA
SONORA
CHIHUAHUA
Austin
Houston
LOUISIANA
New Orleans
San Antonio
COAHUILA
Gulf of California
NUEVO LEON
Monterrey
Gulf of America
BAJA CALIFORNIA SUR
SINALOA
DURANGO
Mexico
TAMAULIPAS
SAN LUIS POTOSI
GUANAJUATO
Merida
Cancún
YUCATAN
Mexico City
MICHOACÁN
CAMPECHE
QUINTANA ROO
PUEBLA
VERACRUZ
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# Hurricanes

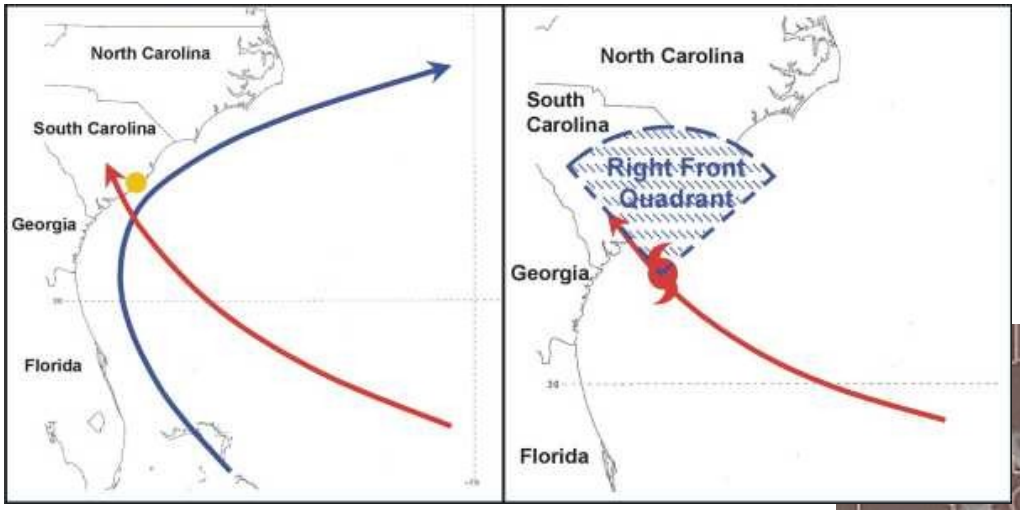

Miles
200
100
0
100
200
15,000 m
10,000 m
5000 m
Sea level
50,000 ft
40,000 ft
30,000 ft
20,000 ft
10,000 ft
Sea level
80 km/hr
(50 mph)
winds
"Eye"
120 km/hr (75 mph)
winds
Heavy
rain
Rain
Warm Ocean
400
300
200
100
0
100
200
300
400
Kilometers
(a)
© Cengage Learning


(b)
ALABAMA
MISSISSIPPI
LOUISIANA
NEW ORLEANS
KATRINA
This is a geographical reference
NOAA-15 RGB= CH(1,2,4) 08/29/2005 11:48 UTC
NOAA
© NOAA

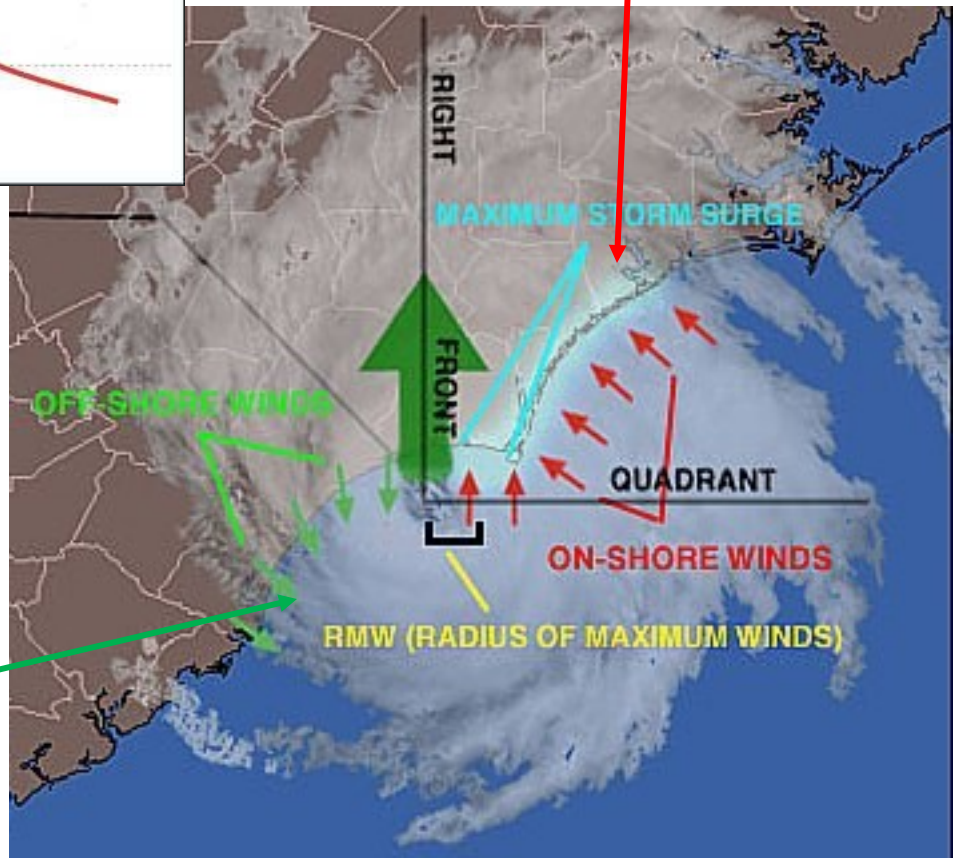
# Hurricane Isaias



# The Right Front Quadrant has the strongest storm surge




Strongest winds and surge



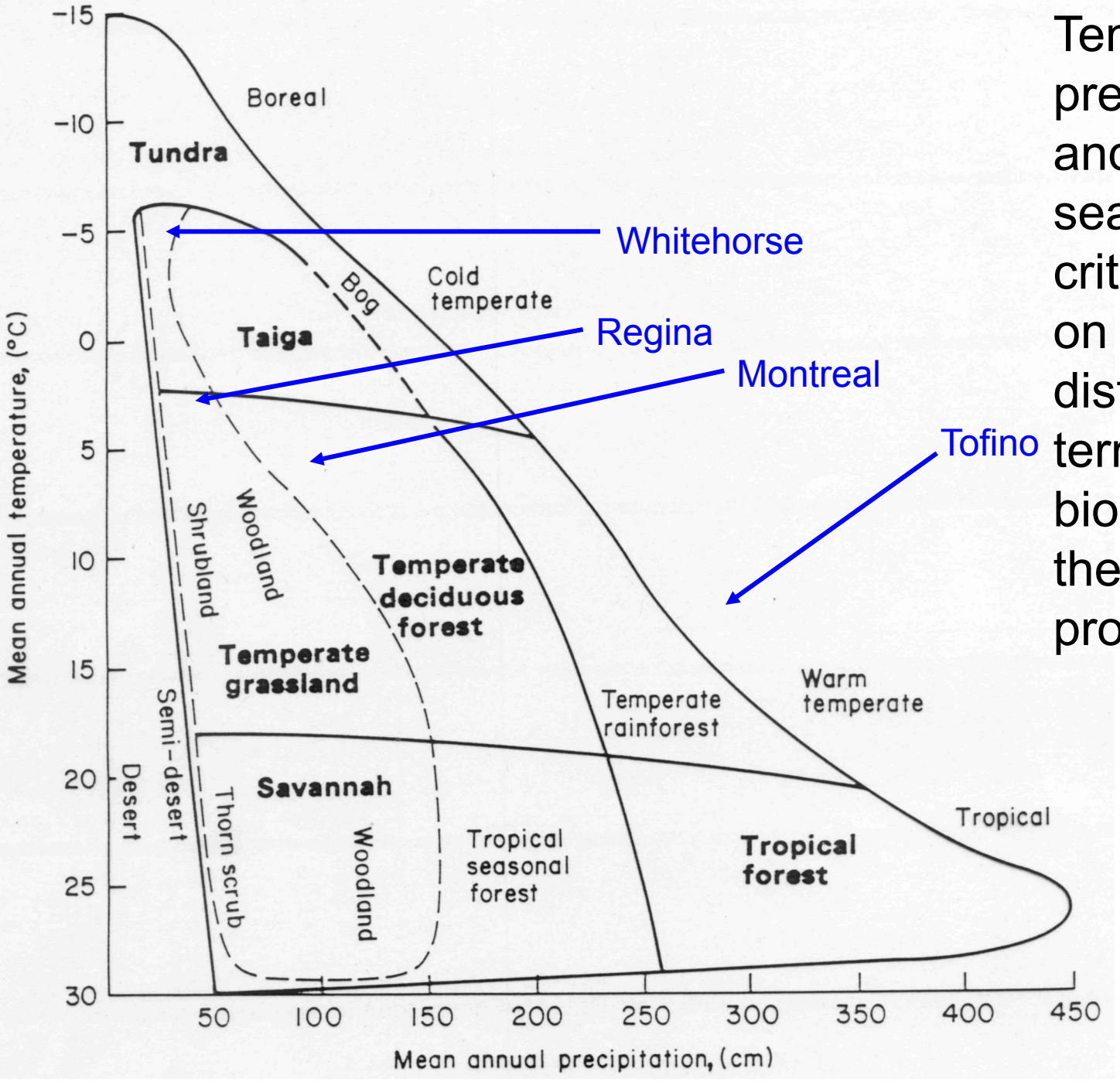

Weakest winds and surge

# Today's Topics

Ecosystem Services
Resilience, Regime Shift
Tipping points, hysteresis
Feedbacks

Lake Ecosystem example
Experimentation
Carrying Capacity
Kelp Forest example

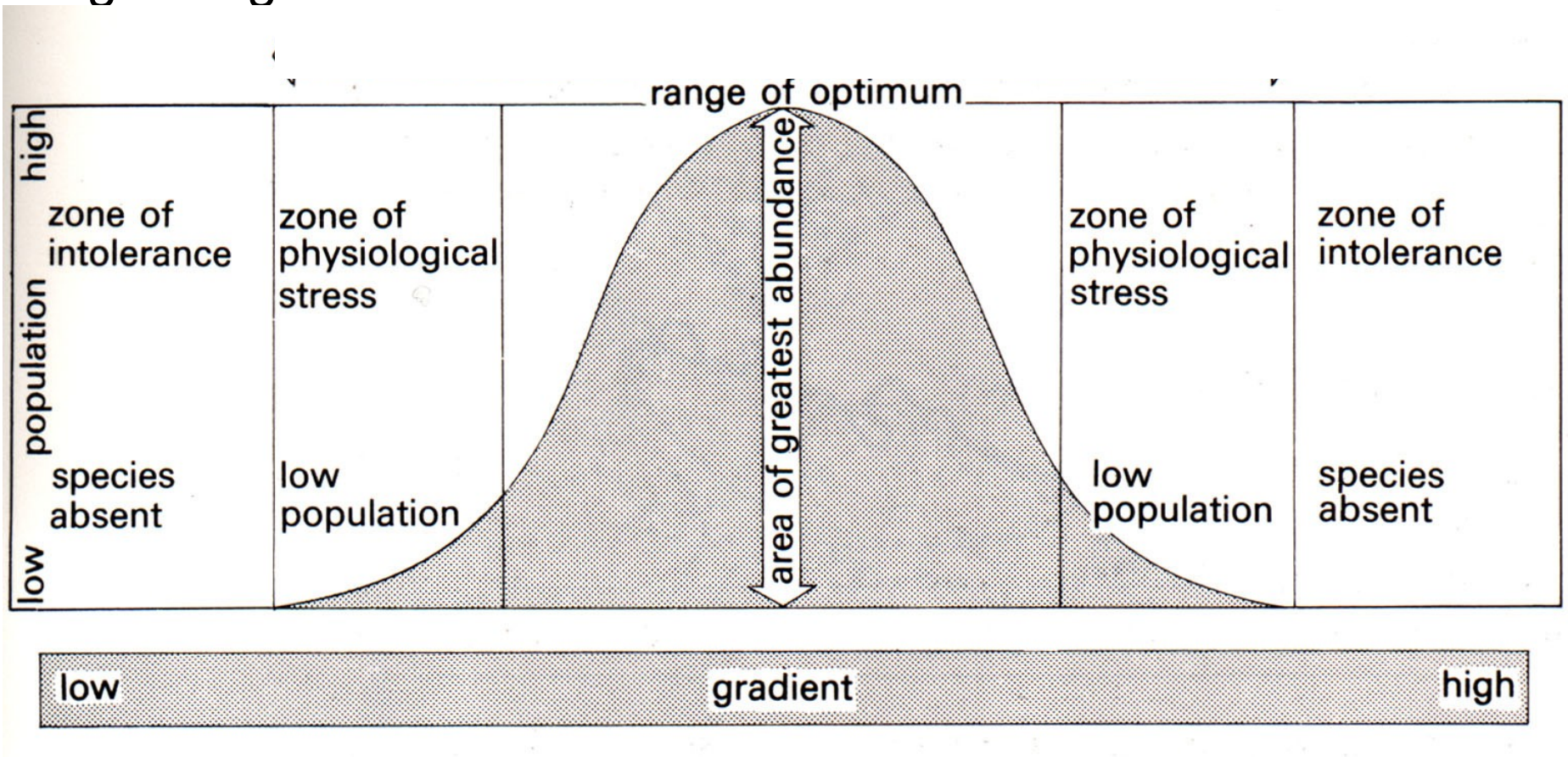
NA Deep Water formation
Cod Fishery




Temperature, precipitation, and their seasonality are critical controls on the distribution of terrestrial biomes and on their productivity.

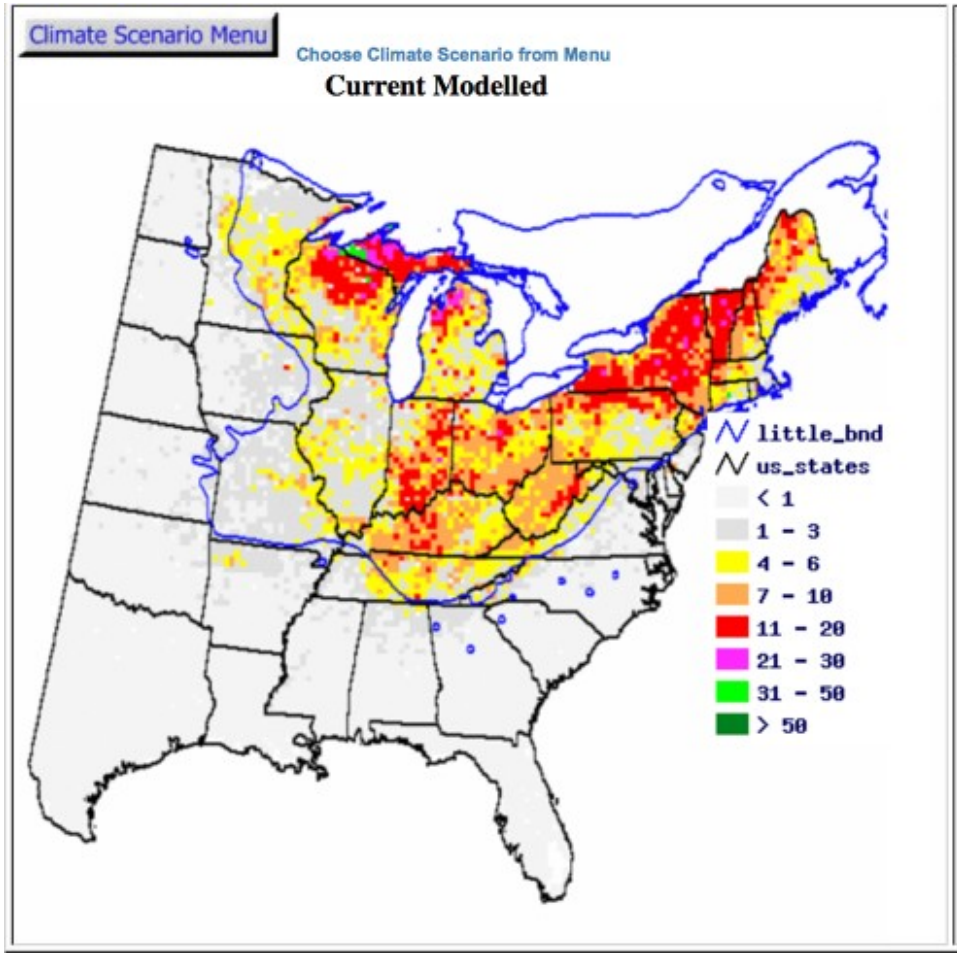
# Plants and abiotic factors: temperature, moisture, Etc.

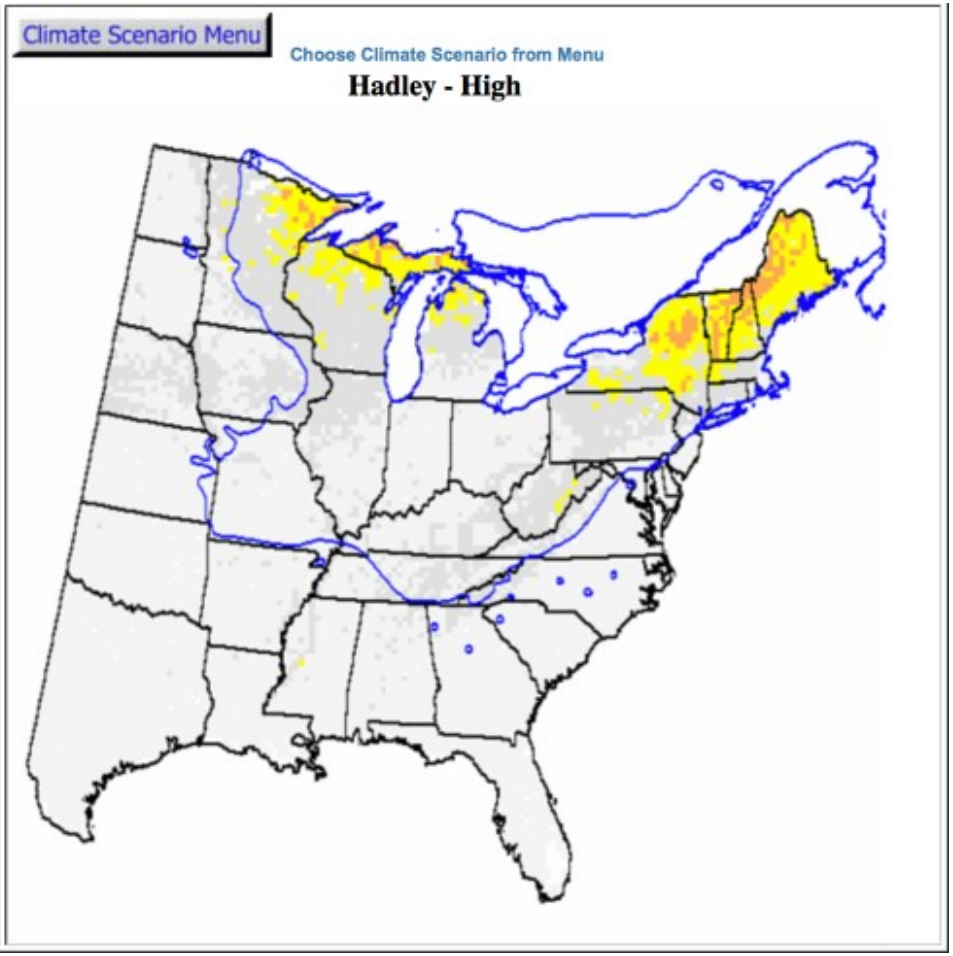
Different species live under a wide variety of conditions. They have different tolerances, and different optimum growing conditions.



2100, business as usual greenhouse gas emissions
USDA Climate Change Tree Atlas

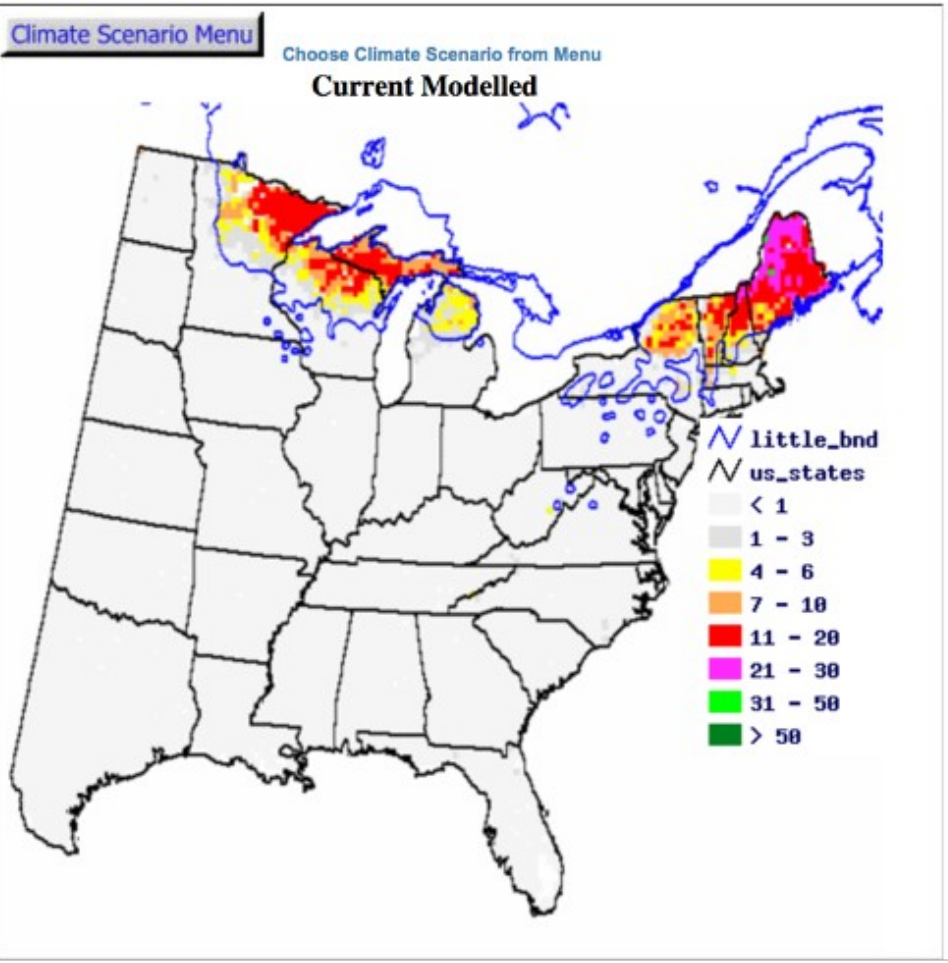
# Sugar Maple

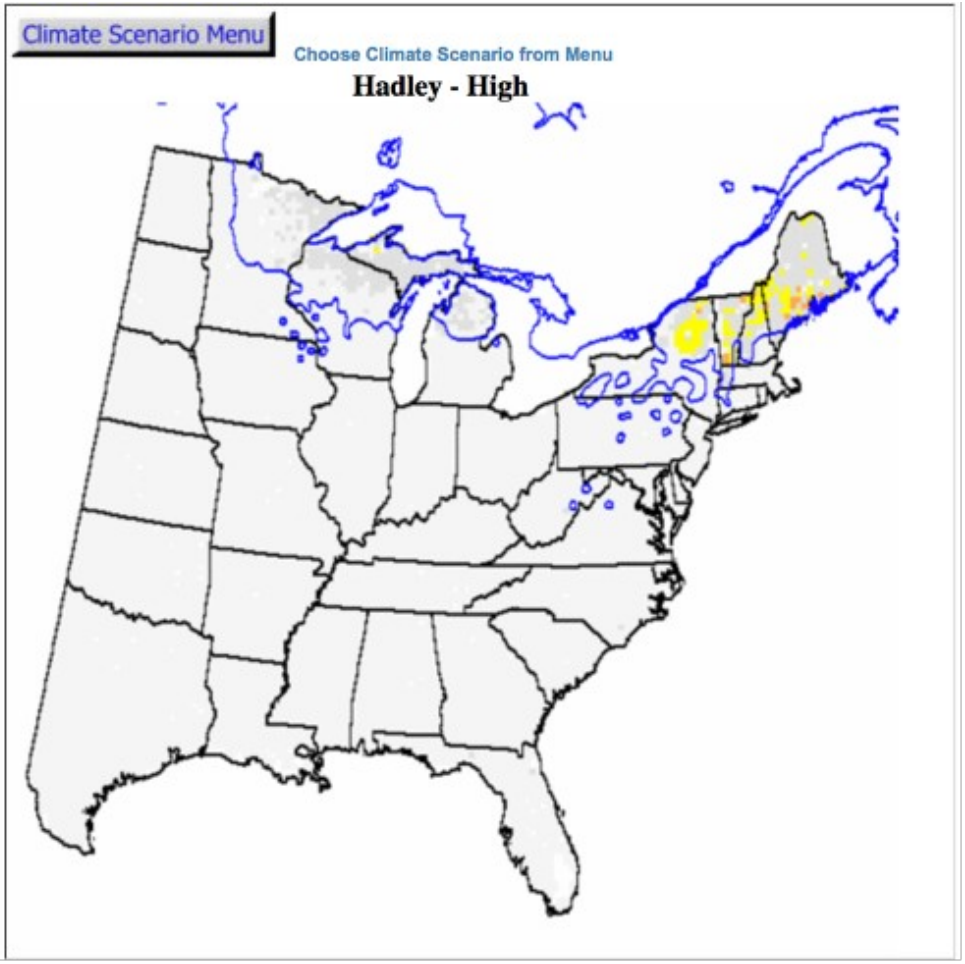


2100, business as usual greenhouse gas emissions
USDA Climate Change Tree Atlas

# Balsam Fir




The tariff threats come as Canadian forestry employers say they're already grappling with challenges including **declining forest health** and existing U.S. tariffs of at least 14 per cent. The economic pressures are spurring a wave of thousands of forestry job losses across the province.

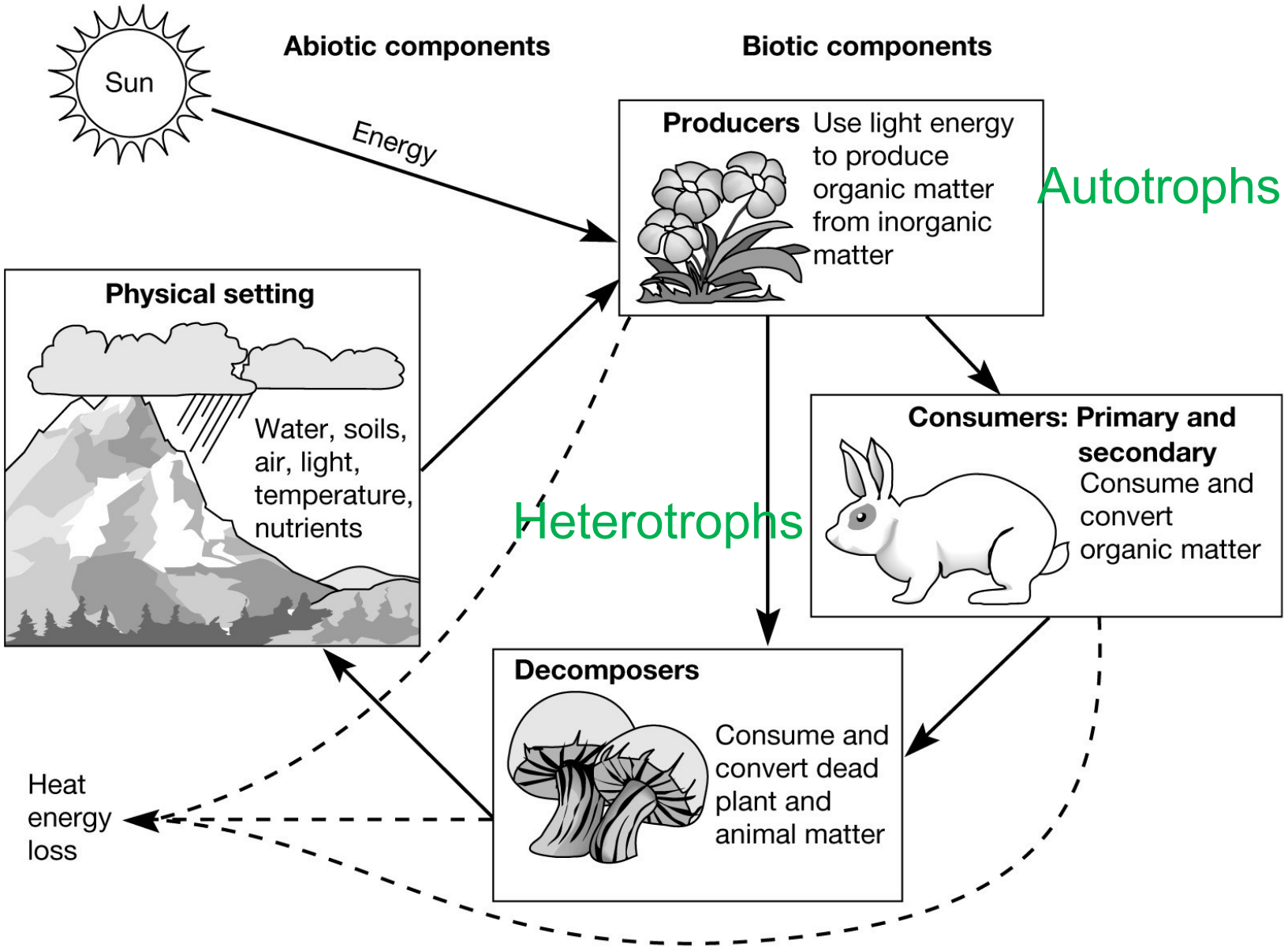
# Ecosystems

- A conceptual tool for ecologists studying an area. It's a bounded ecological system consisting of all the organisms in an area and the physical environment with which they interact

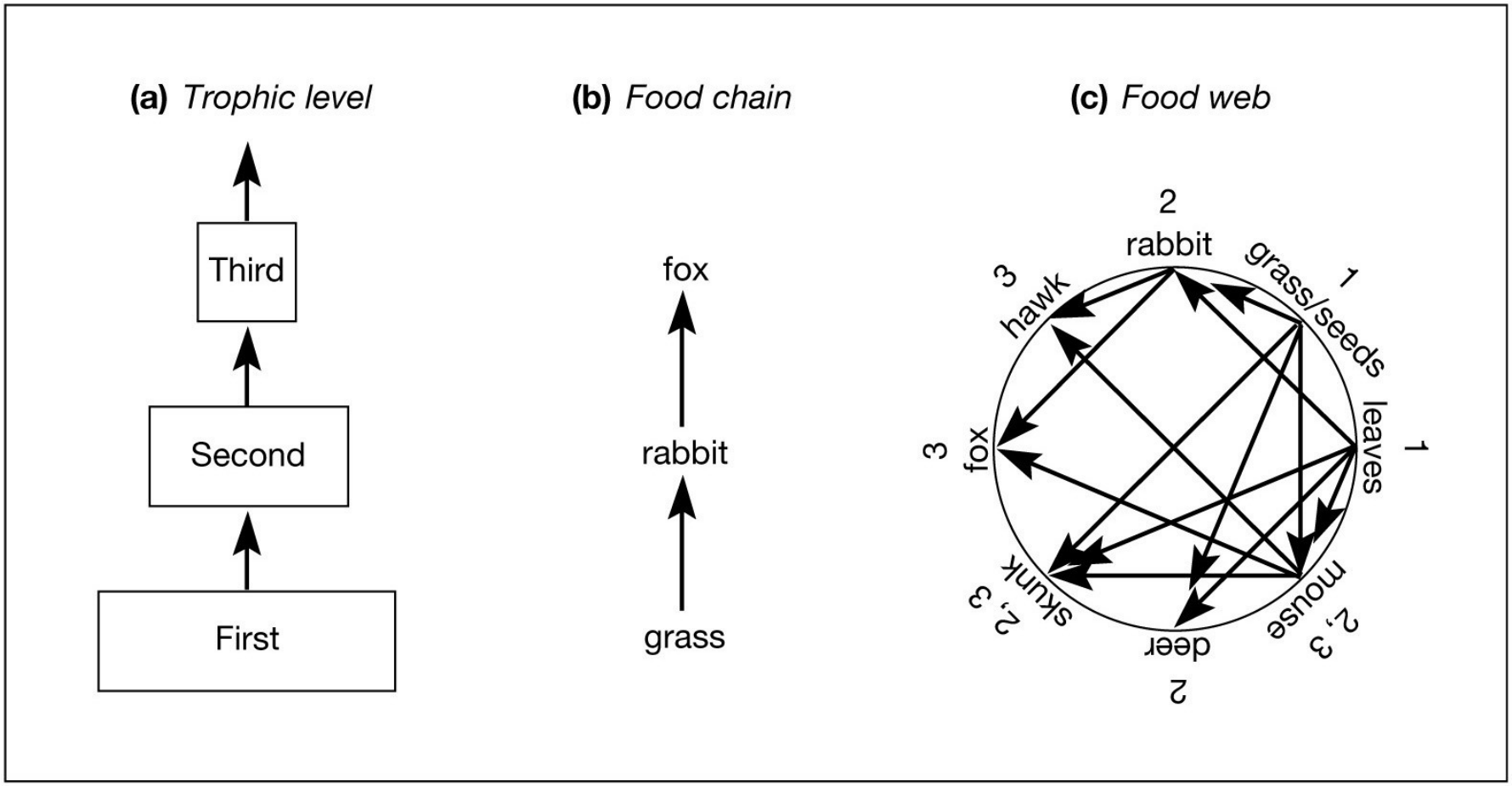
- For example: terrestrial ecosystems (a valley for example) involve topography, runoff, soil, atmosphere, climate as well as animals and plants that are adapted to this environment.

- Aquatic ecosytems could be a lake and its stream inflows and outflows; or a given marsh, etc.

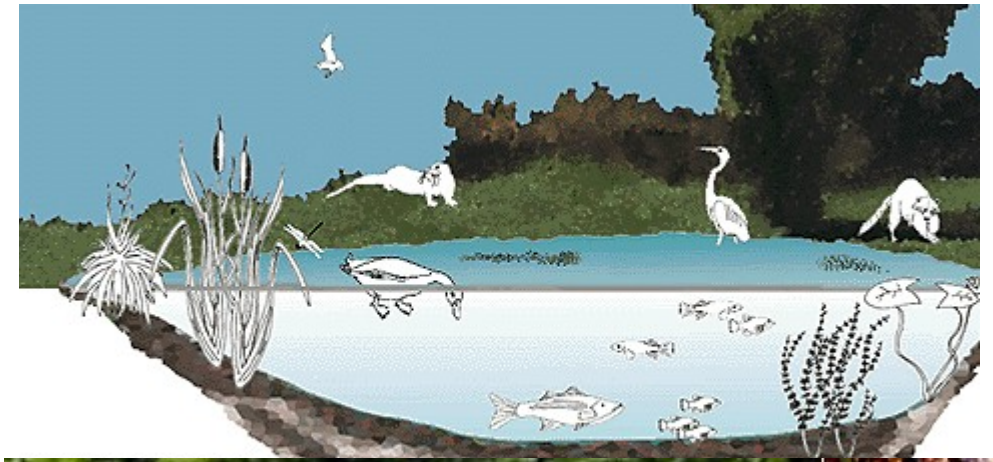
# Ecosystem Components



# Energy and Food Transfers in Food Chains and Food Webs




**Energy transfer efficiency between trophic levels = 10-20%**



# Ecosystem services

Reading from the Ecological Society of America





# Types of Ecosystem Services
# here, examples from wetland ecosystems

## Provisioning Services

Products obtained from wetland ecosystems:

- Food
- Fresh Water
- Fibre & Fuel
- Genetic resources
- Biochemical Products

## Regulating Services

Benefits obtained from regulation of wetland ecosystem processes:

- Climate regulation
- Hydrological regimes
- Erosion Protection
- Reduction of Natural Hazard risk
- Pollution Control & Detoxification processes

## Cultural Services

Material and non-material benefits obtained from wetland ecosystems:[1]

- Spiritual & Inspirational
- Recreational
- Aesthetic
- Educational
- Historical Artifacts
- Traditional Livelihoods & Knowledge

## Supporting Services

Services necessary for the production of all other ecosystem services:

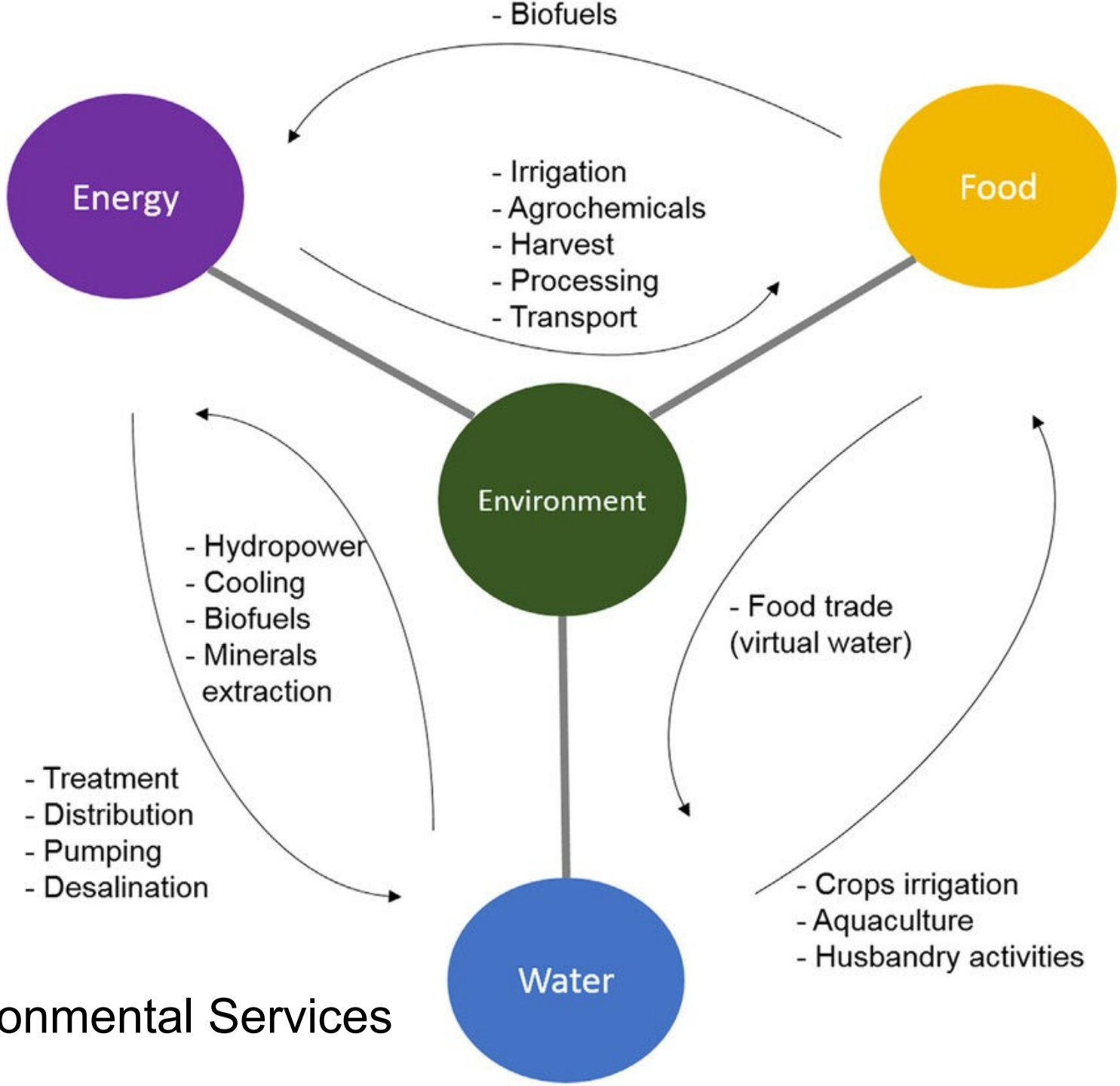
- Soil formation
- Nutrient cycling
- Primary production

[1] Note that "Cultural Services" includes material and non-material cultural values, benefits and functions (as identified in COP8 DOC.15. *Cultural aspects of wetlands*).

# Ecosystem Services




Carmine BeeEater (Tessa Murray)




Environmental Services

# Resilience and Regime Shift

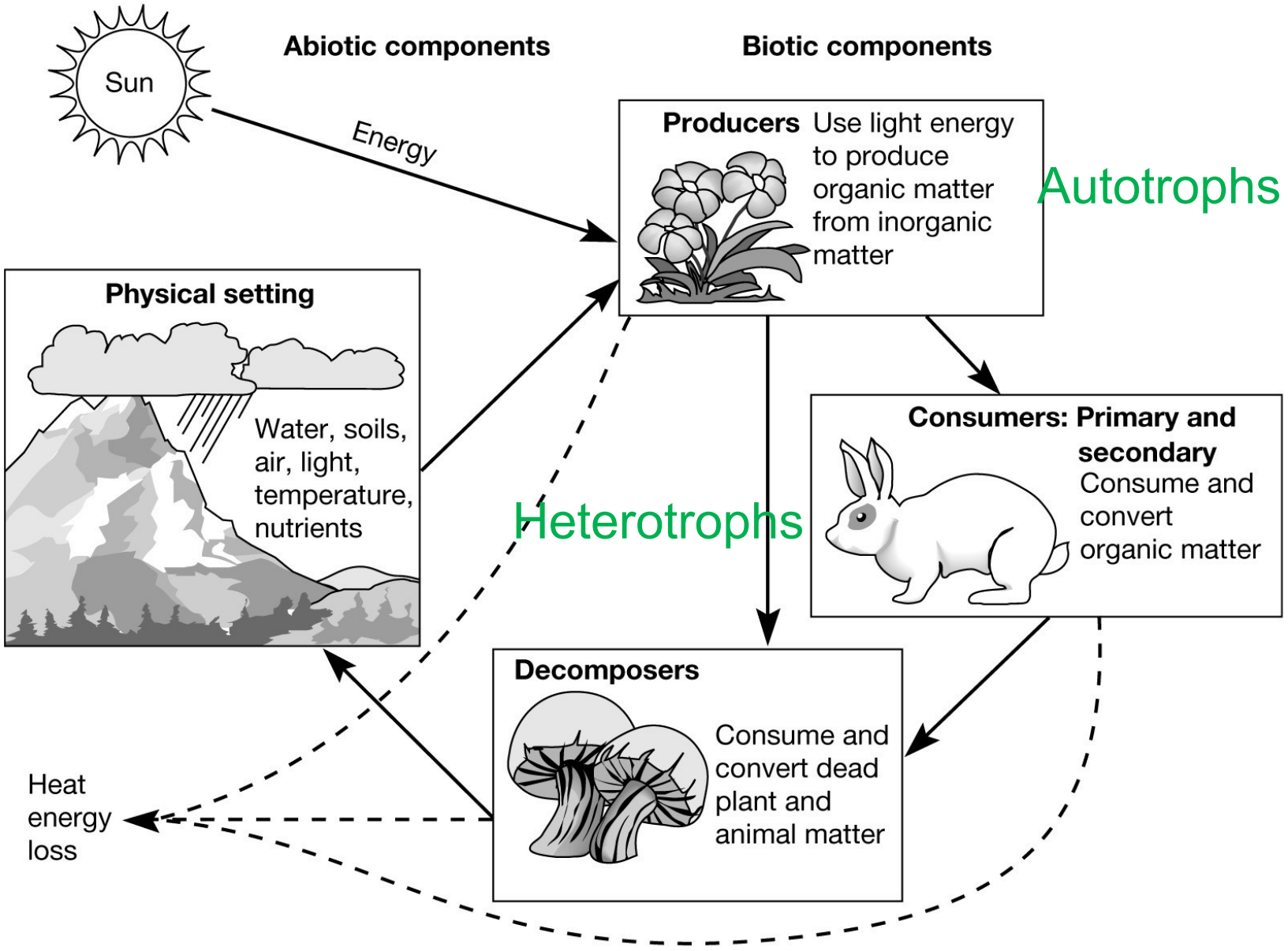
Folke et al., ”Regime shift, resilience…”,
In ‘Readings’ on myCourses

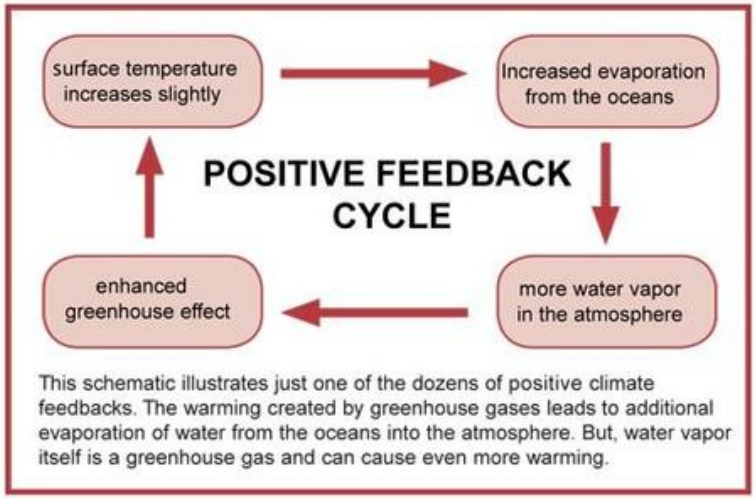
(this reading illustrates key concepts in this lecture: ecosystem regime shifts & resilience; applied to different ecosystems…)

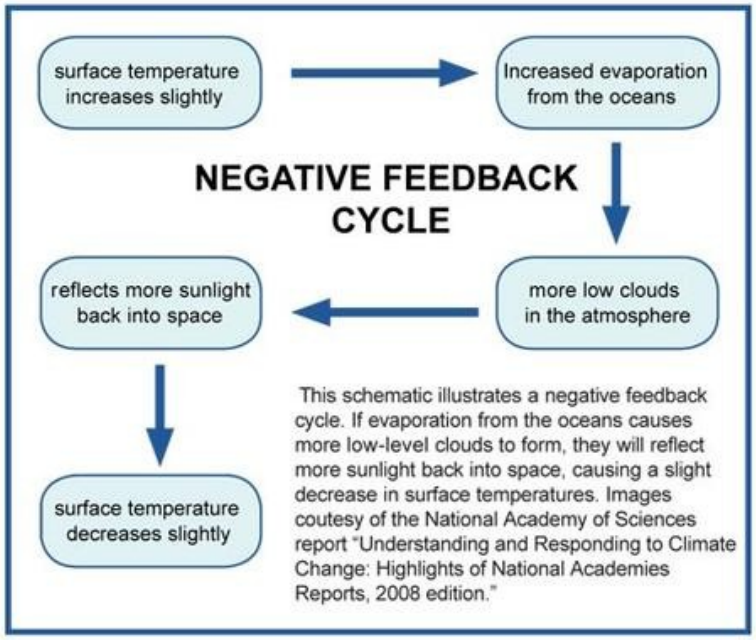
Concept of ‘Regime shifts’: sudden, large scale and hard to reverse transformations of a system, in response to slowly varying changes in some ‘driver(s)’

The role of human agents interacting with natural processes in triggering such ‘unexpected shifts’

# Ecosystem Components: but Feedbacks

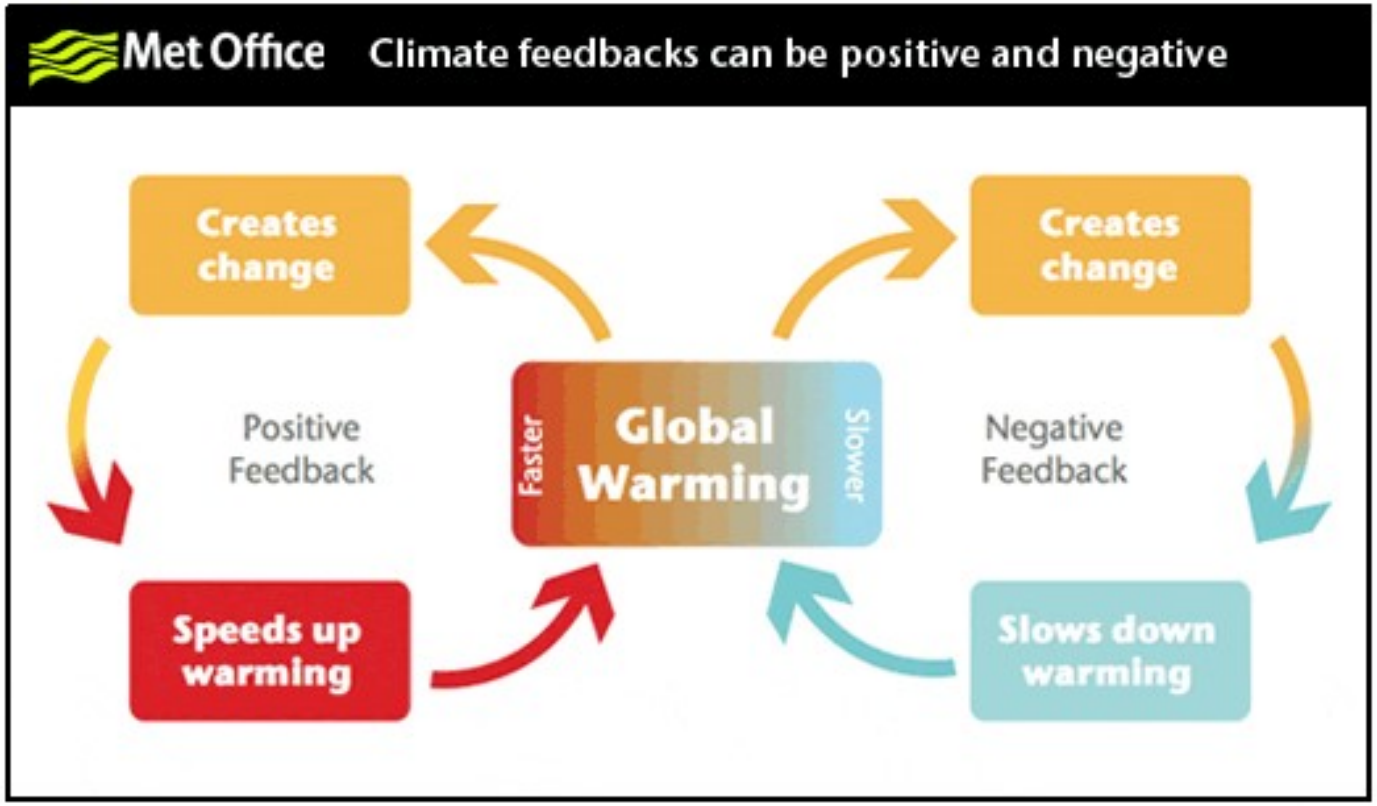




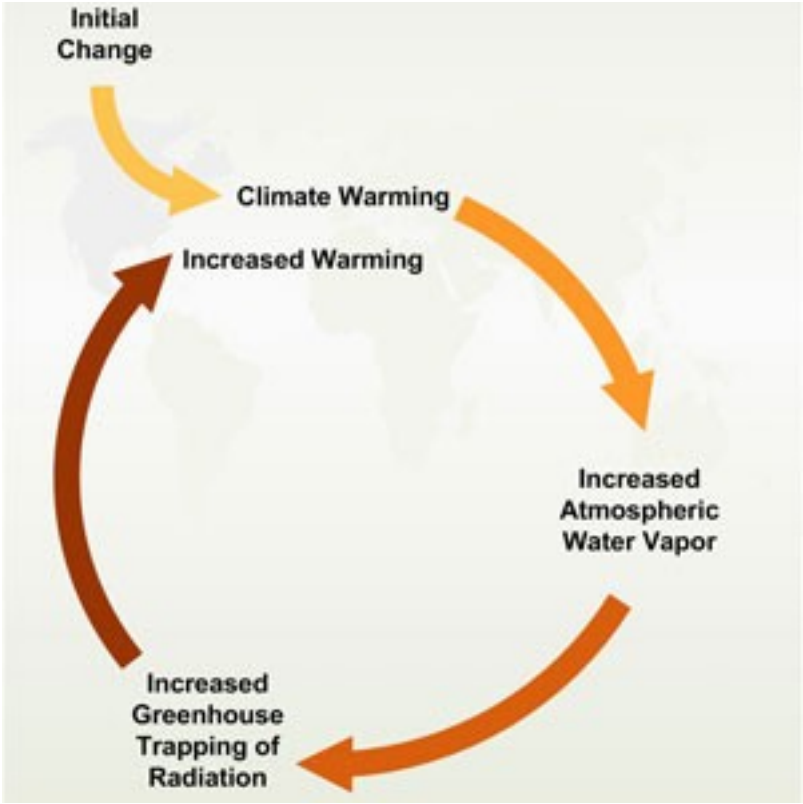



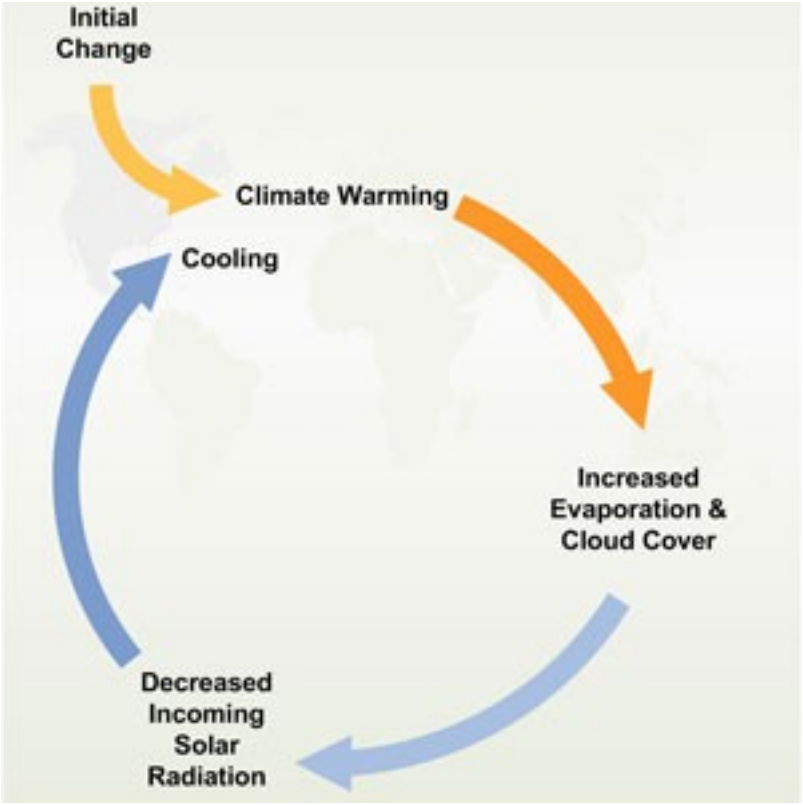

**Feedback Example: Earths climate system**: Sun insolation is an external driver, but the system state at any time involves feedback responses by oceans, land cover (albedo), human fossil fuel consumption, clouds, weathering of rocks , etc.

# Stability of any system depends on the interaction between a number of positive and negative internal feedbacks

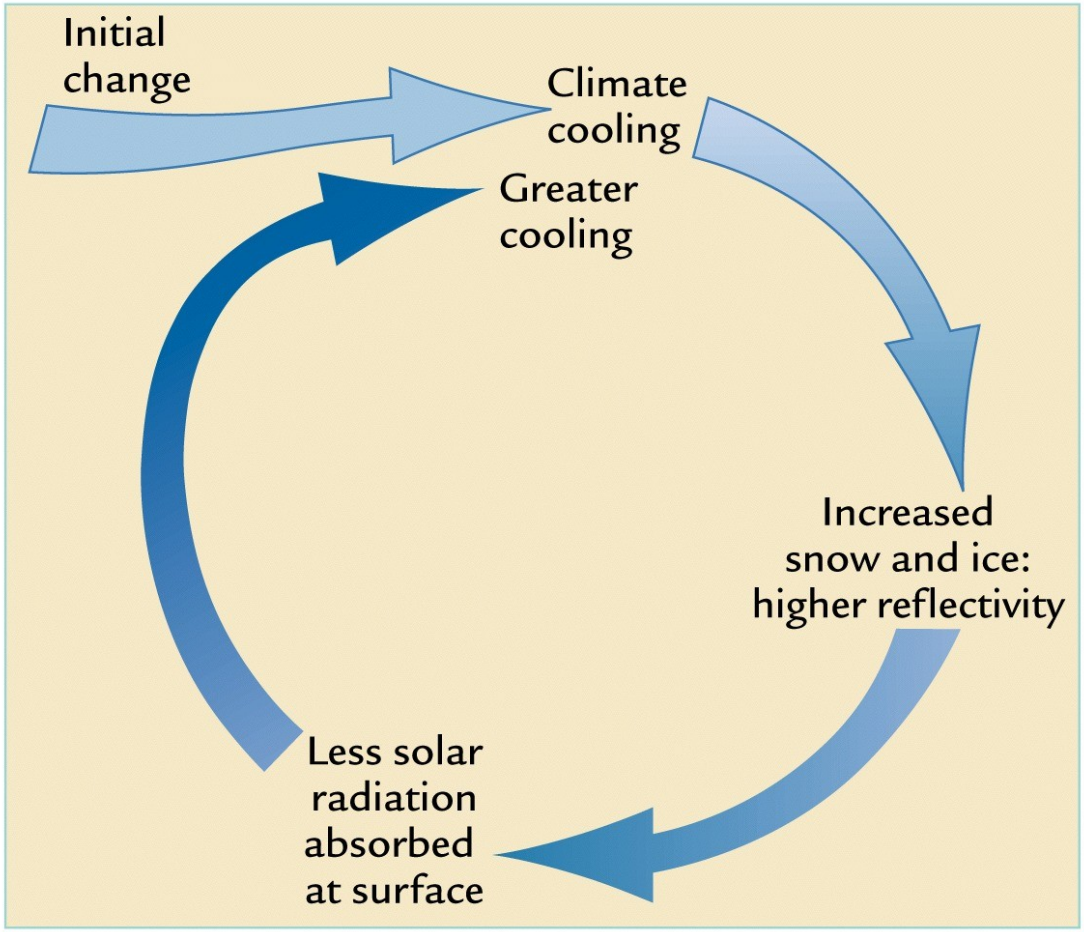


# There are dozens of other + and – feedbacks loops operating in the climate system, some will be explained in coming lectures

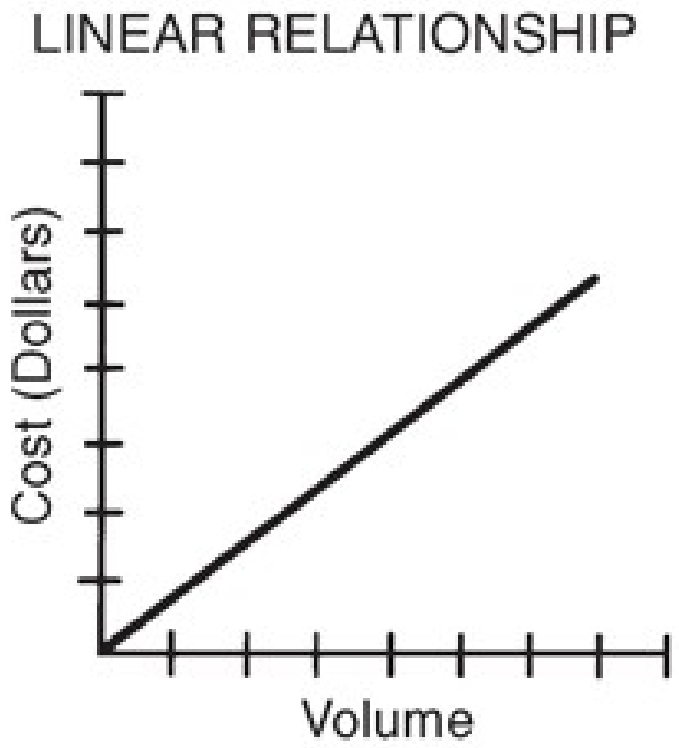


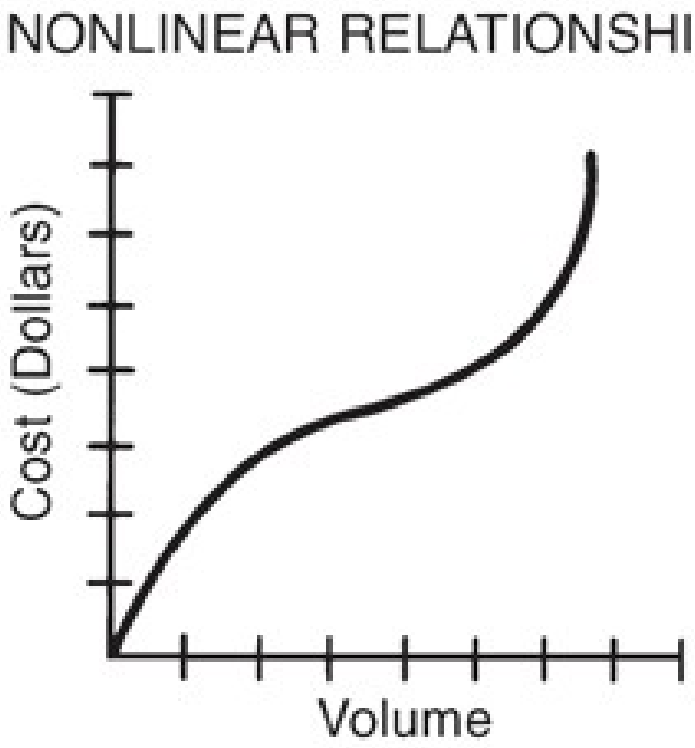



# Many other types of feedbacks govern climate at different time scales



Real system relations are not necessarily “linear”.
Changes may kick in more strongly only at certain levels



## Short TASK: Due Thursday after break before midnight

Think of two feedbacks in your experience:

A Positive (re-enforcing) Feedback

A Negative (diminishing, restoring) Feedback

Post them in 'Assignments' -we'll compile the early submissions and discuss them in class on Monday after break. Creativity is nice, but the feedback examples won't be graded –they are for discussion.

Some feedbacks discussed in class:
El Niño - La Niña southern oscillation
Ice-albedo feedback
Clouds (shortwave versus longwave behavior)

How stable are lake ecosystems? (how unchanging are there chemical and biological characteristics?)





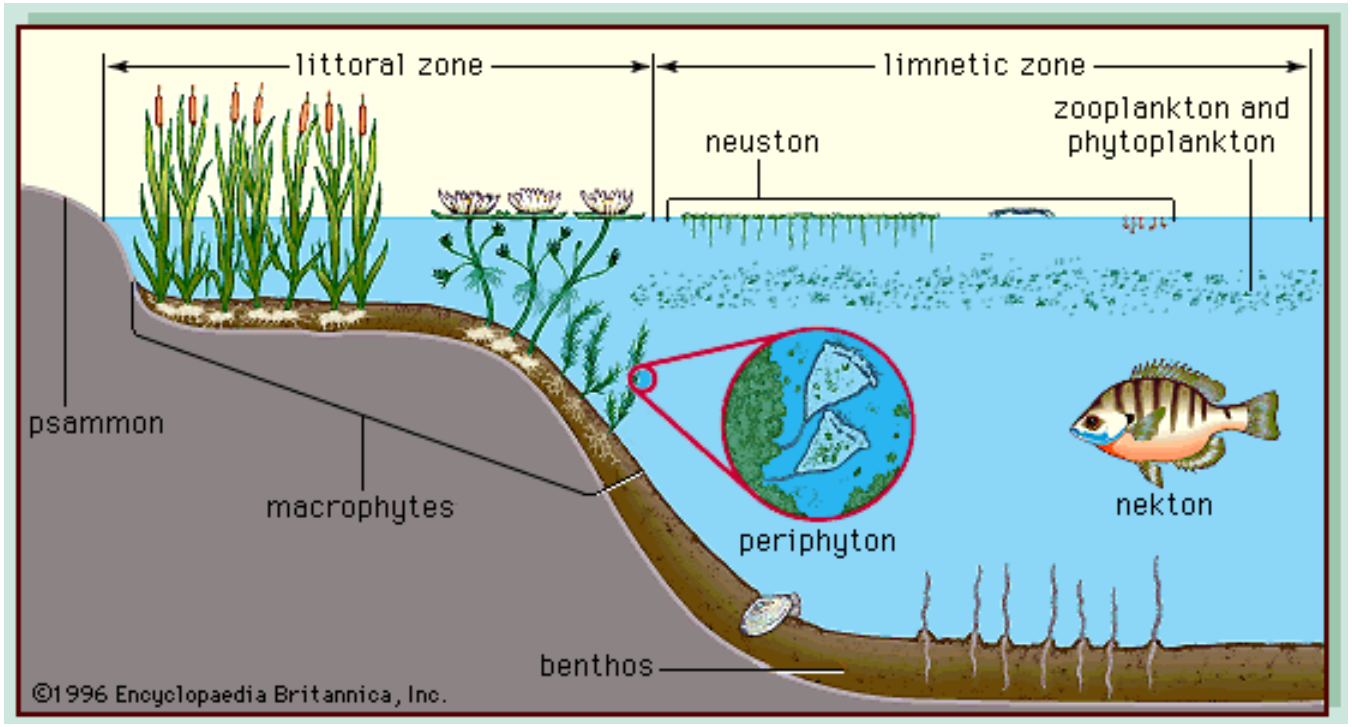
Healthy lakes with good dissolved oxygen levels and diversity of fish species





No sudden change had occurred  in the watershed, so that the ‘culprit’ driver of the change was long mysterious

# Lake Ecosystem

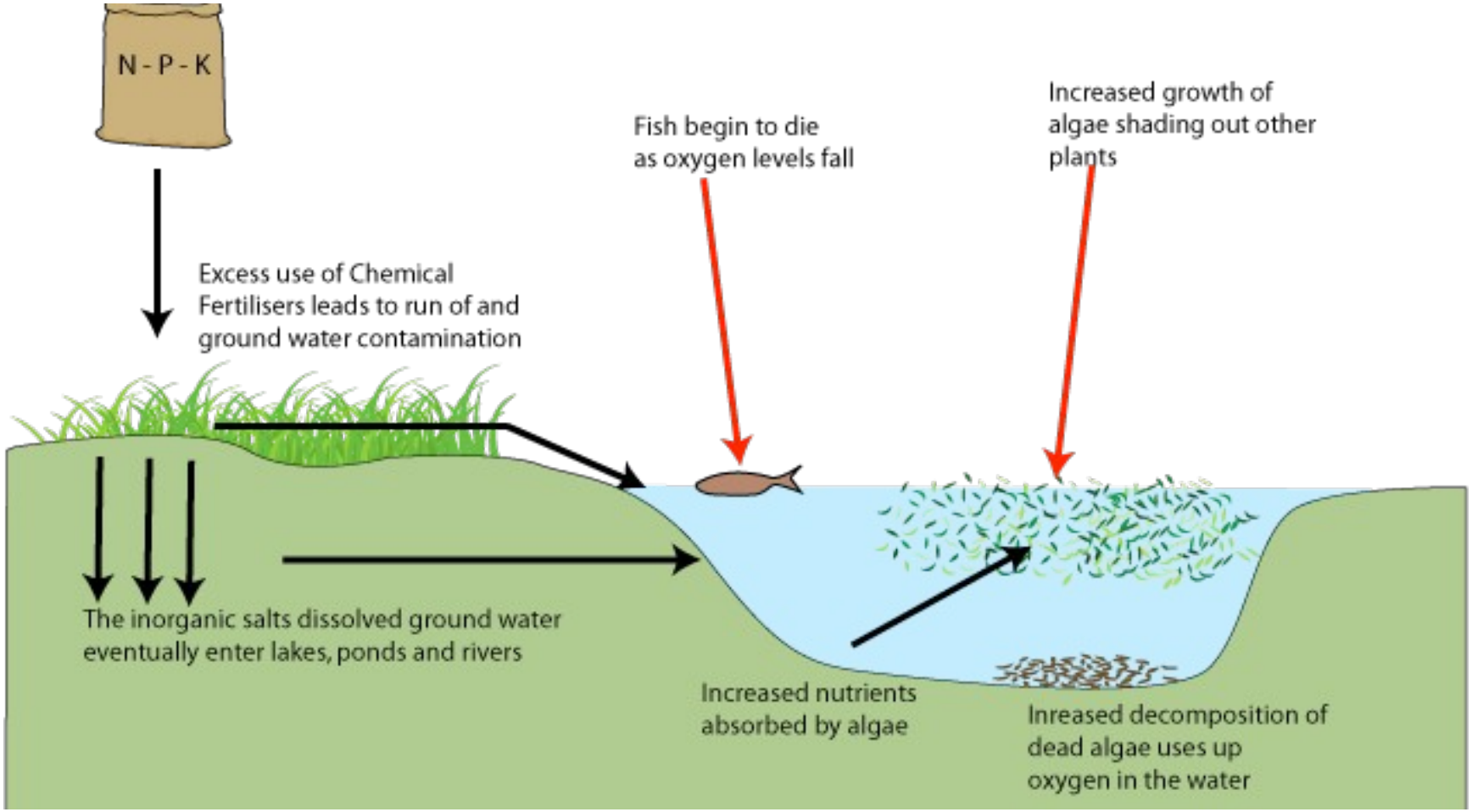




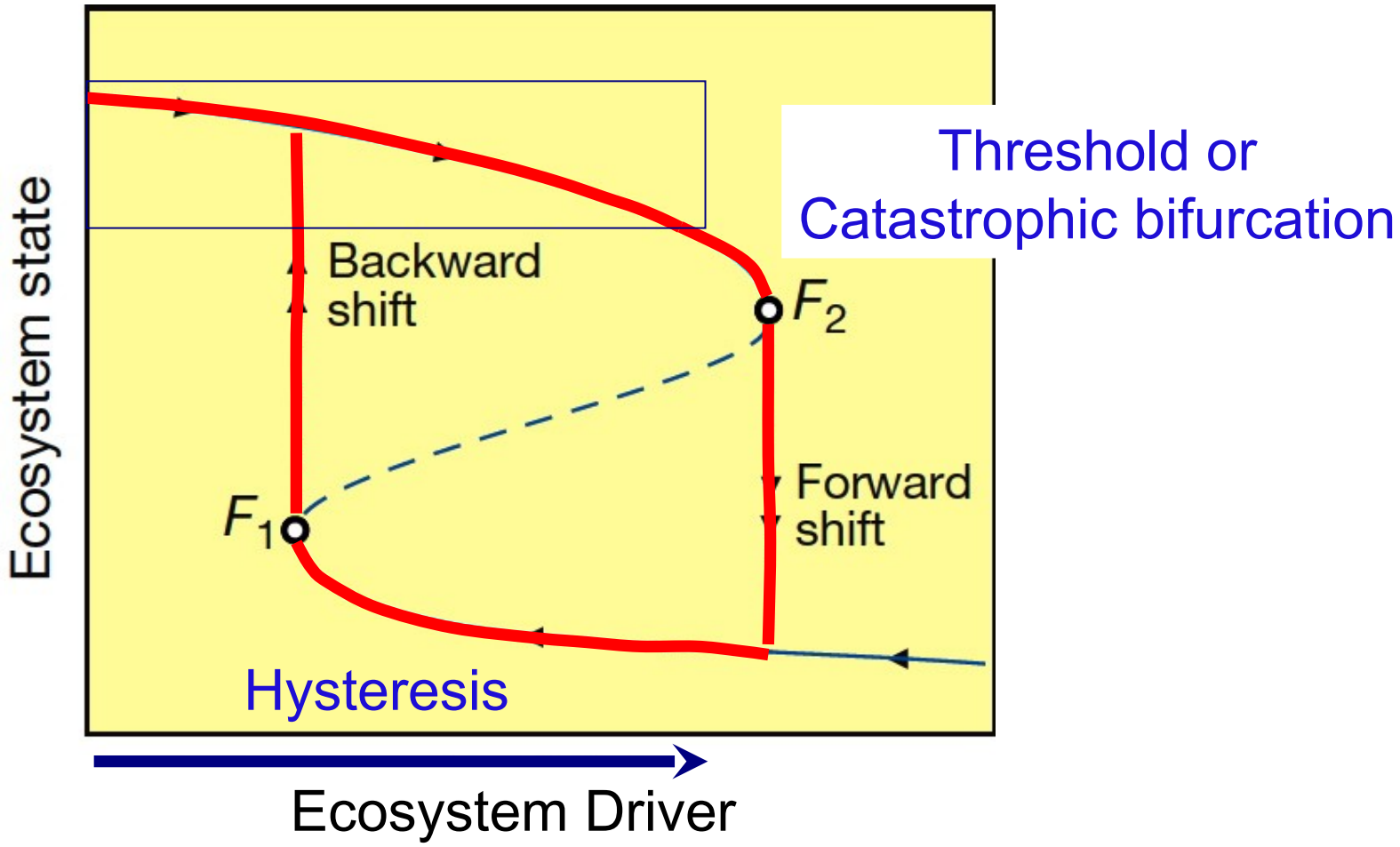

Plankton

Photosynthesis of rooted plants (in shallow zones) and of phytoplankton (tiny alga, in water column) provide the food energy for the ecosystem: for tiny animals (zooplankton, also in water column) and ultimately for fish.

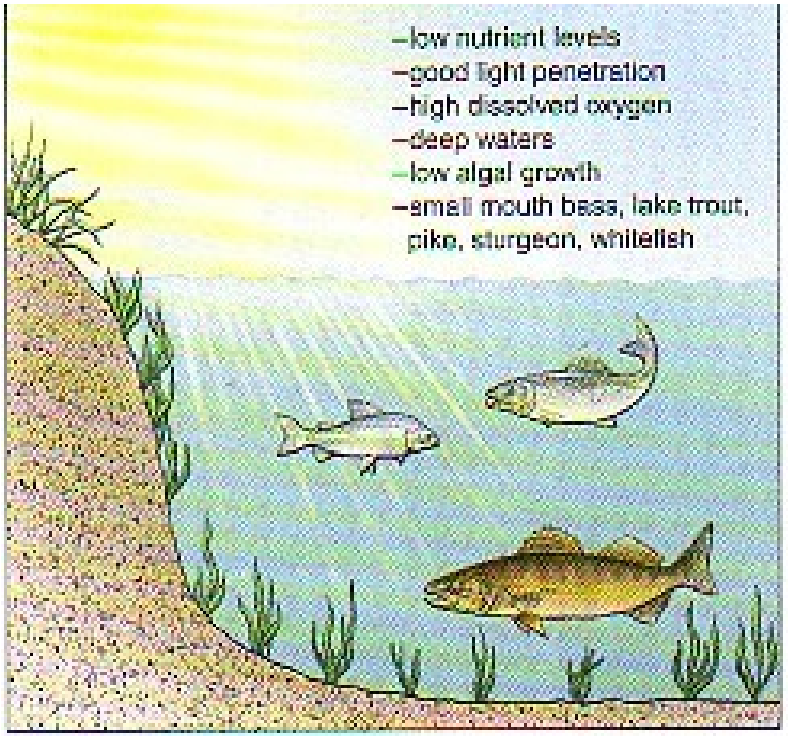
# Schematic: main drivers and main effects of anthropogenic eutrophication



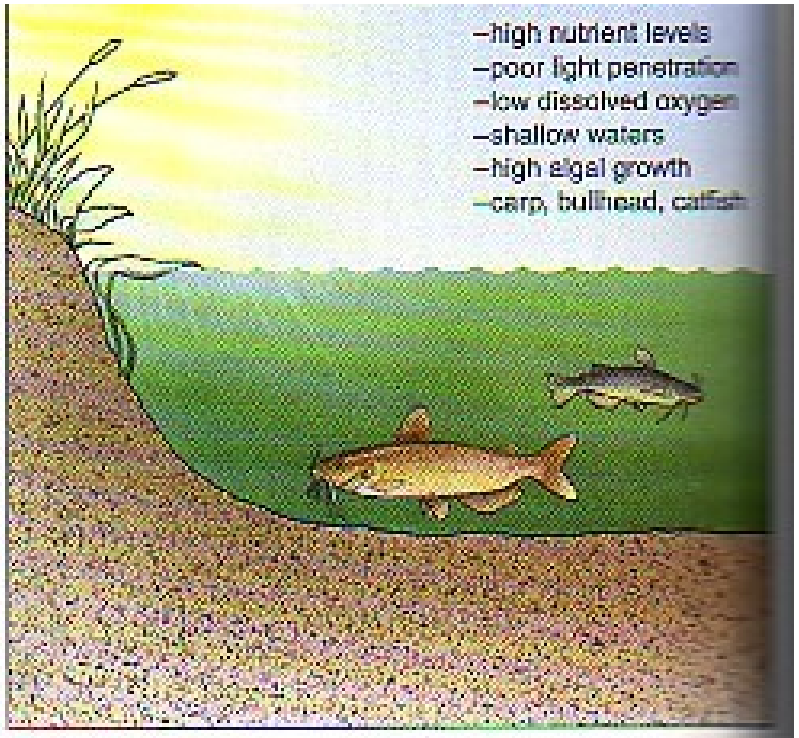
# Alternative Stable States and Catastrophic Bifurcations



# Two stable lake regimes:




(a) Oligotrophic lake




(b) Eutrophic lake

Clear Water Lakes

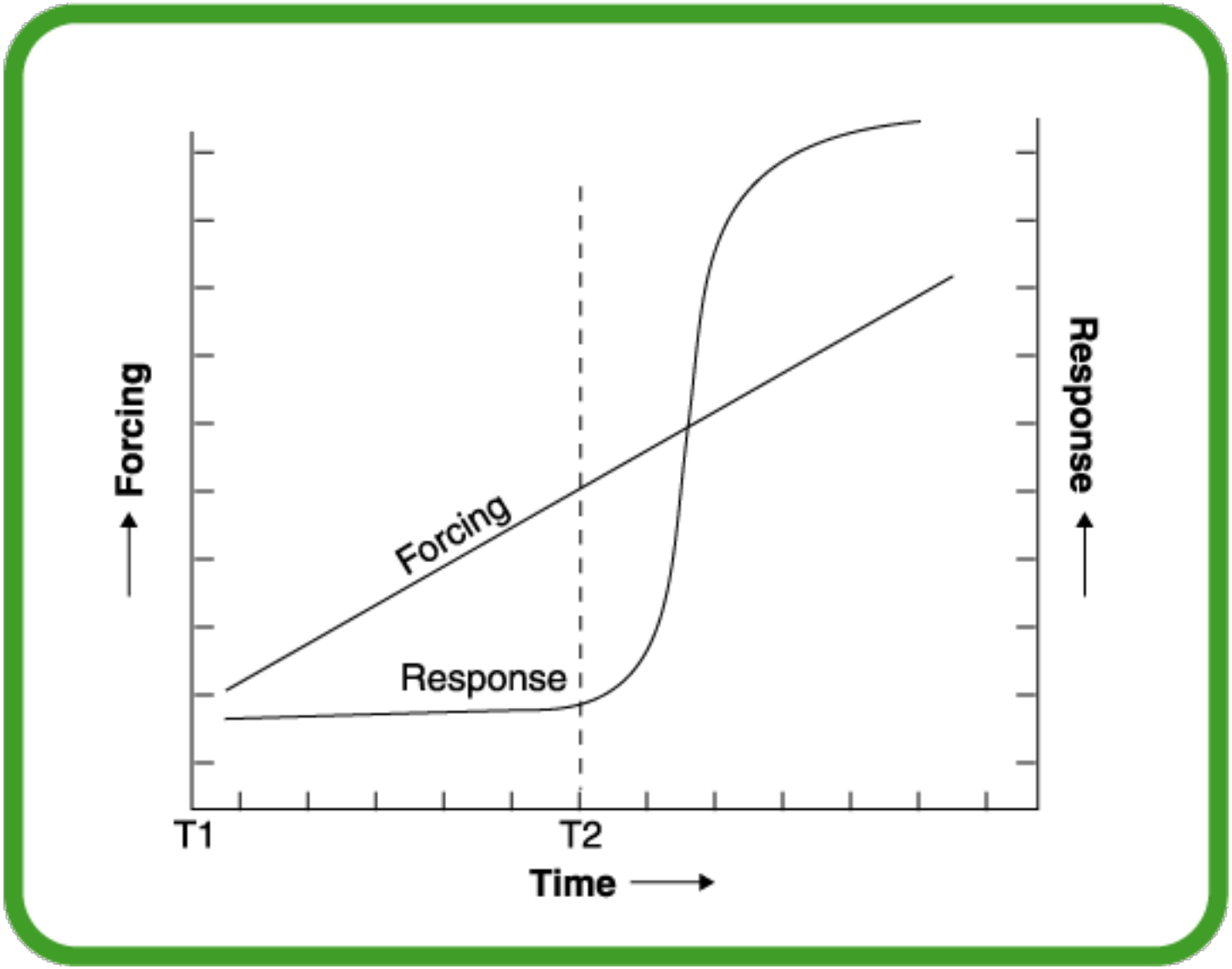
And

Eutrophic Lakes

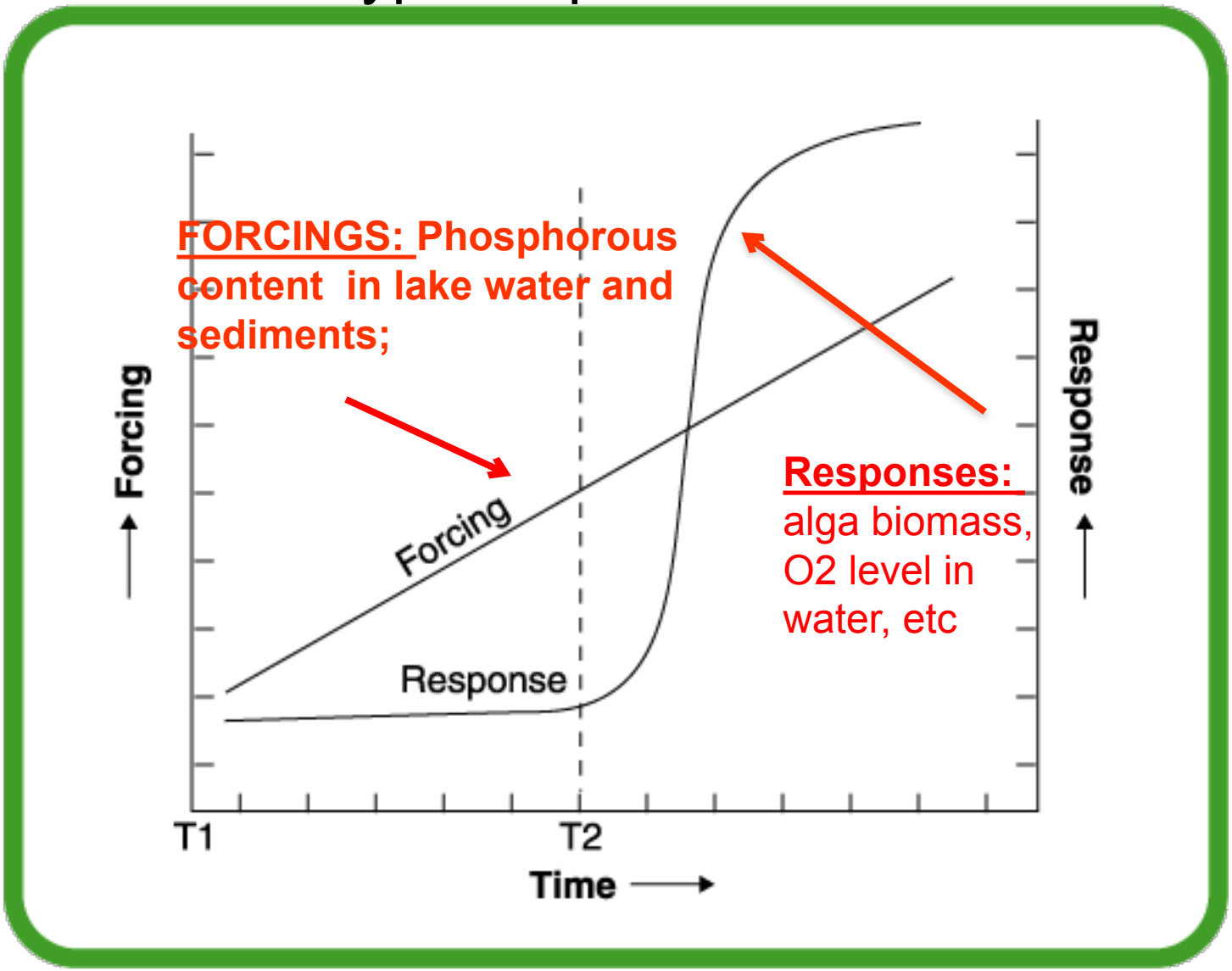




# A Characteristic of regime shift

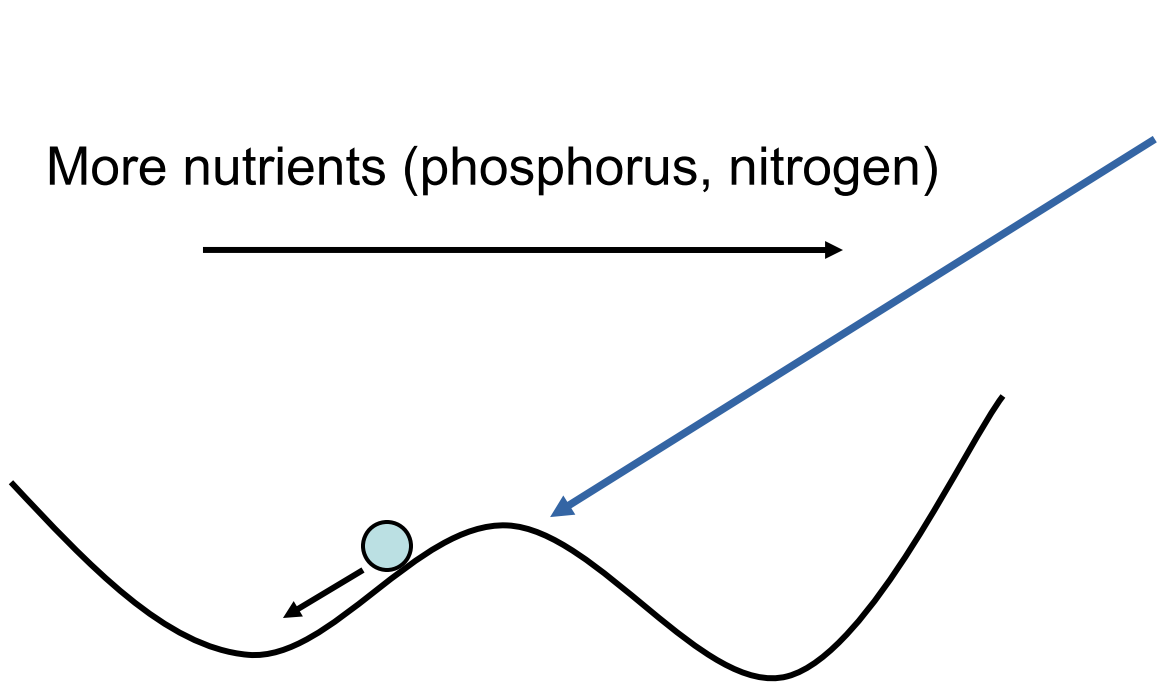


# Characteristic of regime shift: a sudden, threshold type response

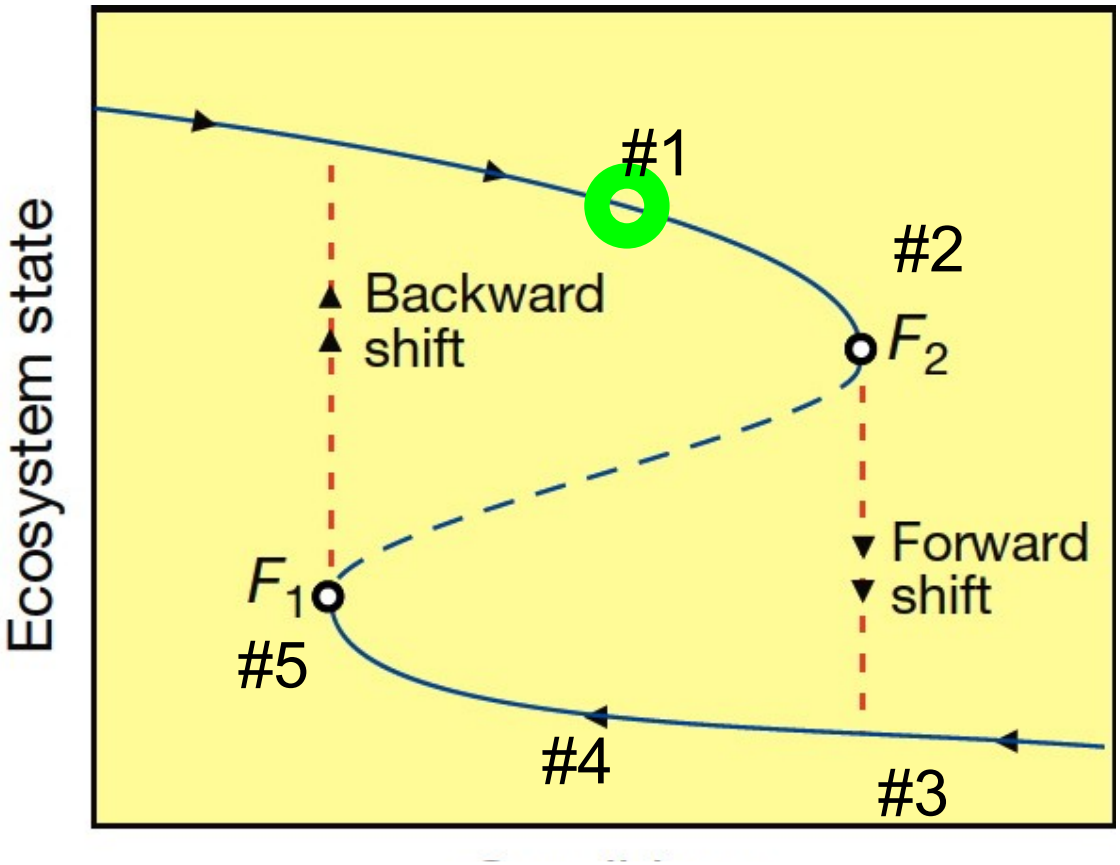


# Regime Shift

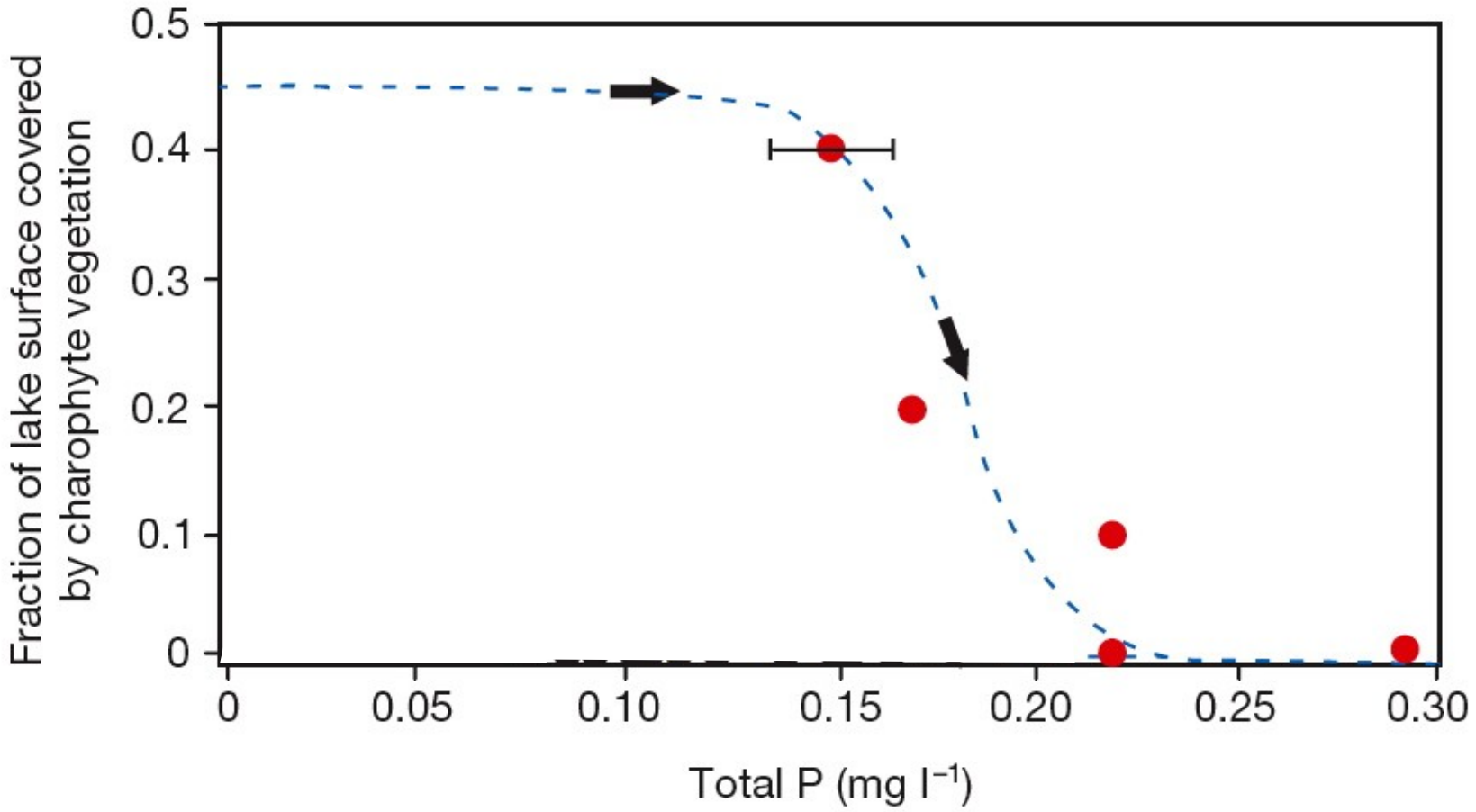
A Conceptual model of twin regimes in the lake



# Alternative Stable States and Catastrophic Bifurcations

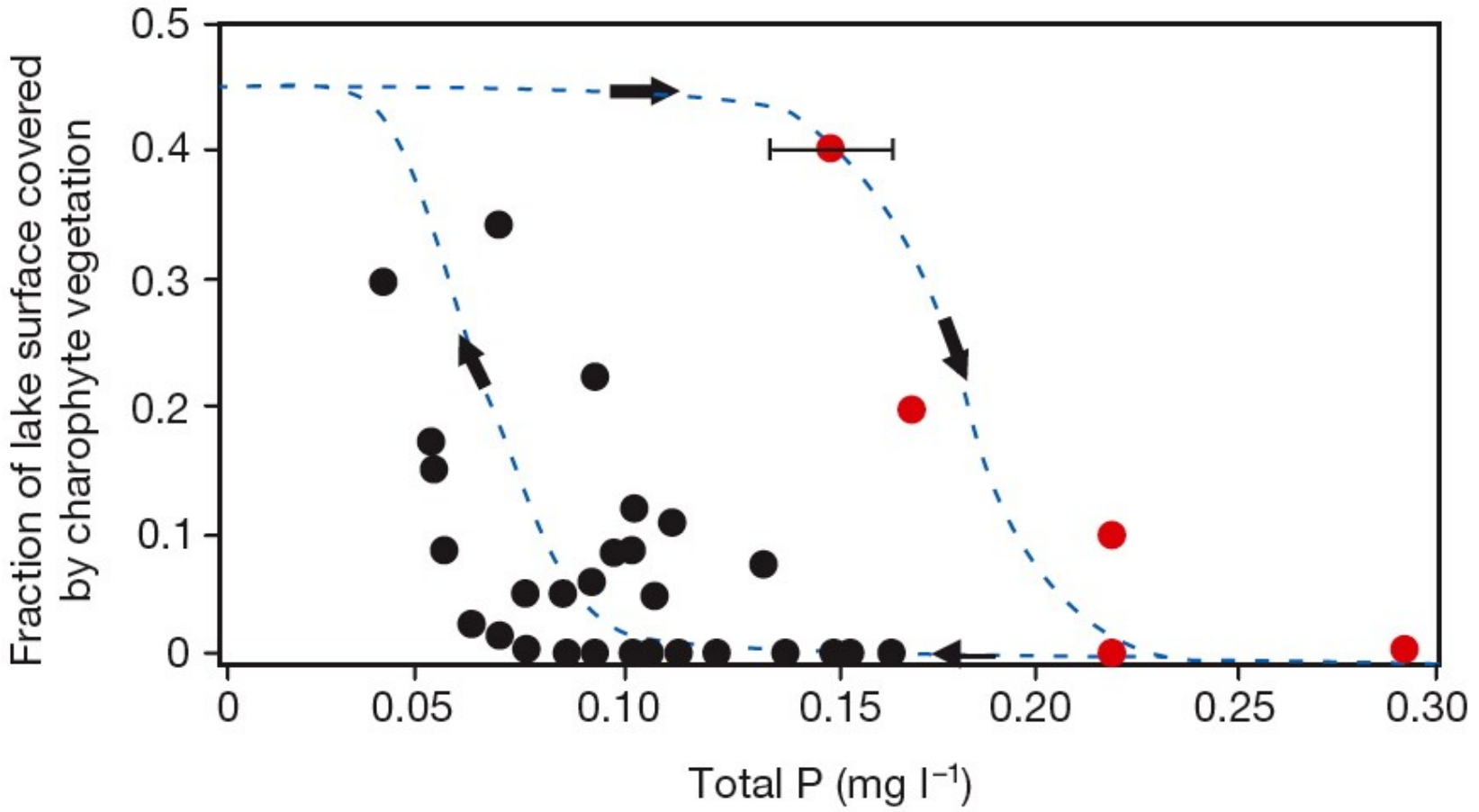


# Lake Eutrophication Example




Scheffer et al 2001

# Lake Eutrophication Example




Scheffer et al 2001

# Sudden transitions and Hysteresis

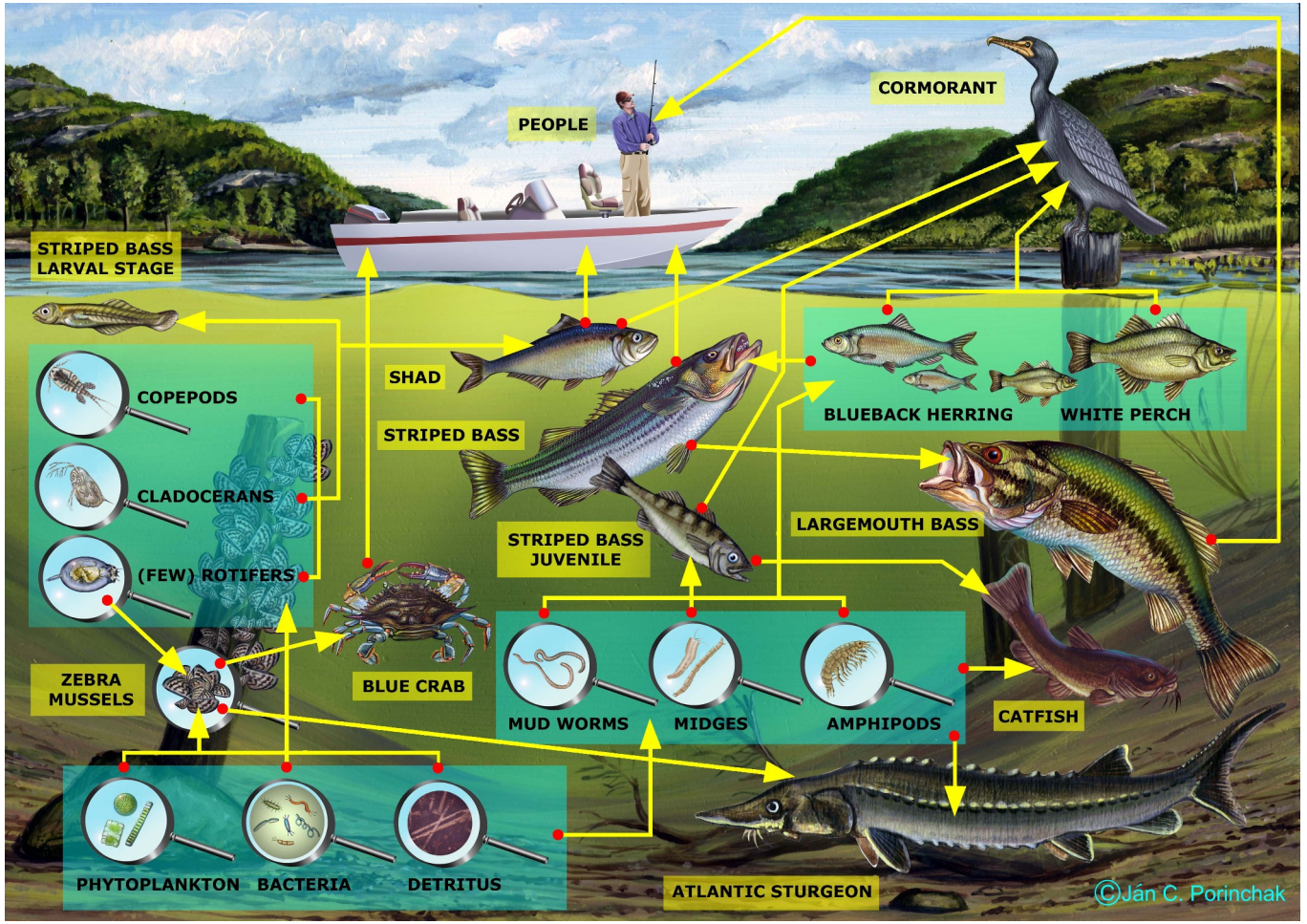
When you add the straw that breaks the camel's back (i.e. reach a threshold)….



1. The break occurs suddenly (often a surprise)

2. Removing that straw, will not heal its back (hysteresis)

# Freshwater ecosystem




But the ecosystem nutrient web is complex. How do we know the limits (resiliency) of the system?

# Experimental Lakes Area



ELA




Pioneering Science of
David Schindler

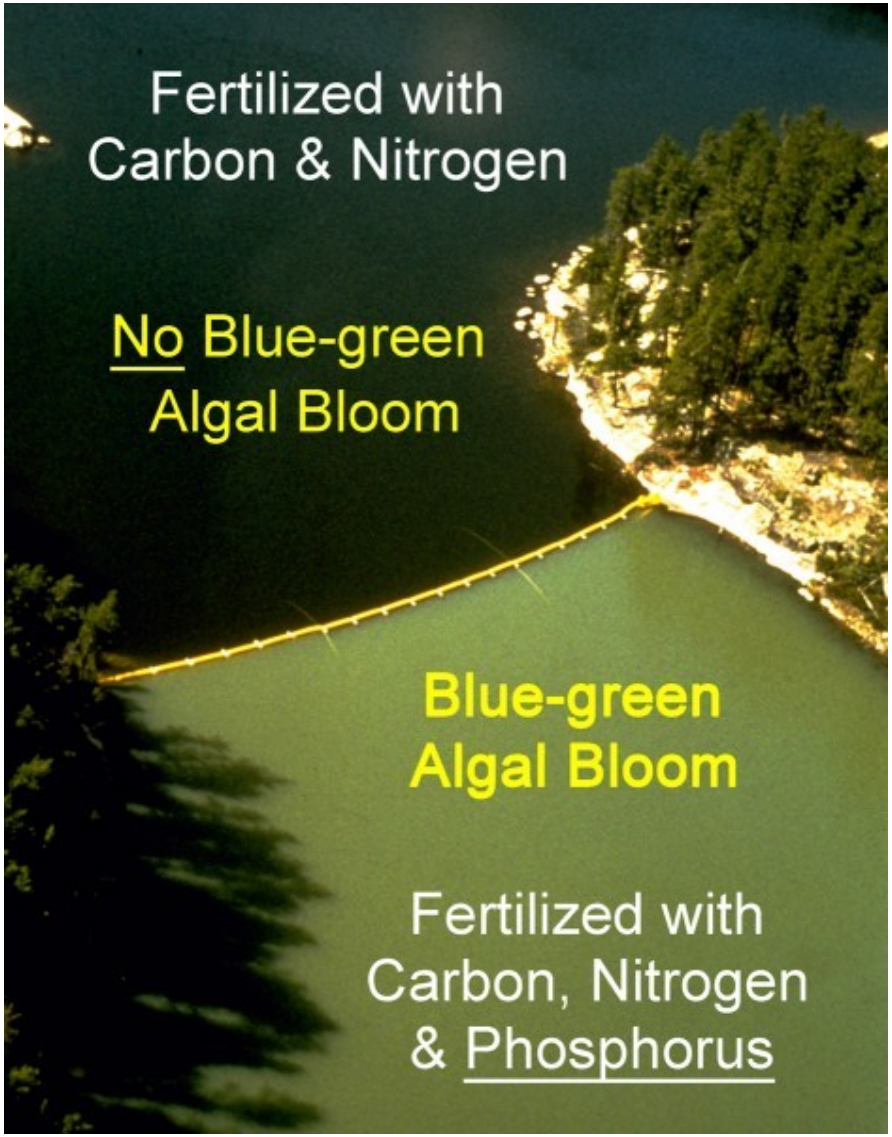
# Experimental manipulation: healthy (oligotrophic) lake (top) vs Eutrophic lake (bottom)



Experimental Lakes Area
(near Kenora, Ontario)

Pioneering Science of
David Schindler

# Lake 226



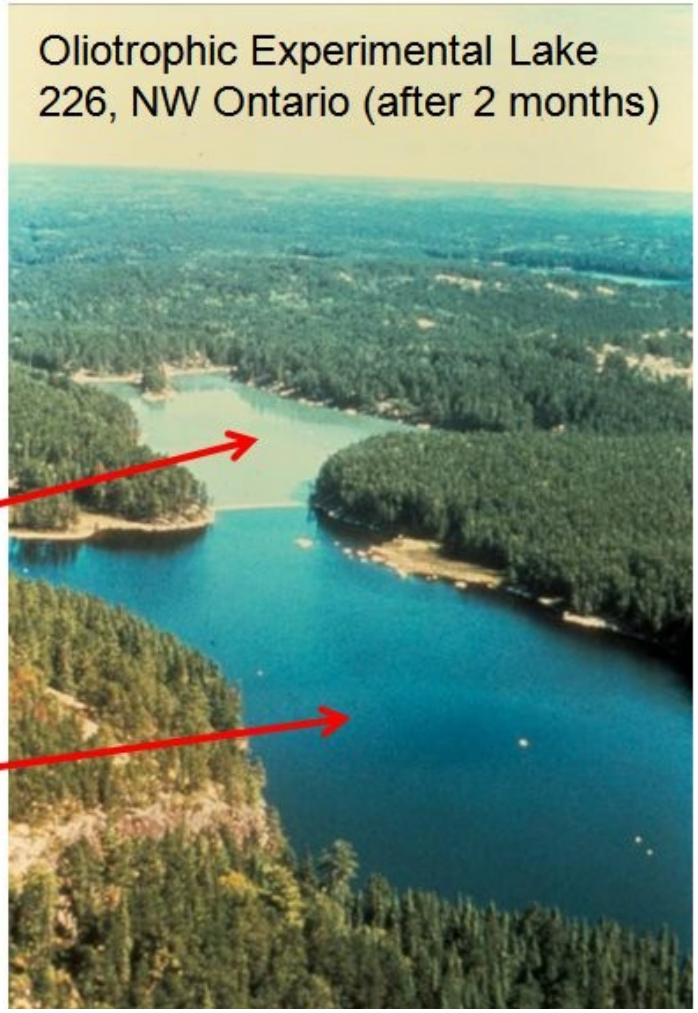

Algae overwelmed the fertilized side

# Experiments leading to policy

- Schindler's (1974) study gave most compelling evidence for phosphorus being the cause of man-made eutrophication
- Legislation was later adopted limiting P in detergents and effluents

fertilized with P, N and C

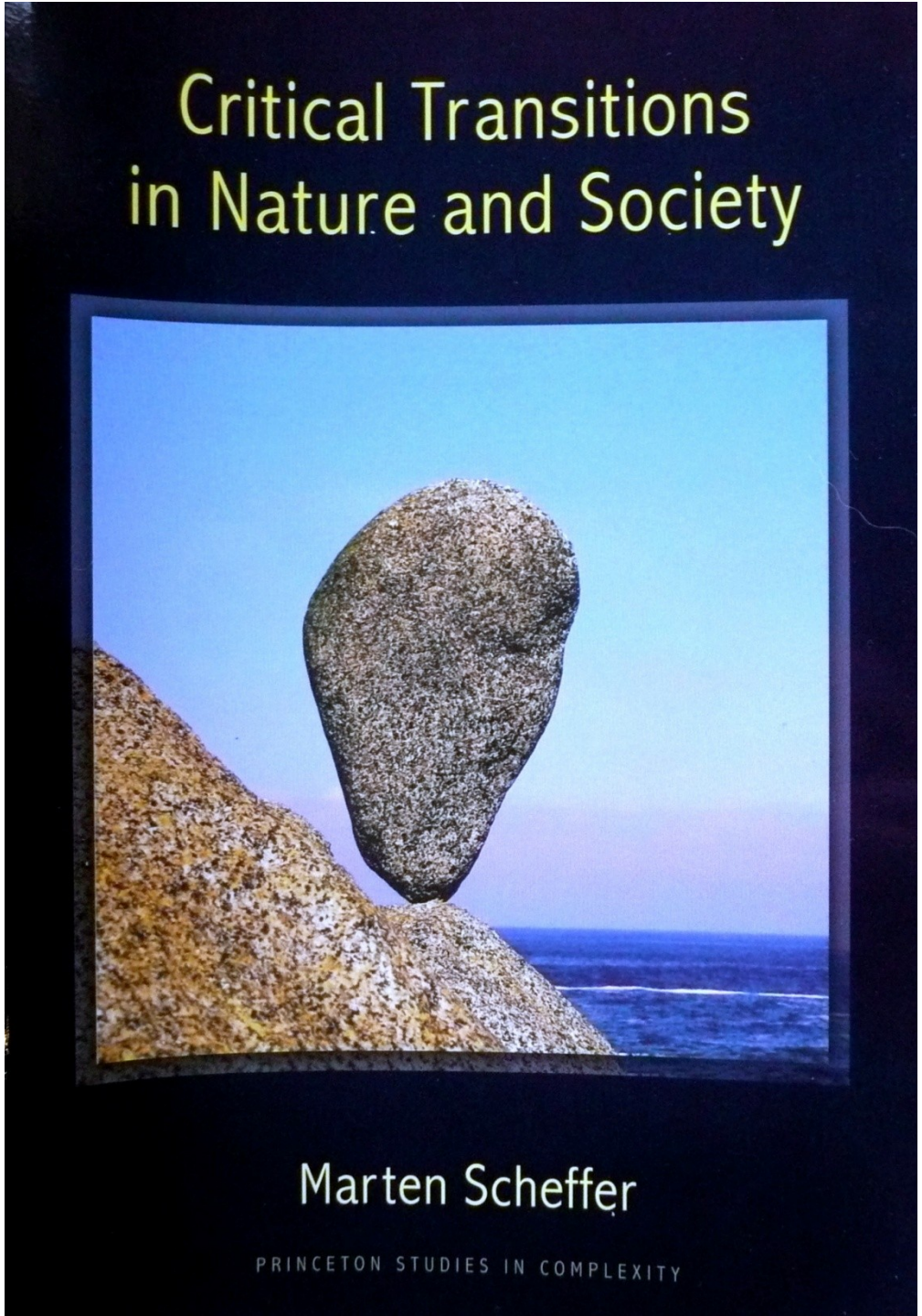
Basin fertilized with only C and N




The Experimental Lakes area defunded by the Harper government; saved by the International Institute for Sustainable Development (UN and private funding, now with some government funding.)

# Estérel Resort, north of Montreal






Equilibrium

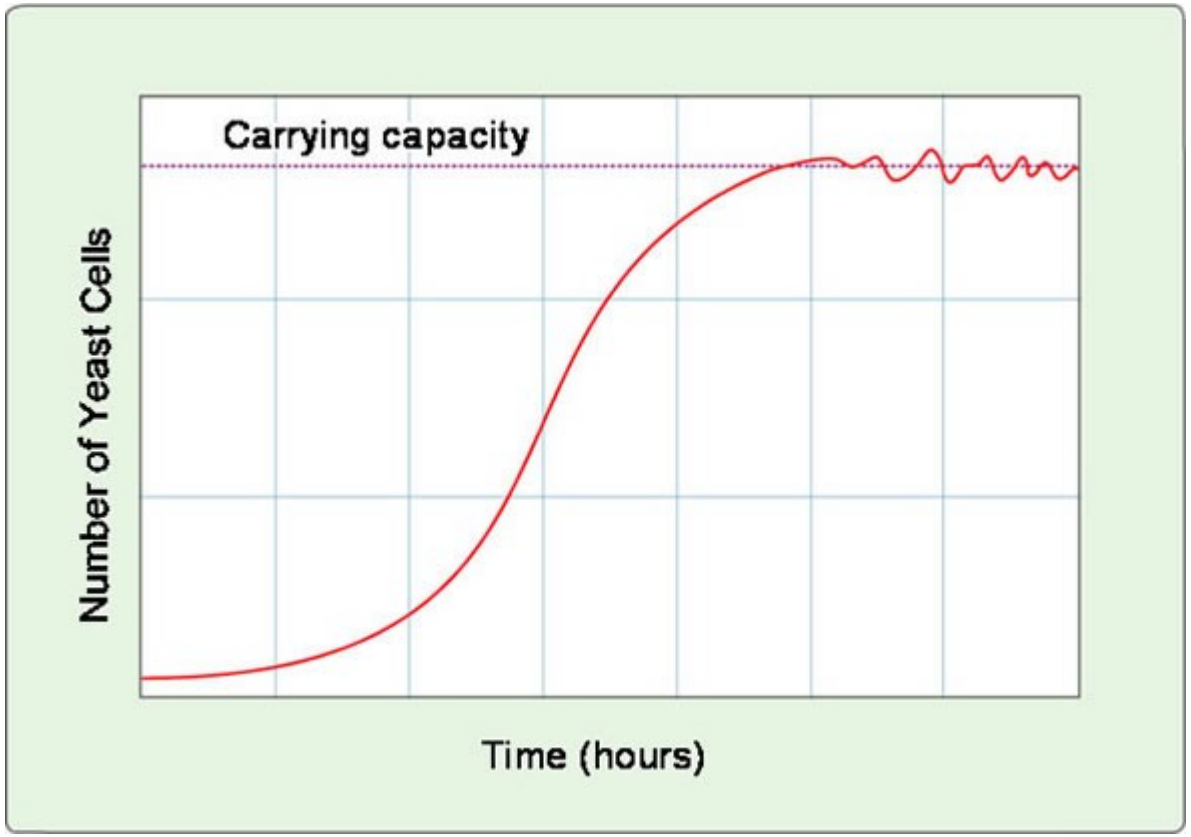
Competing Equilibrium

Tipping Point or Regime Shift

Resilience

# Carrying Capacity

For most organisms in nature, population growth slows to zero as population approaches a limit: the 'carrying capacity'. This is a dynamic equilibrium.

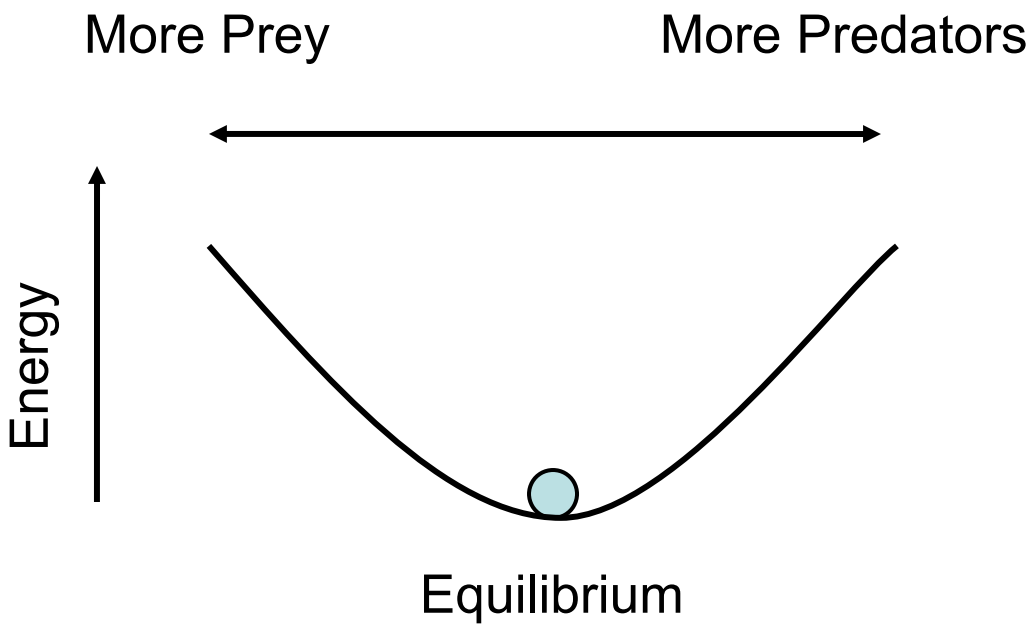


# Predator-Prey relationships

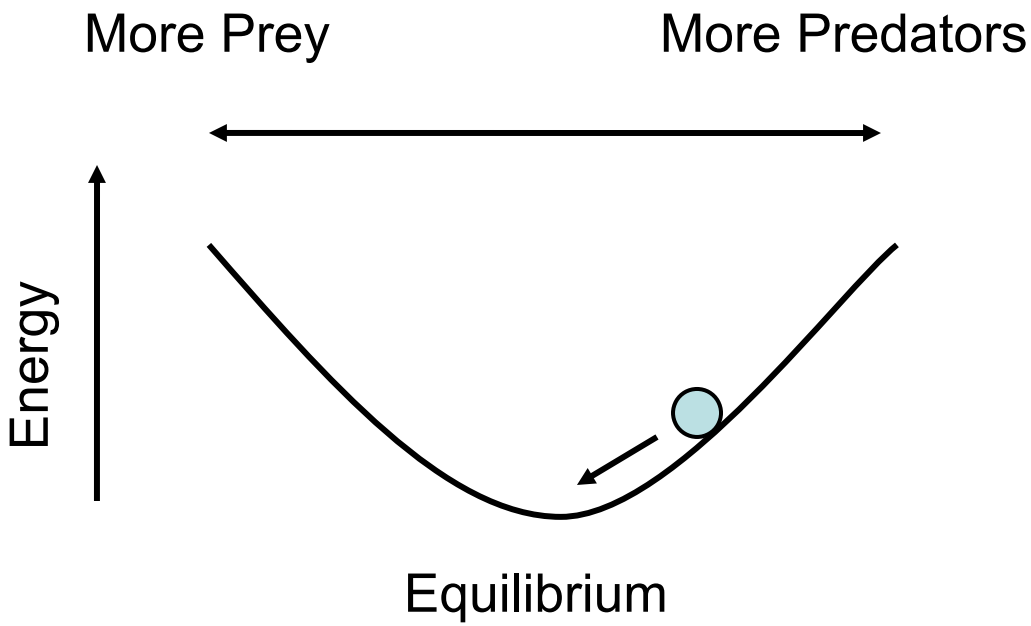


Each depend upon the other: the prey is food, but the predator keeps the population in control.

# Negative Feedbacks

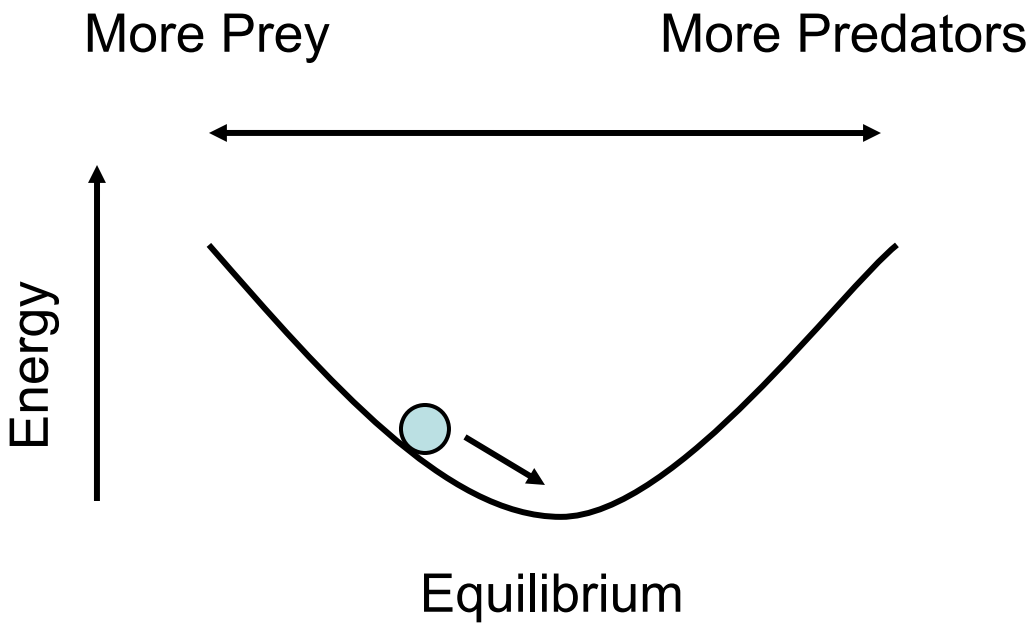


# Negative Feedbacks



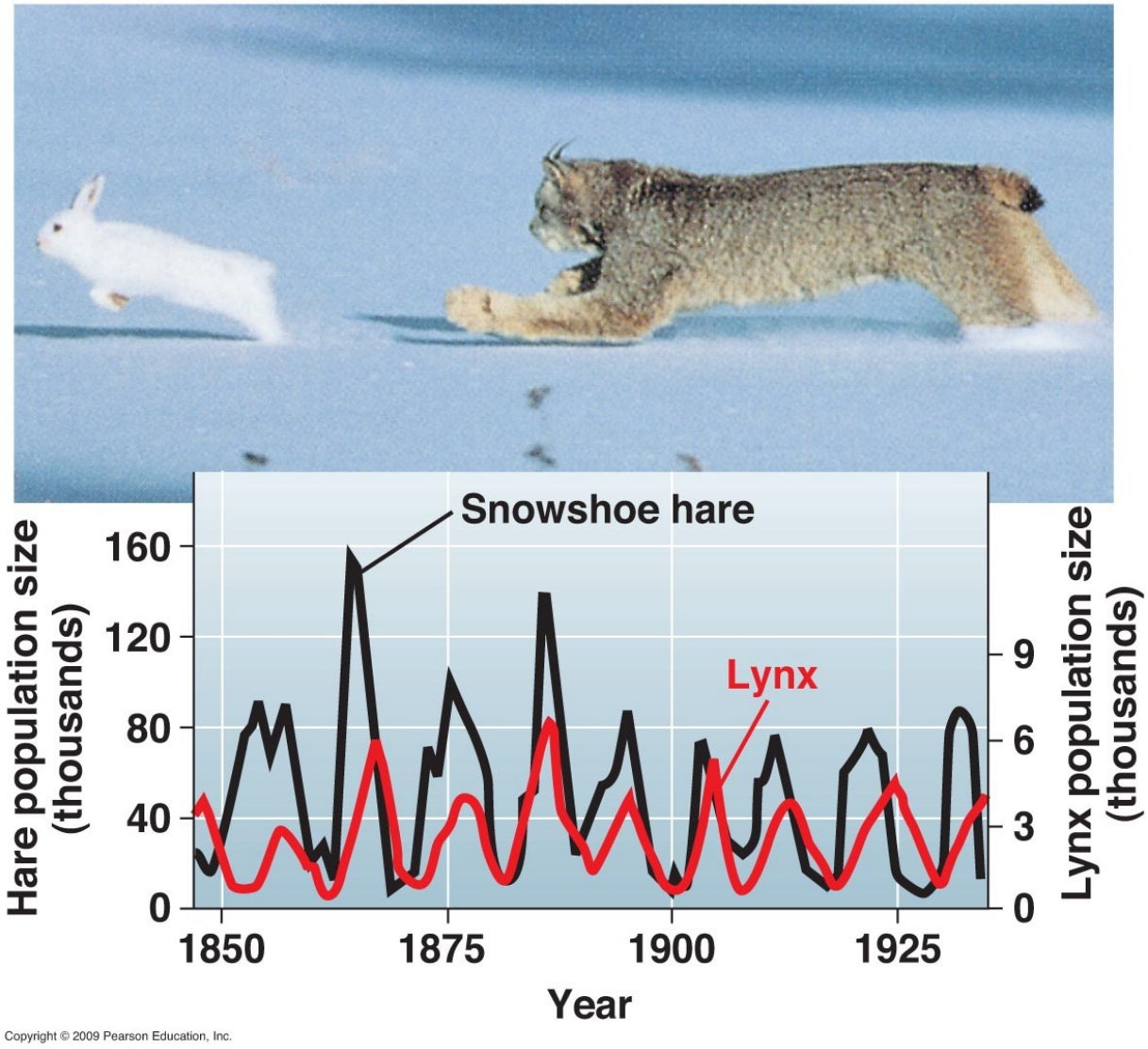

This is a dynamic equilibrium

# Negative Feedbacks




This is a dynamic equilibrium

# Steady state

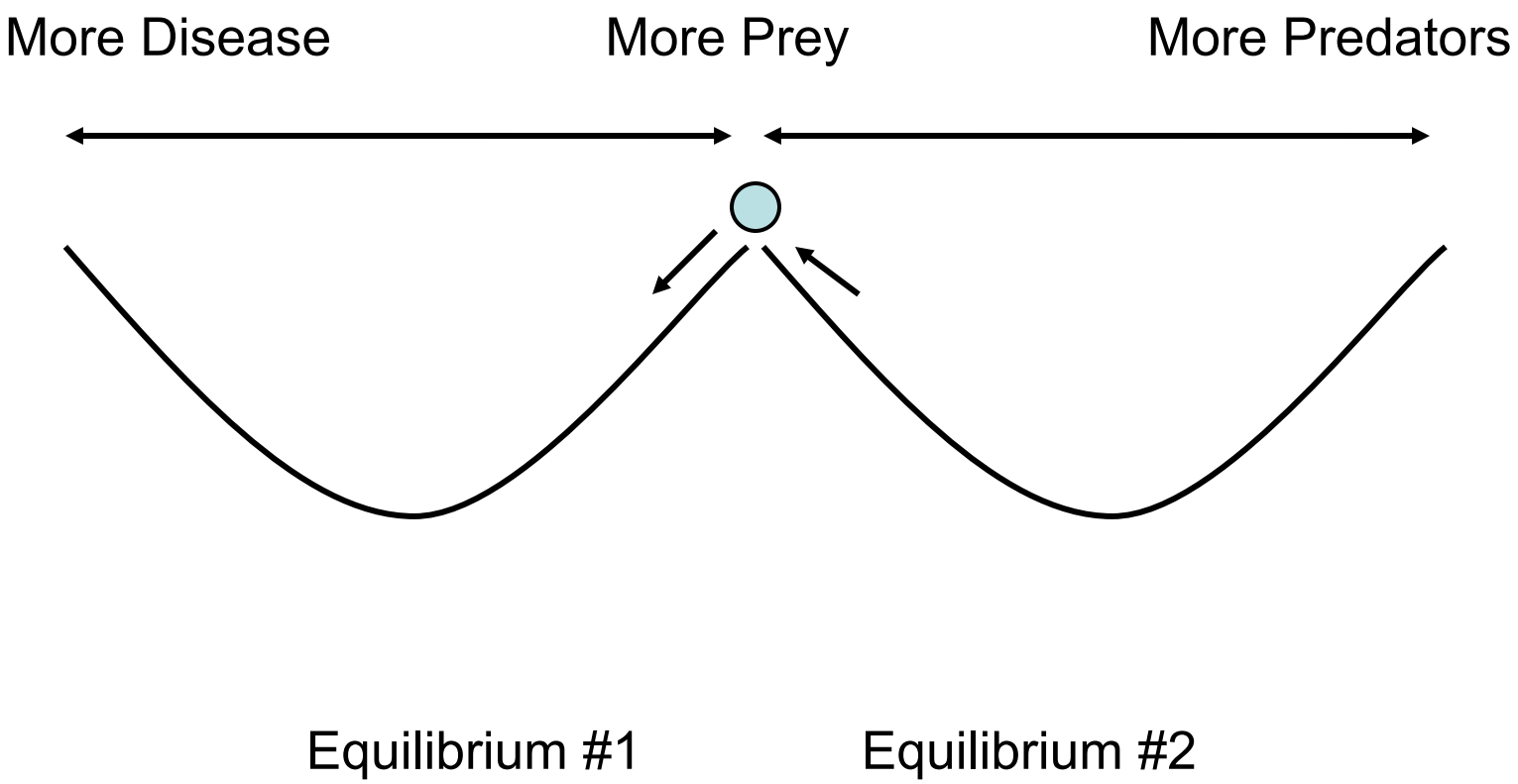



Copyright © 2009 Pearson Education, Inc.

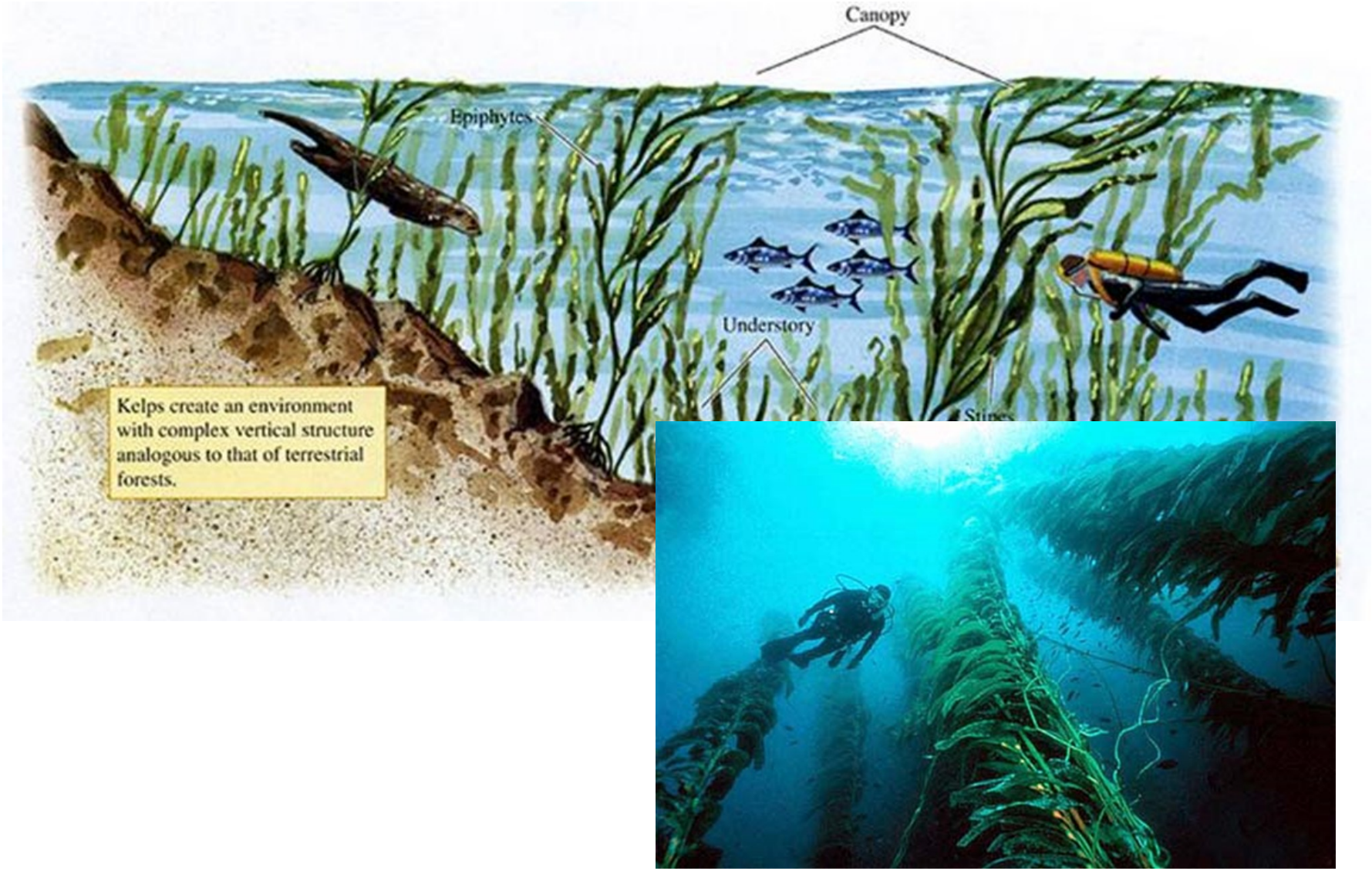
A simple dynamic equilibrium: negative feedback

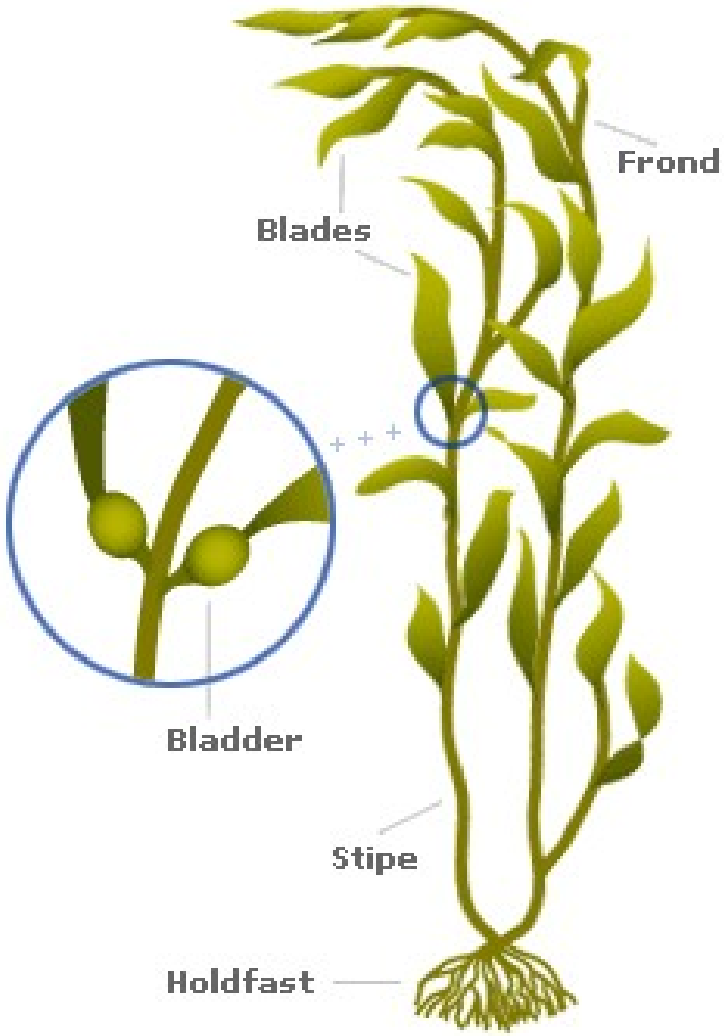
# Regime Shift

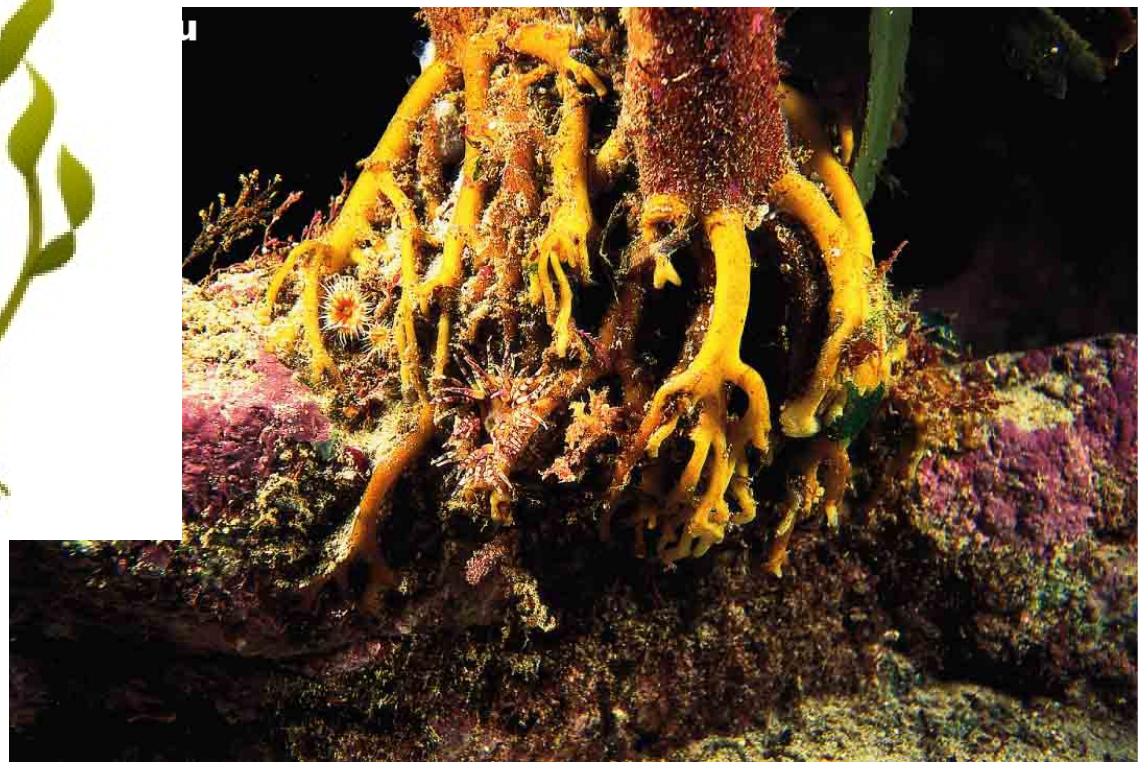
A Conceptual model of twin ‘system’ equilibria



# Rocky , cold temperate Sea Coasts and kelp ‘forests’




Frond
Blades
Bladder
Stipe
Holdfast



‘Regime shift’: Some ‘kelp forests’ can suddenly become ‘barren’ (at right). How???



# Urchin “barren” state :  kelp all gone



A diverse community of  organisms find feeding, hiding and reproductive habitats in this kelp forest canopy



Starfish and sea urchin species avidly eat into the kelp holdfasts, (which hold the kelp frond to bottom)

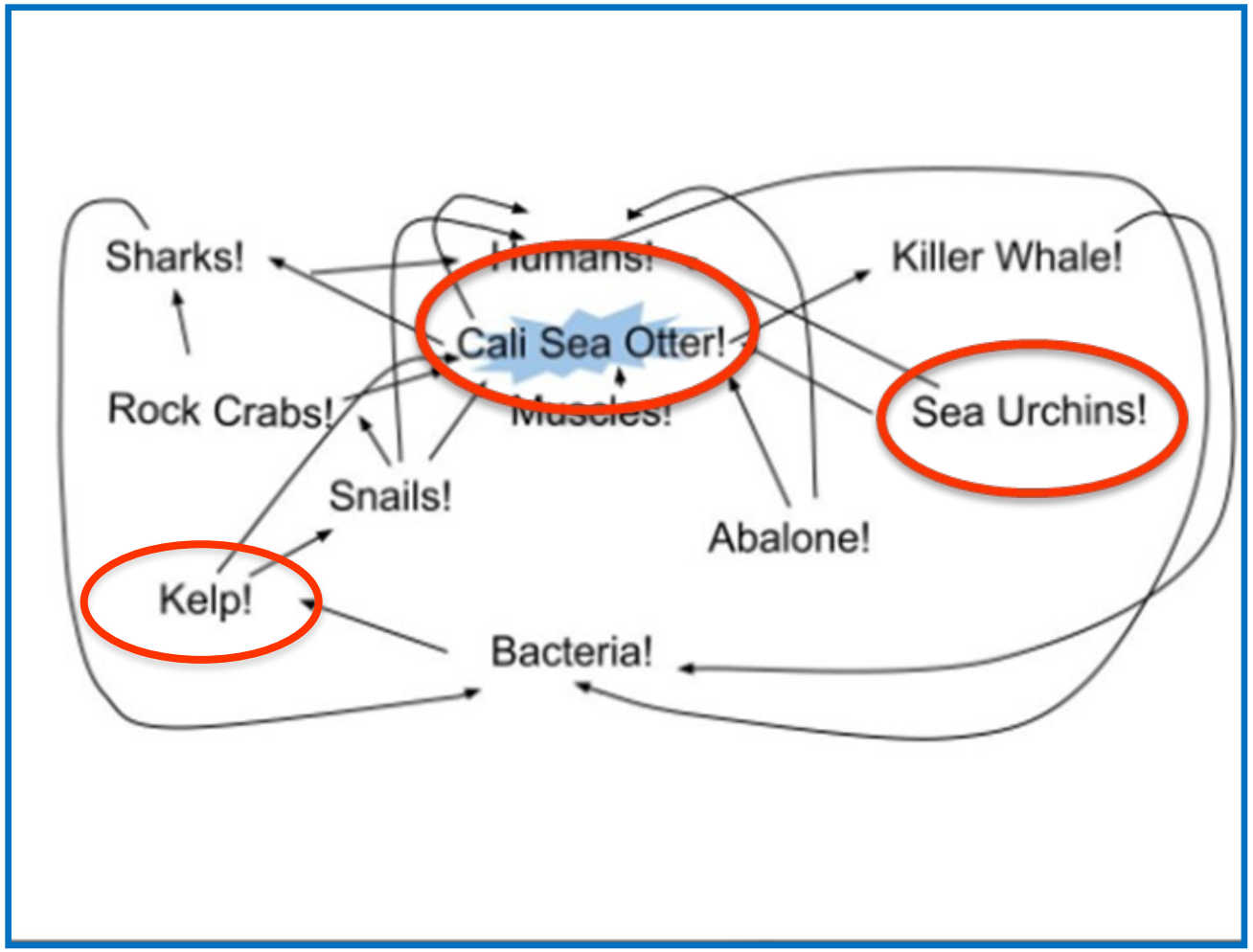
but sea urchins, themselves, are prey for other coastal animals, such as otters, which can control urchin populations



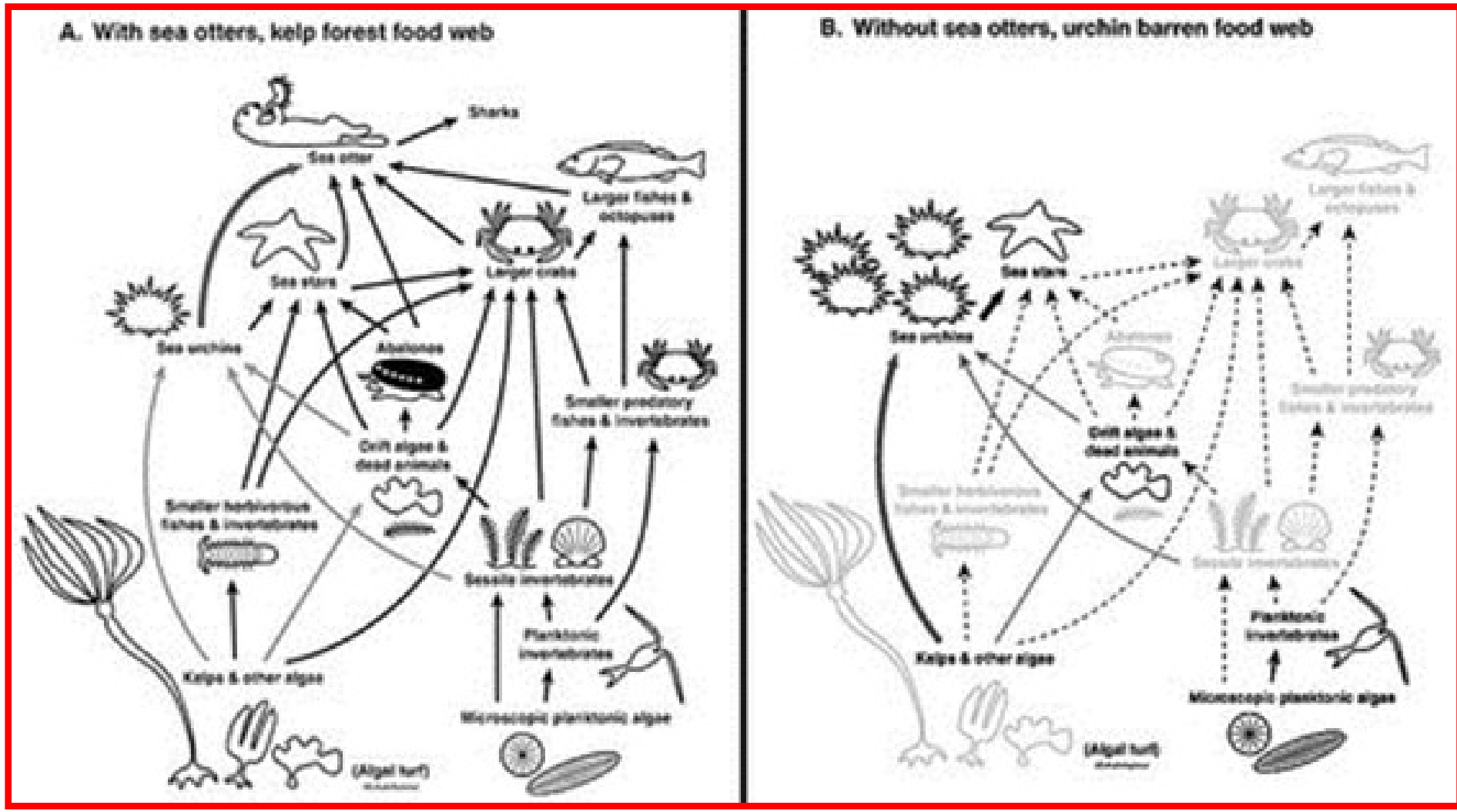
# Otter hunting for urchin

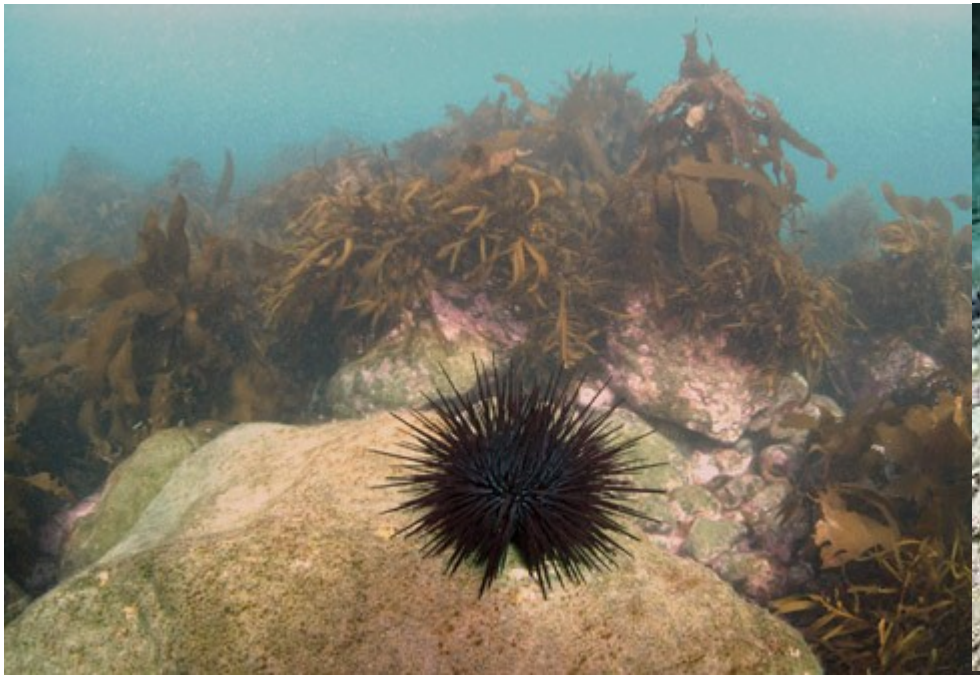


# Kelp forest food web:
# Sea Otter are a ‘keystone’ species in the kelp forest ecosystem



# Food webs of each system state:
# Kelp forest state , with otters (left);
# urchin barren state , without otters (r)

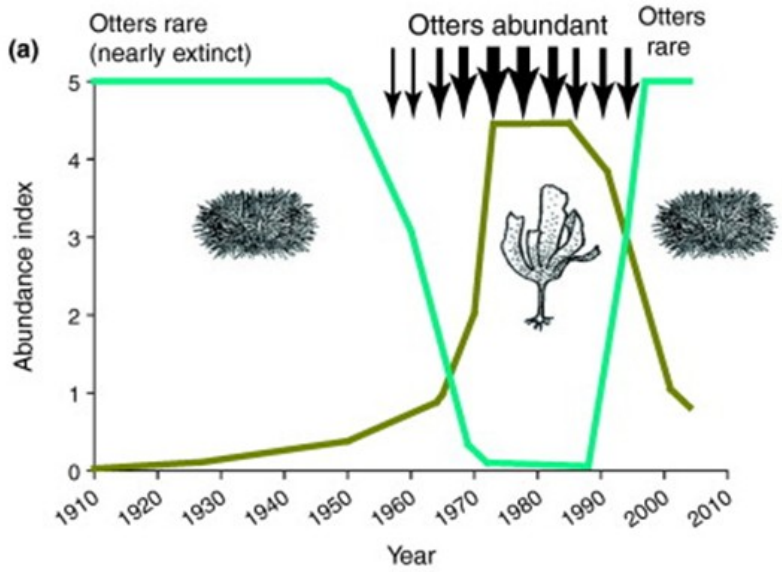








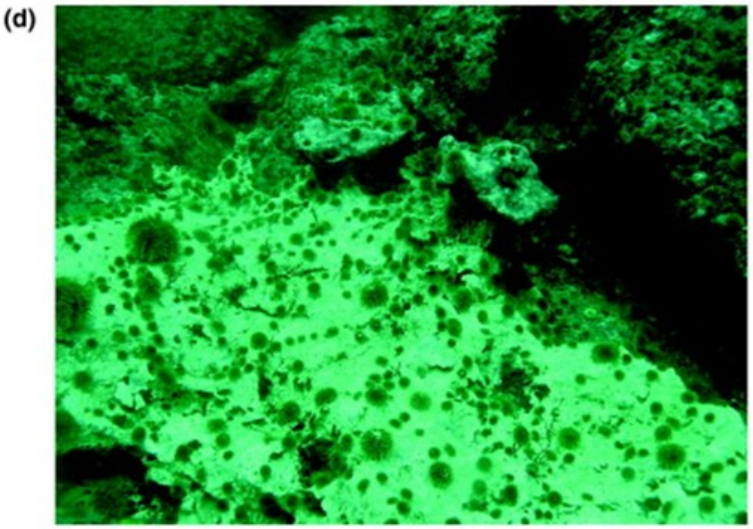
# Switches from kelp to barrens have been correlated with otter and urchin population sizes



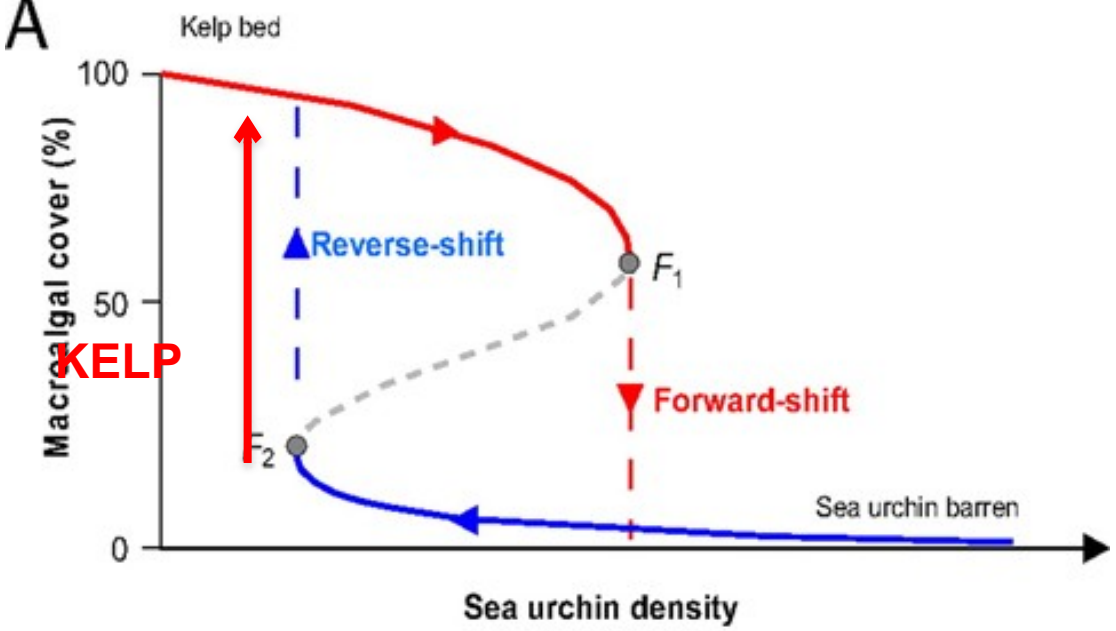


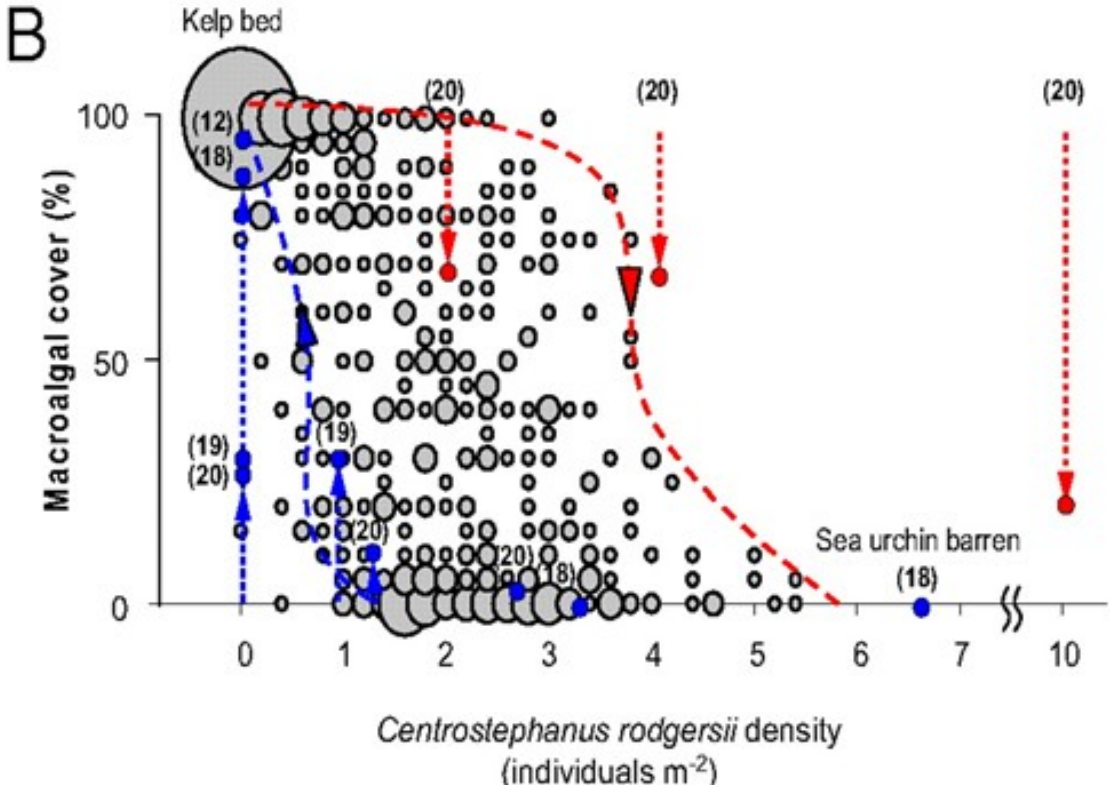




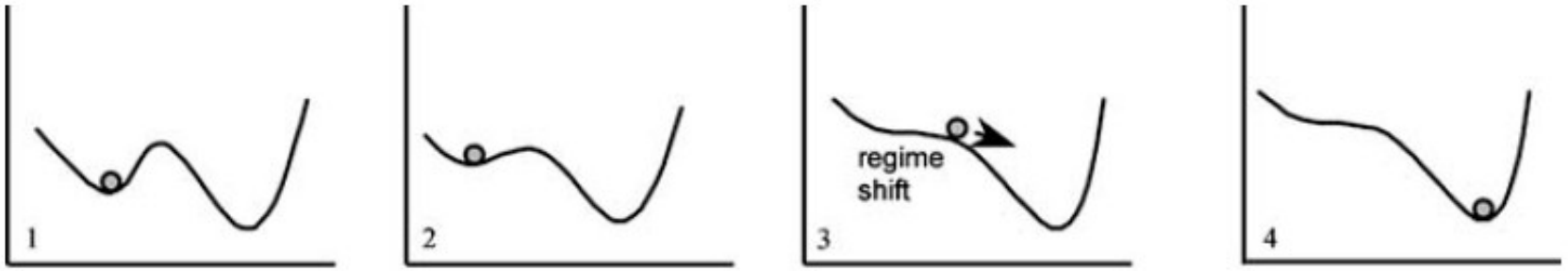

TRENDS in Ecology & Evolution



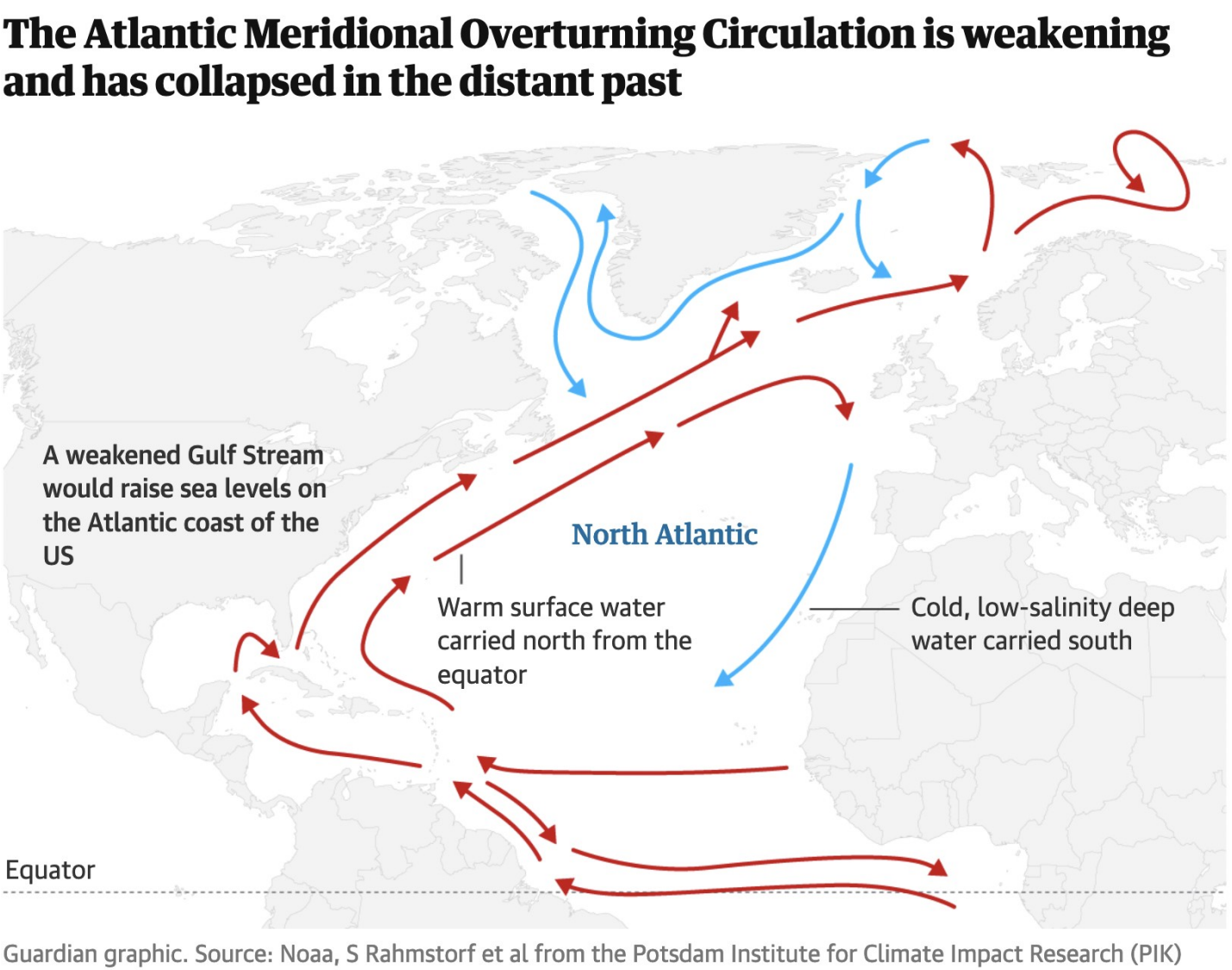




Observational data supporting the threshold effect of urchin populations on kelp abundance

# Examples of other ecosystem regime shifts (discussed in Folke: « Regime shifts, resilience… »)




| | | | |
|---|---|---|---|
| clear-water lakes | phosphorous accumulation in agricultural soil and lake mud | flooding, warming, overexploitation of predators | turbid-water lakes |
| coral-dominated reefs | overfishing, coastal eutrophication | disease, bleaching hurricane | algae-dominated reefs |
| grassland | fire prevention | good rains, continuous heavy grazing | shrub-bushland |
| grassland | hunting of herbivores | disease | woodland |




The system is being eroded by the faster-than-expected melt-off of Greenland's glaciers and Arctic ice sheets, which pours freshwater into the sea and obstructs the sinking of saltier, warmer water from the south.

# Flow in Sverdrups (Sv)

One Sverdrup is the flow of 1 million cubic meters per second.

Example: 1 km x 1 km square of water flowing at 1 meter/second

Or: 100 km x 100 meter rectangle flowing at 10 cm/second (etc.)

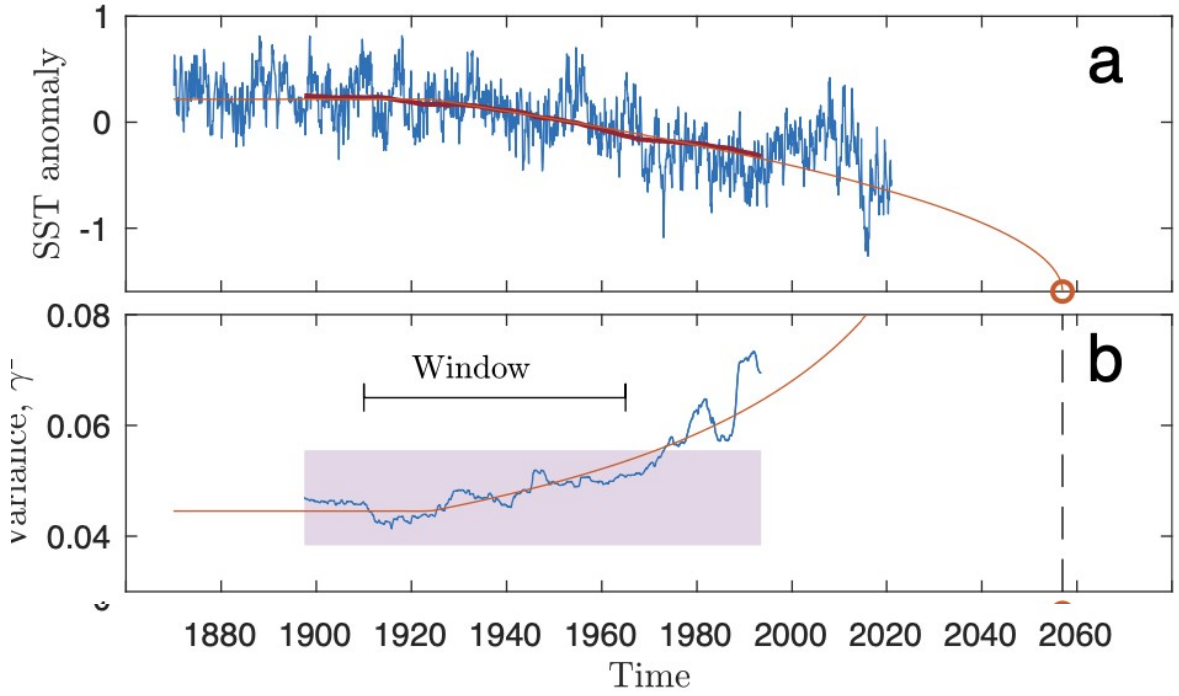
The water transport in the Gulf Stream gradually increases from 30 Sv in the Florida Current to a maximum of 150 Sv south of Newfoundland.

The Antarctic Circumpolar Current, at approximately 125 Sv, is the largest ocean current.

The entire global input of fresh water from rivers to the ocean is approximately 1.2 Sv

The Amazon is 0.2 Sv

Collapse Soon




AMOC has declined 15% since 1950 and is in its weakest state in more than a millennium, according to previous research that prompted speculation about an approaching collapse.

Until now there has been no consensus about how severe this will be. One study last year, based on changes in sea surface temperatures, suggested the tipping point could happen between 2025 and 2095.

However, the UK Met Office said large, rapid changes in AMOC were “very unlikely” in the 21st century. (Spring 2024)

JANUARY 15, 2025 The GIST Editors' notes

Collapse Not Soon

# Critical ocean current has not declined in the last 60 years, AMOC study finds

by Woods Hole Oceanographic Institution

Jens Terhaar et al, Nature Communications (2025). DOI: 10.1038/s41467-024-55297-5

**Fig. 2: AMOC anomalies at 26.5°N vs Atlantic subpolar gyre SST anomalies across climate models from CMIP6 at various timescales.**

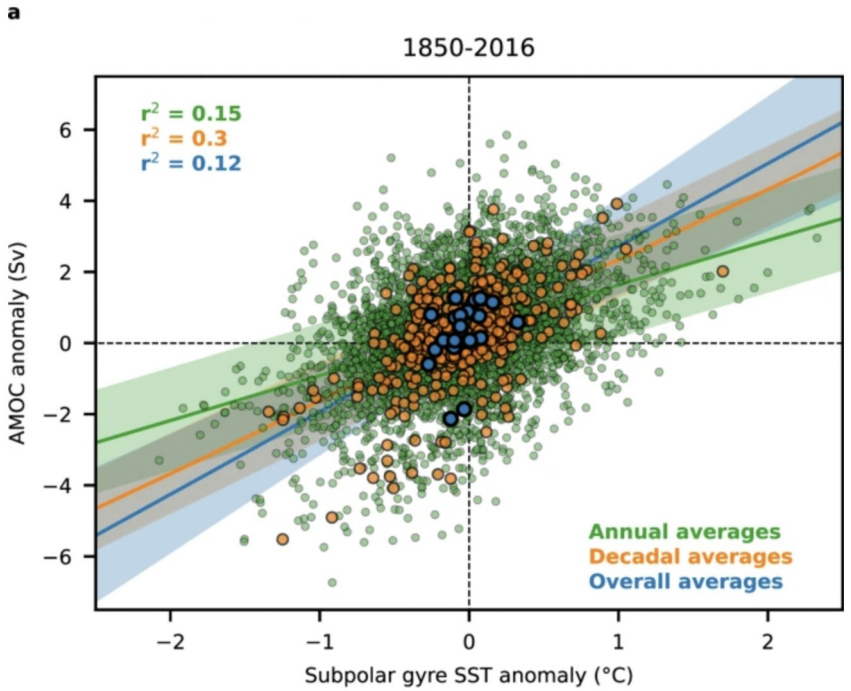
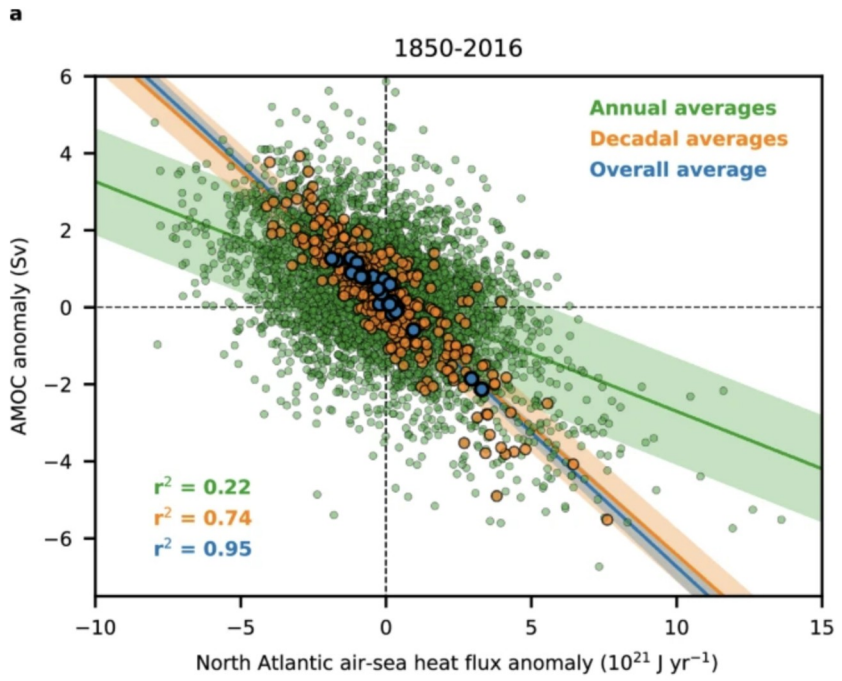
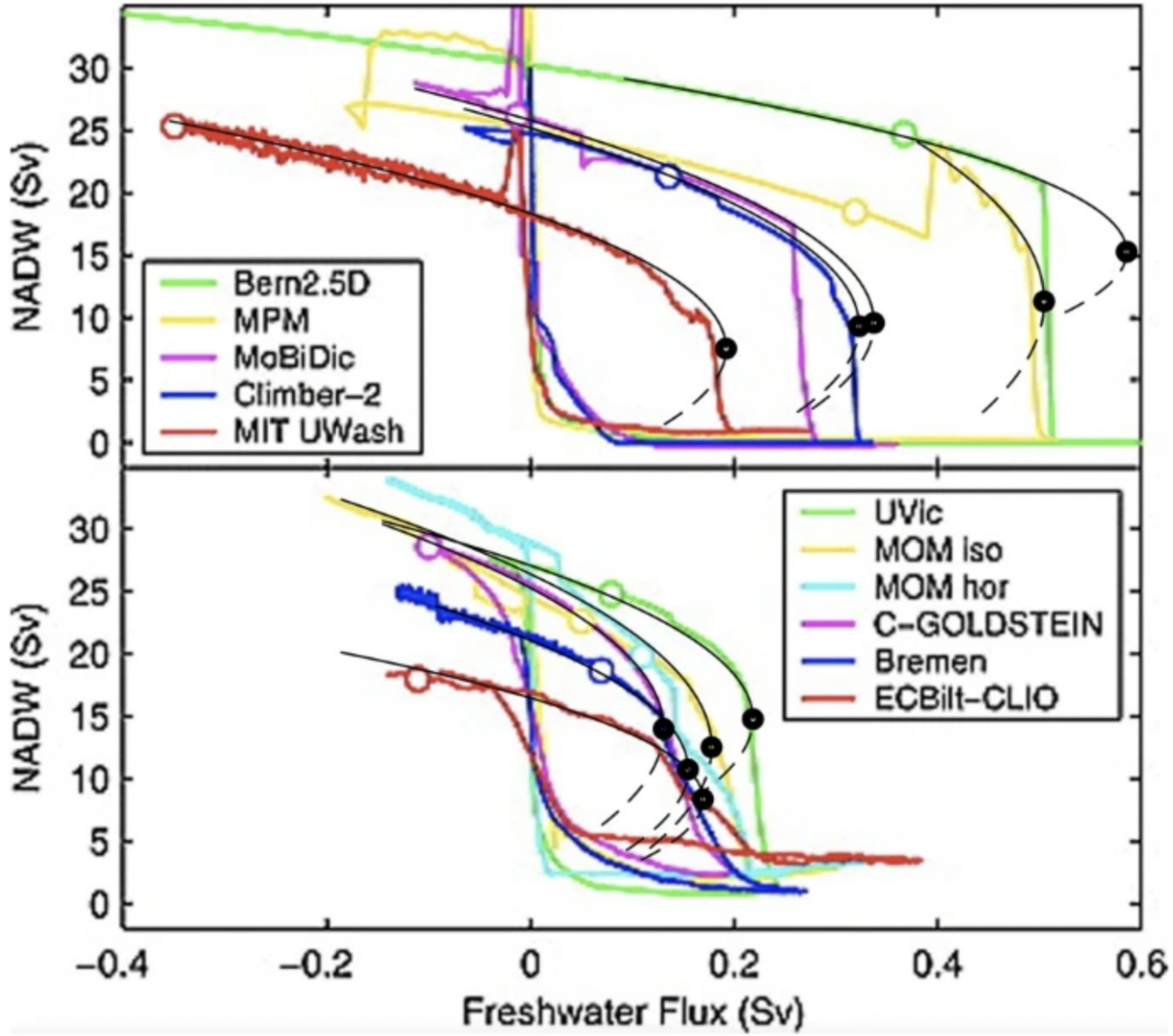



**Fig. 3: AMOC anomalies at 26.5°N vs air-sea heat flux anomalies north of 26.5°N across climate models from CMIP6 at various timescales.**




Sea Surface Temperature Anomalies not a good predictor –instead look at total heat transfer

**Fig. 2: Steady state curves from climate model simulations of the North Atlantic Deep Water (NADW), with a very slowly changing control parameter (freshwater forcing).**



SCIENCE ADVANCES | RESEARCH ARTICLE

OCEANOGRAPHY

# Physics-based early warning signal shows that AMOC is on tipping course

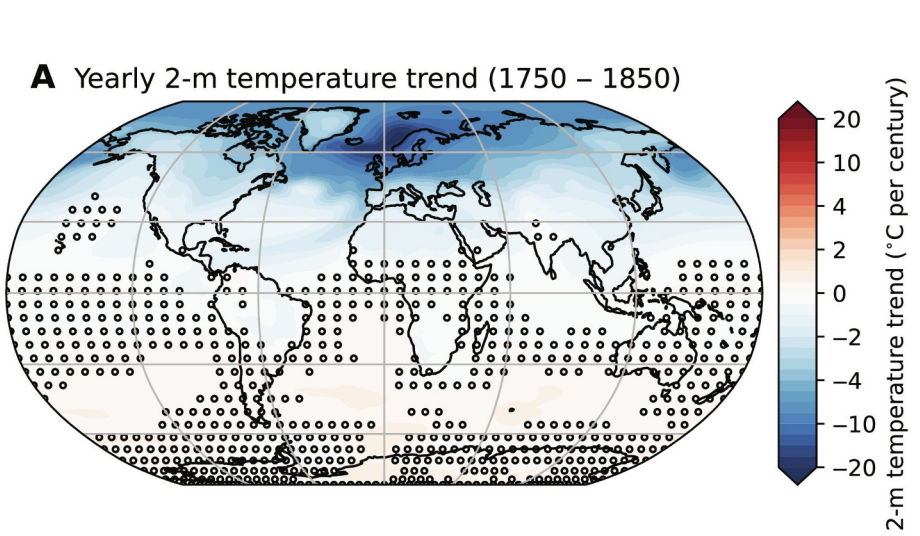
**René M. van Westen*, Michael Kliphuis, Henk A. Dijkstra**

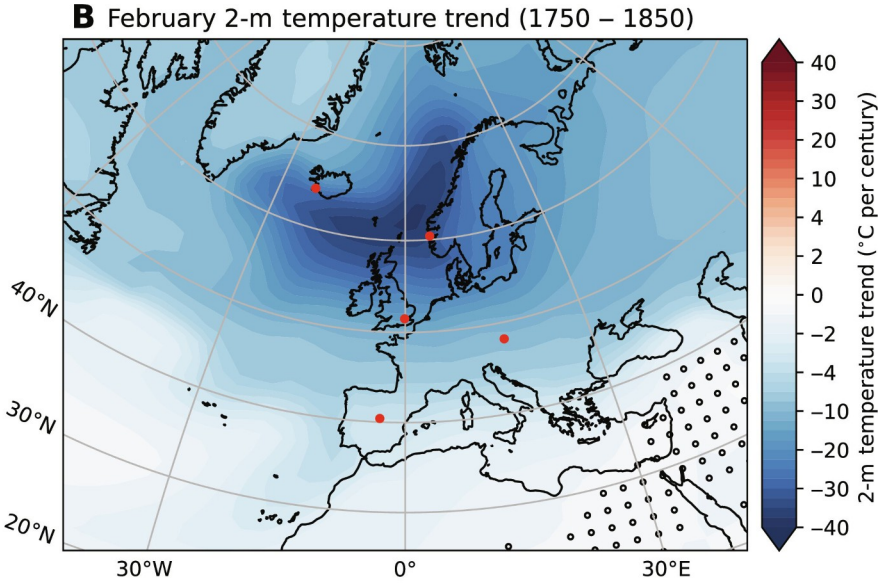
**One of the most prominent climate tipping elements is the Atlantic meridional overturning circulation (AMOC), which can potentially collapse because of the input of fresh water in the North Atlantic. Although AMOC collapses have been induced in complex global climate models by strong freshwater forcing, the processes of an AMOC tipping event have so far not been investigated. Here, we show results of the first tipping event in the Community Earth System Model, including the large climate impacts of the collapse. Using these results, we develop a physics-based and observable early warning signal of AMOC tipping: the minimum of the AMOC-induced freshwater transport at the southern boundary of the Atlantic. Reanalysis products indicate that the present-day AMOC is on route to tipping. The early warning signal is a useful alternative to classical statistical ones, which, when applied to our simulated tipping event, turn out to be sensitive to the analyzed time interval before tipping.**

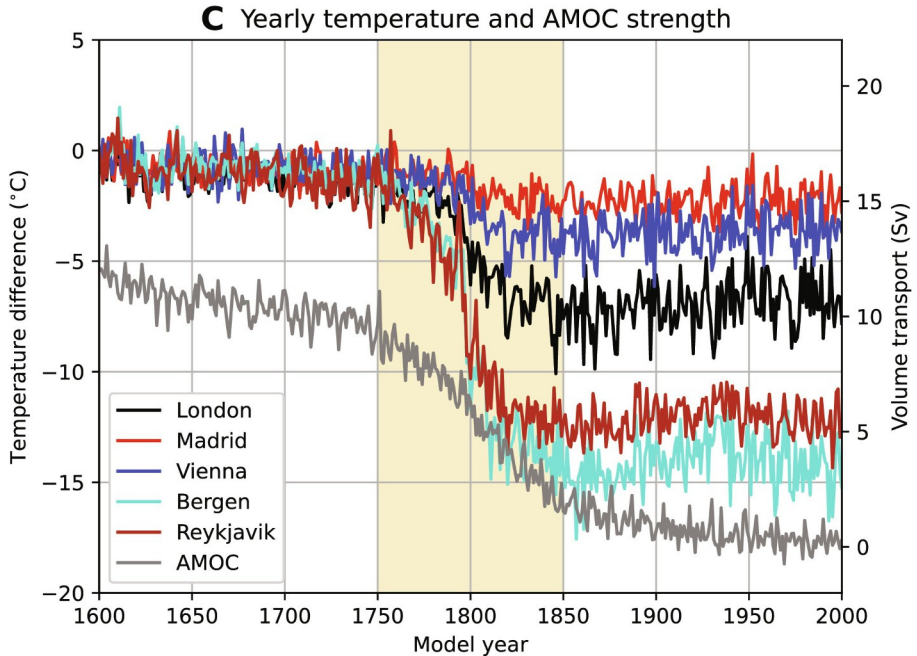
The new paper said the results provided a “clear answer” about whether such an abrupt shift was possible: “This is bad news for the climate system and humanity as up till now one could think that AMOC tipping was only a theoretical concept and tipping would disappear as soon as the full climate system, with all its additional feedbacks, was considered.”

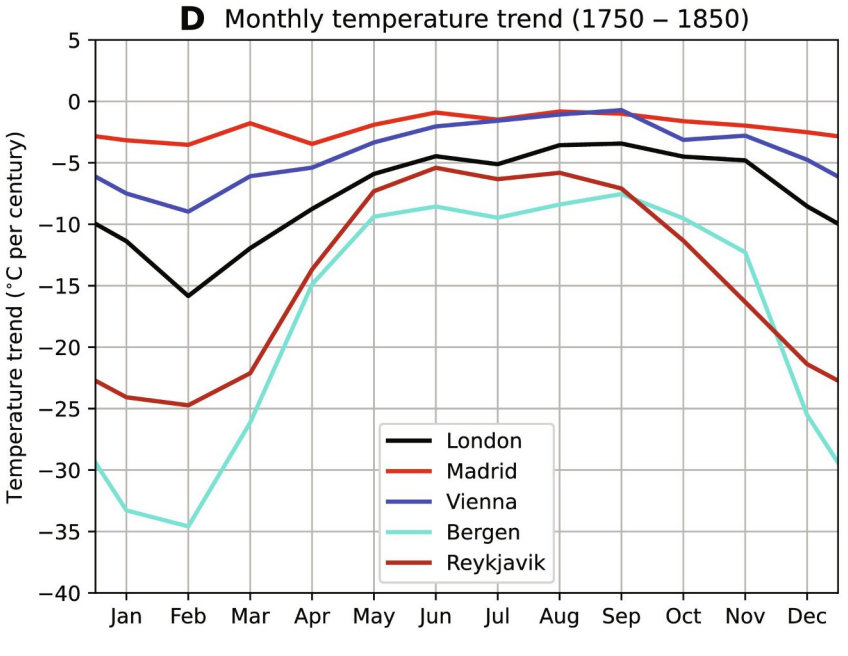
**He said there was not yet enough data to say whether this would occur in the next year or in the coming century, but when it happens, the changes are irreversible on human timescales.**

# AMOC Tipping Point

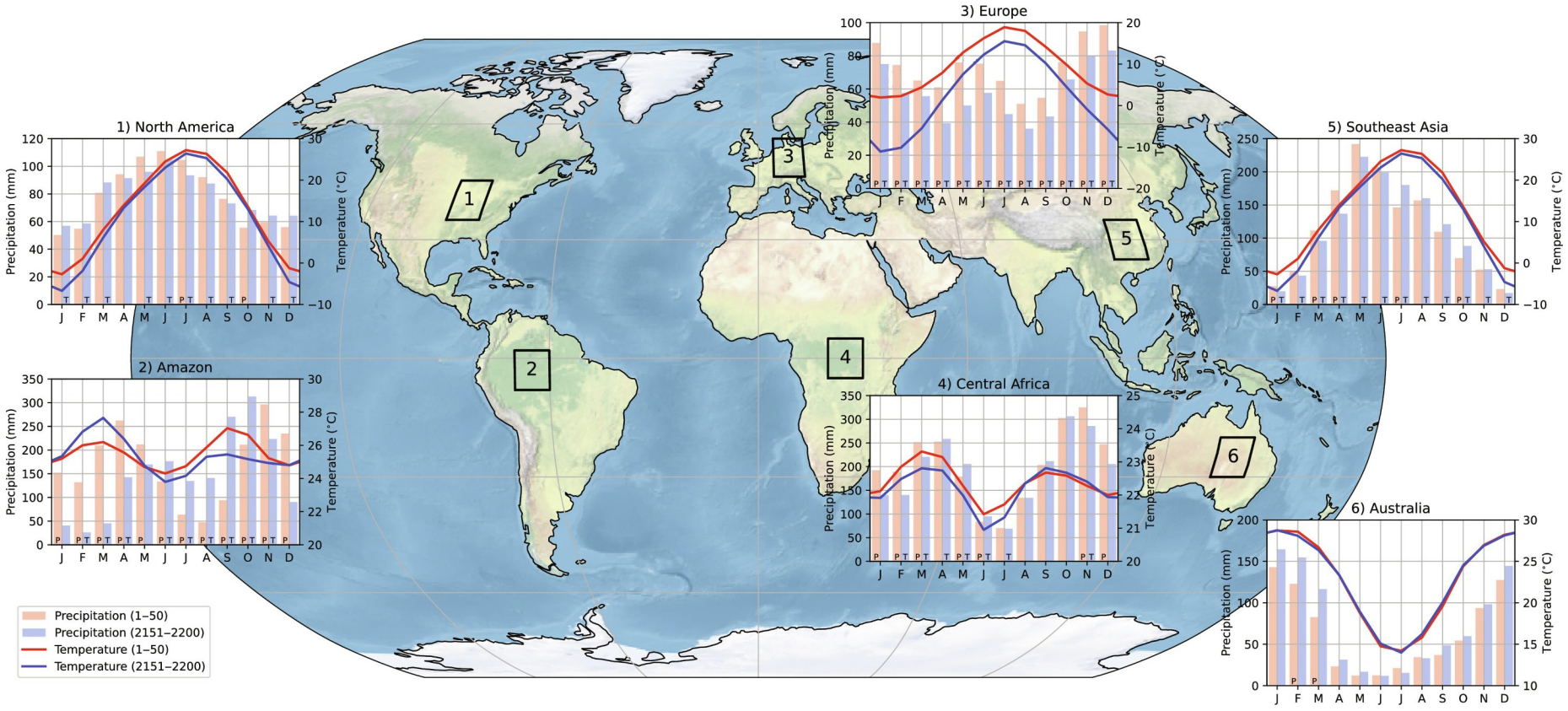


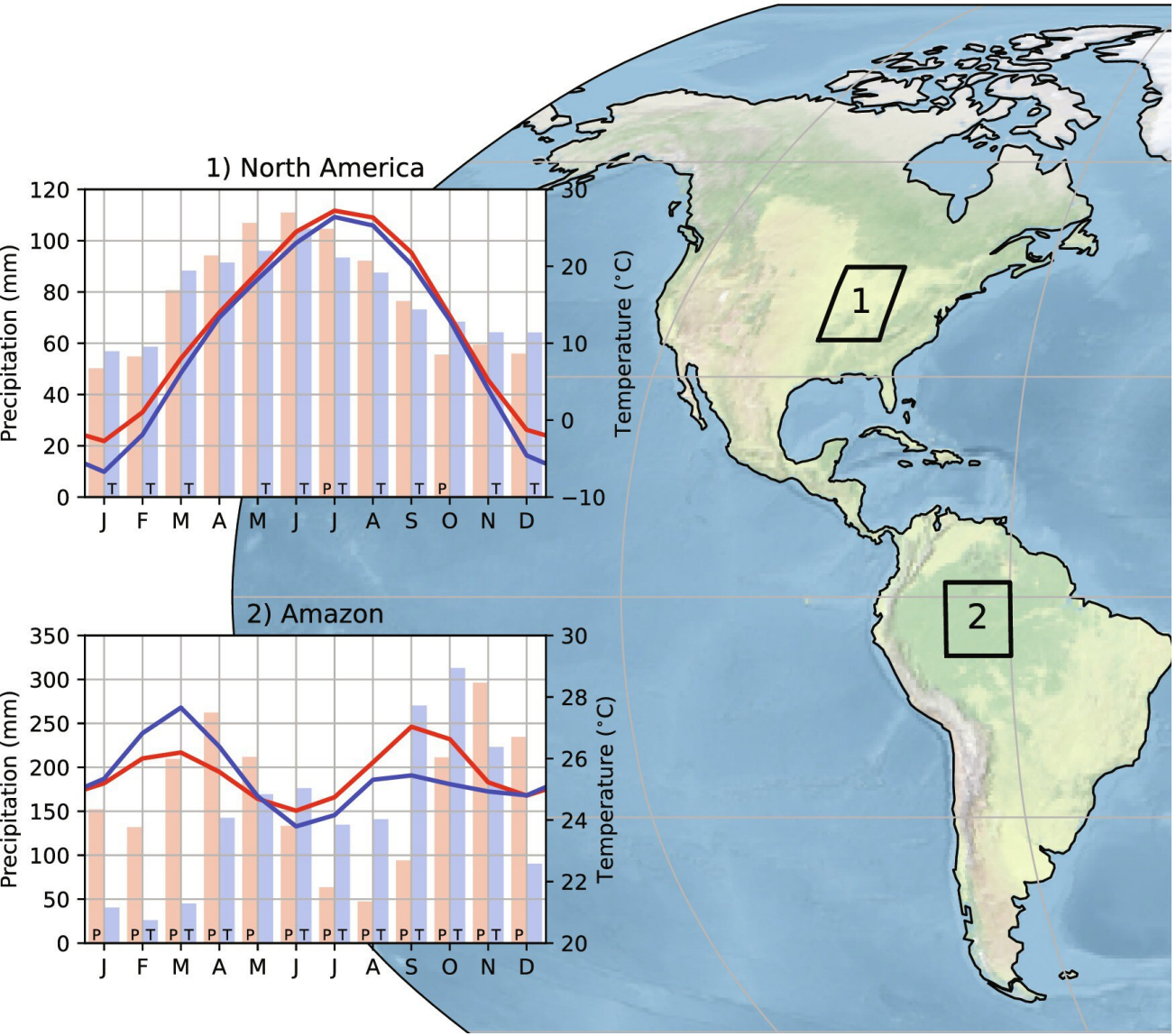



# AMOC Tipping Point



# vanWestin et al (2024)

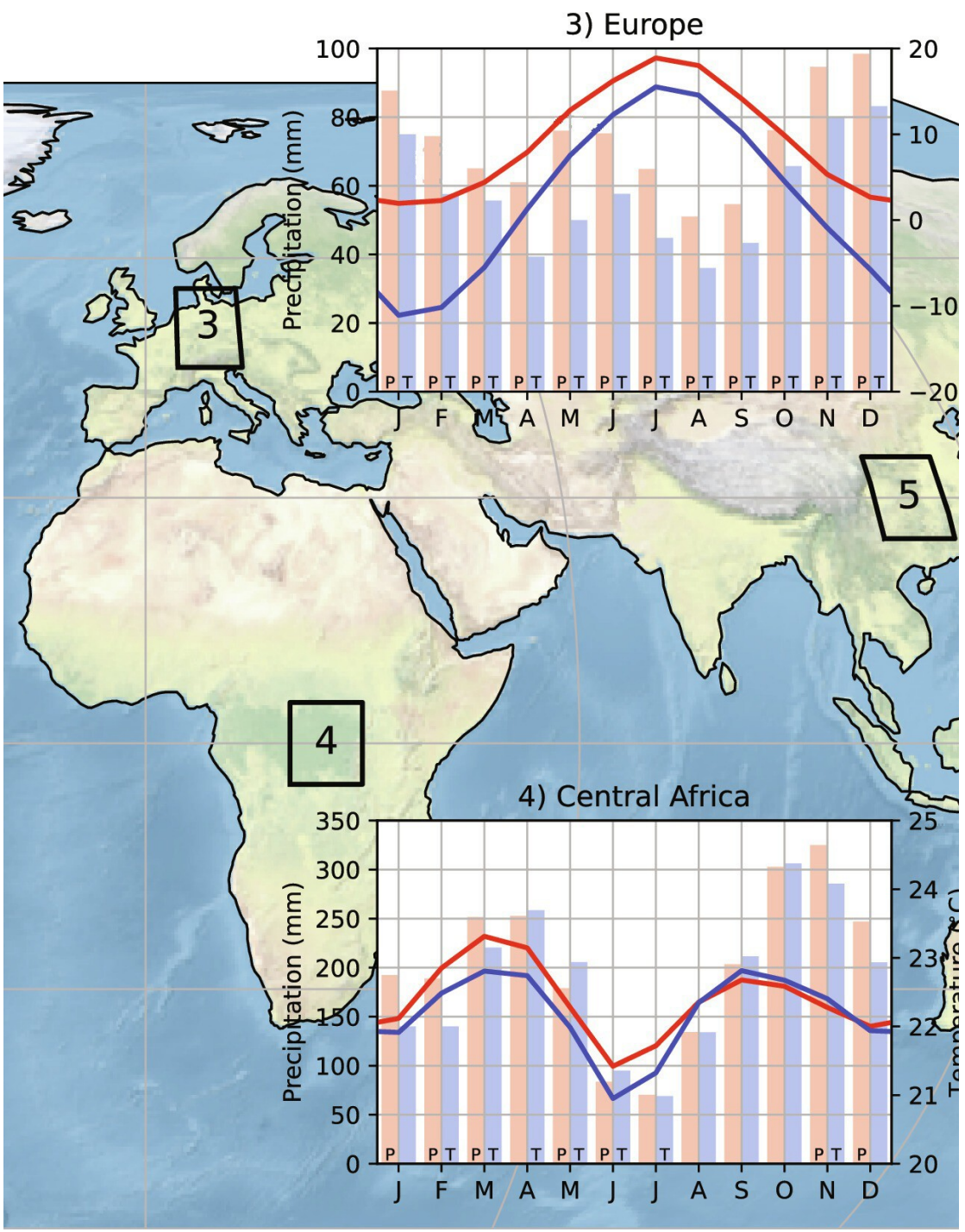



Temperature and Precipitation

Red: before

Blue: after

# vanWestin et al (2024)

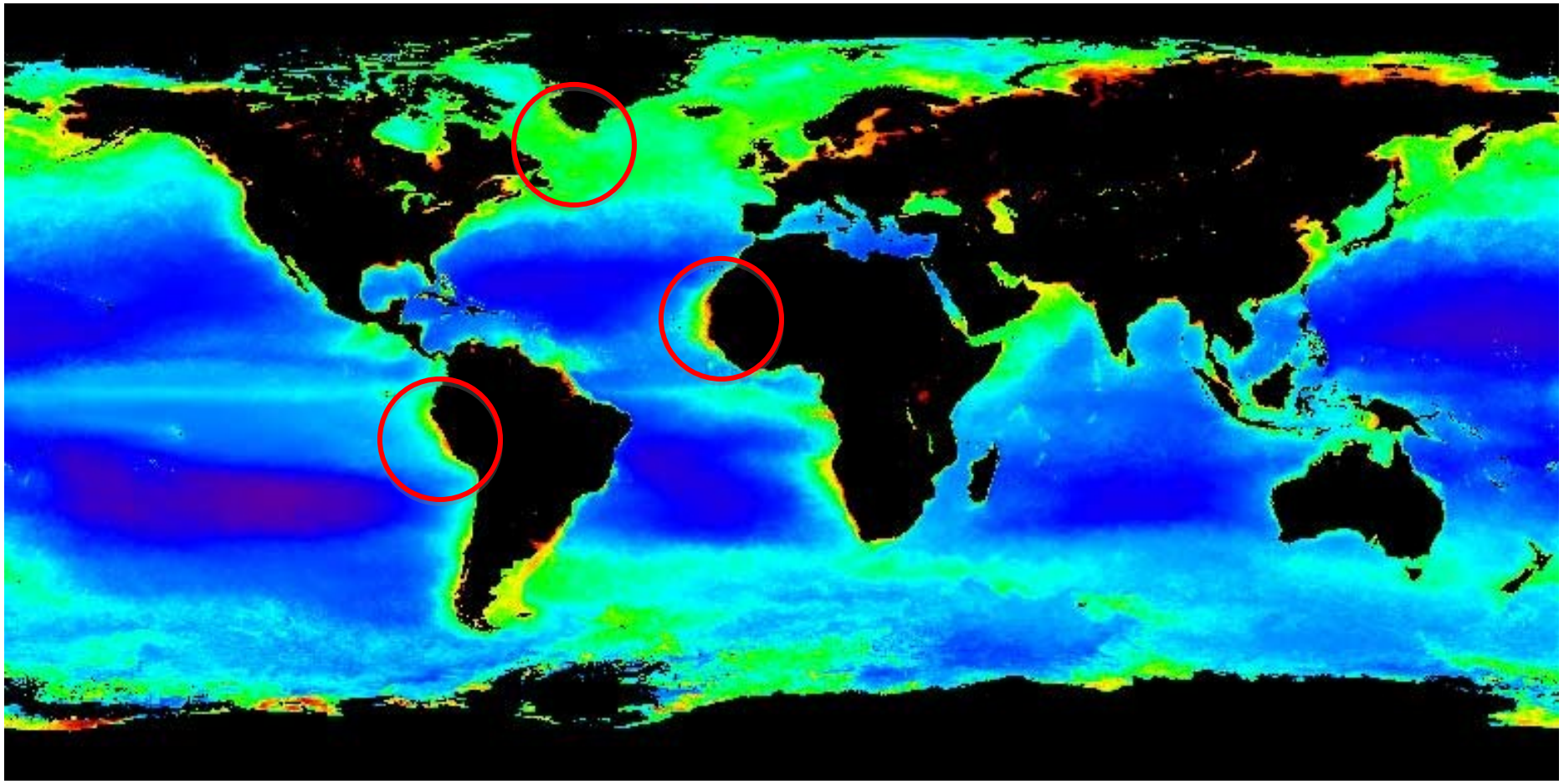



Temperature and Precipitation

Red: before

Blue: after

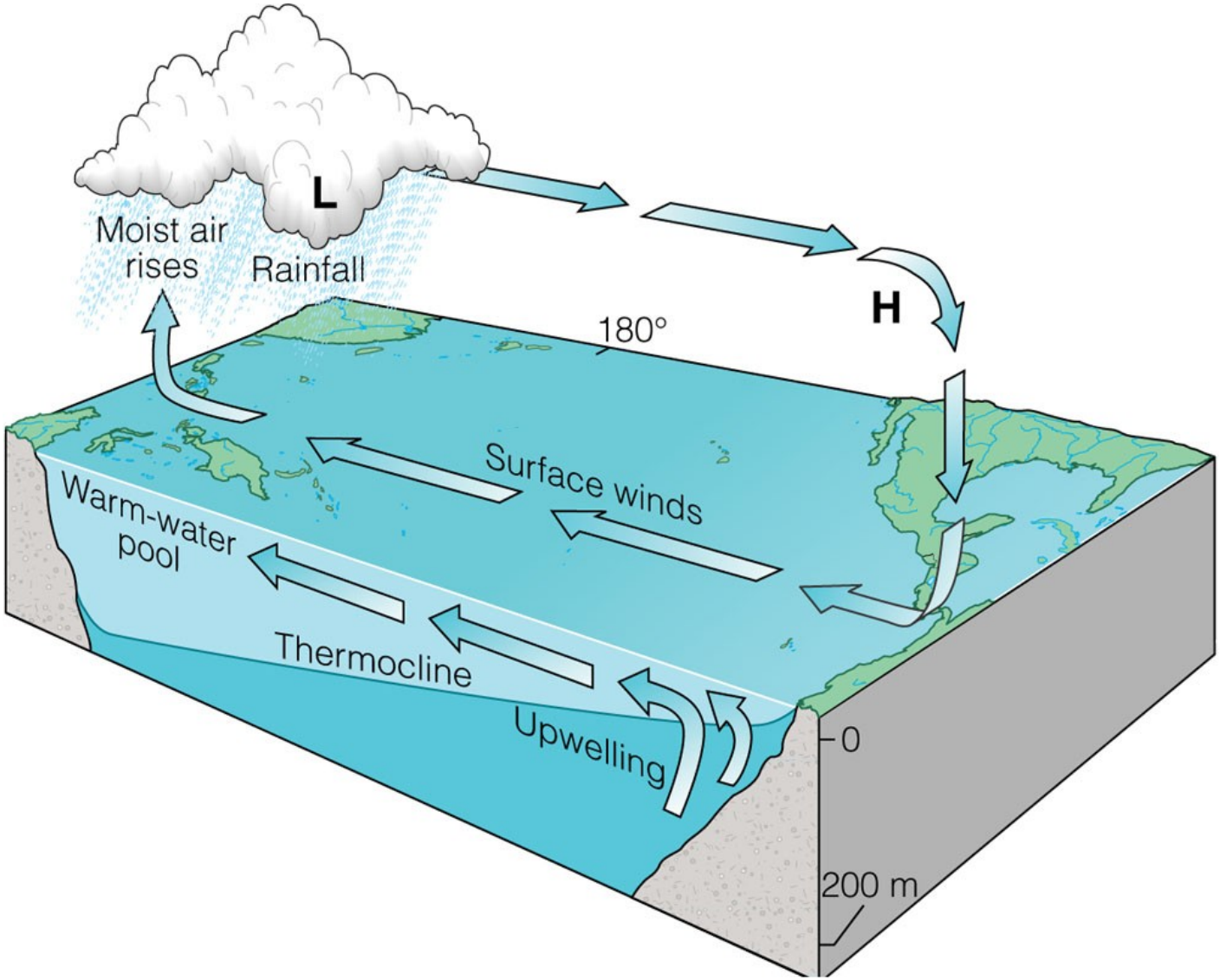
Global Chlorophyll Distribution



Where are the nutrients coming from in these three places?

NASA

Surface winds Drive Ocean Currents

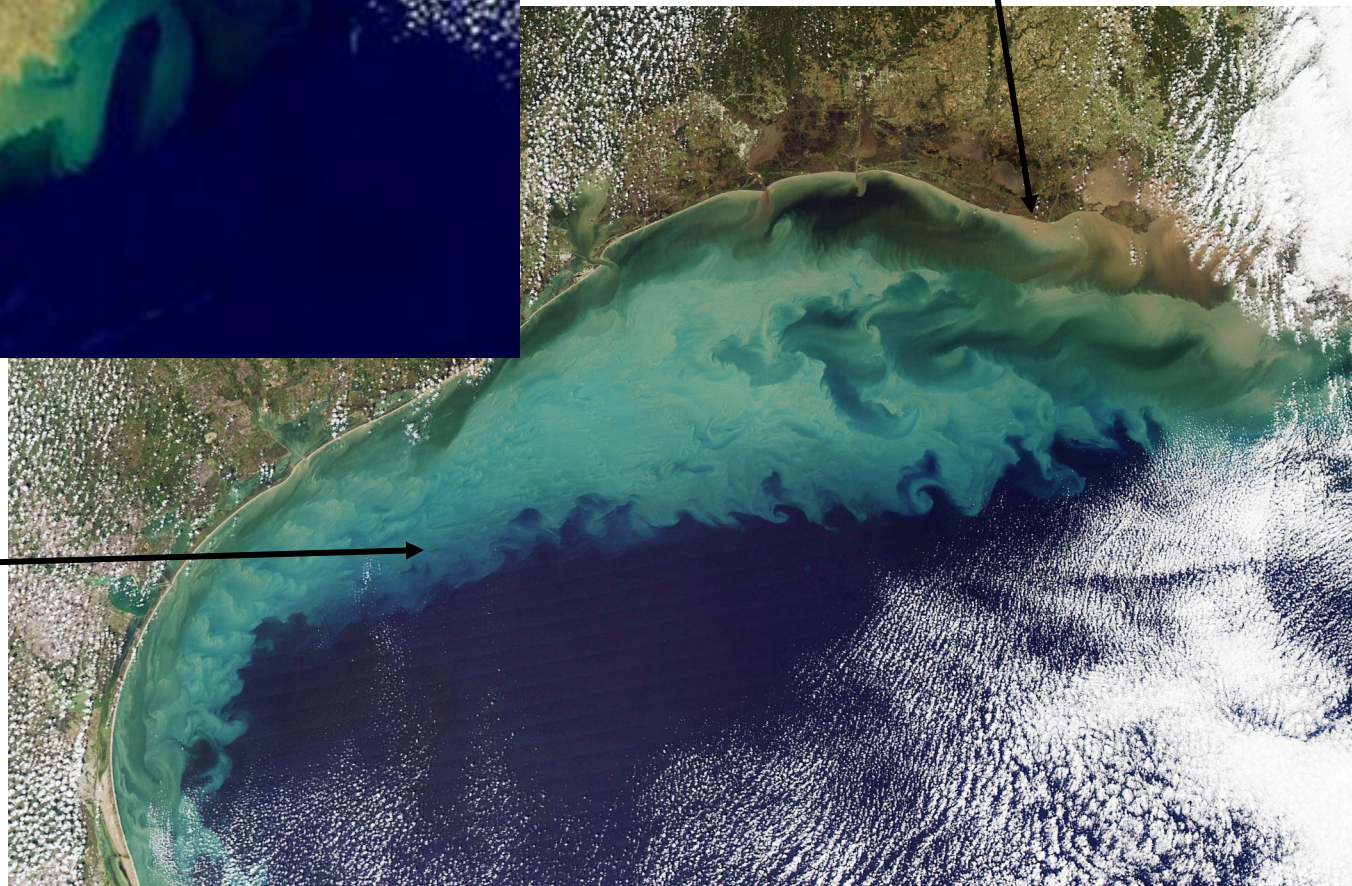



© 2005 Brooks/Cole - Thomson

Phytoplankton Bloom: Gulf of Mexico



Sediment input



Chlorophyll

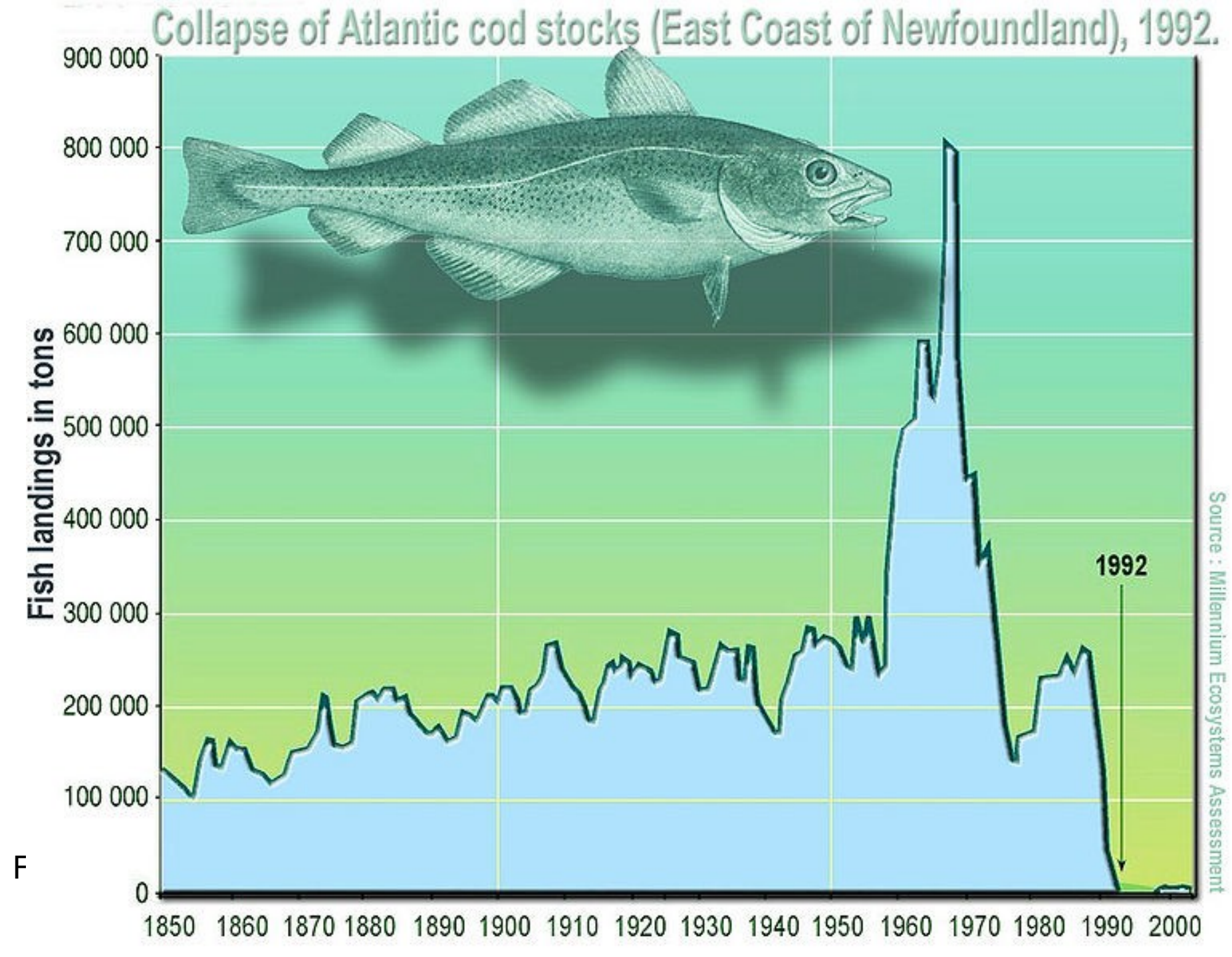
NASA

Phytoplankton Bloom: Vancouver Island



MODIS

Nutrients limit the ocean productivity, not water, sunlight, or $CO_2$


Collapse of Atlantic cod stocks (East Coast of Newfoundland), 1992.
Fish landings in tons
900 000
800 000
700 000
600 000
500 000
400 000
300 000
200 000
100 000
0
1850 1860 1870 1880 1890 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000
1992
Source : Millennium Ecosystems Assessment

F

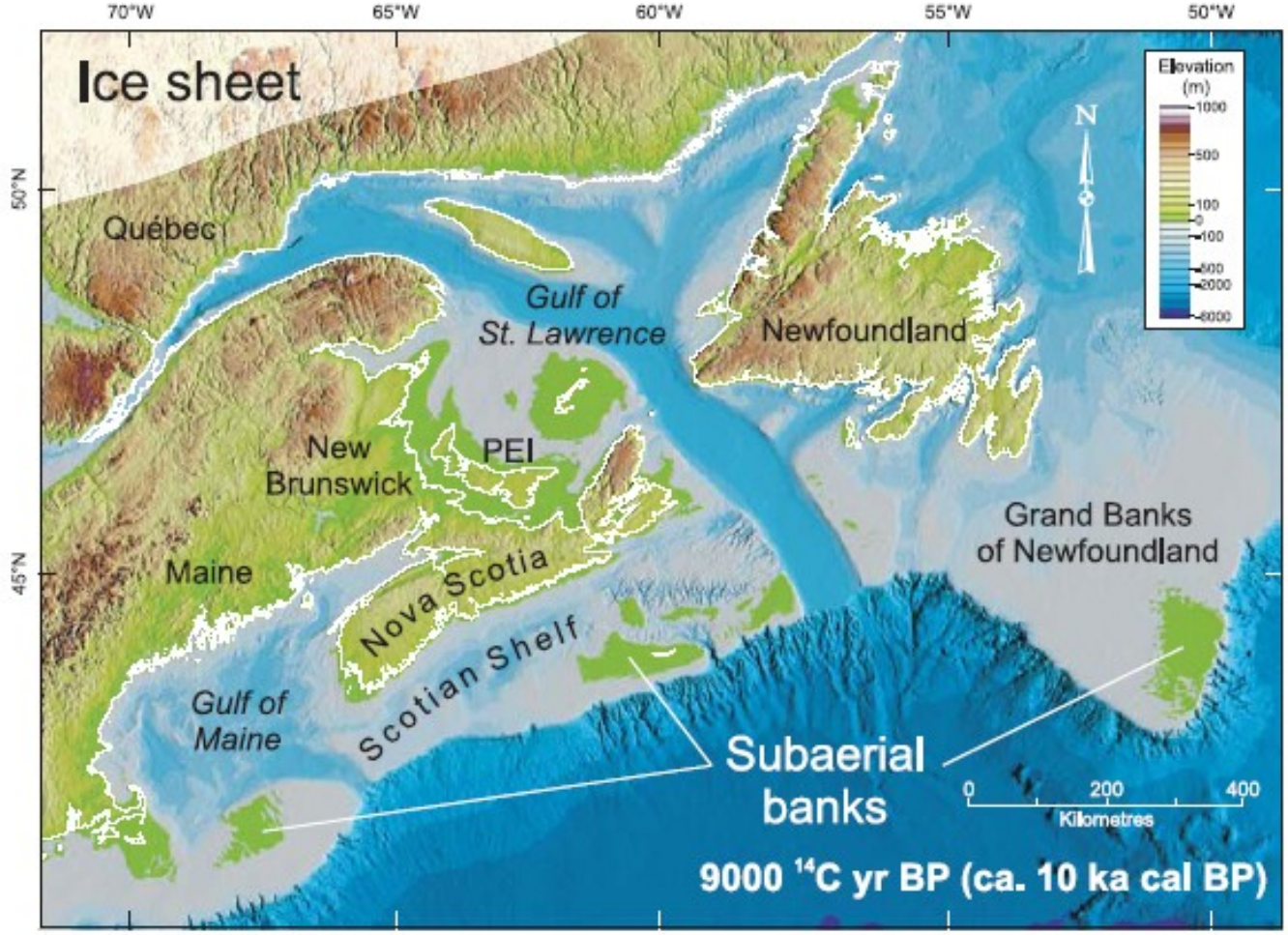
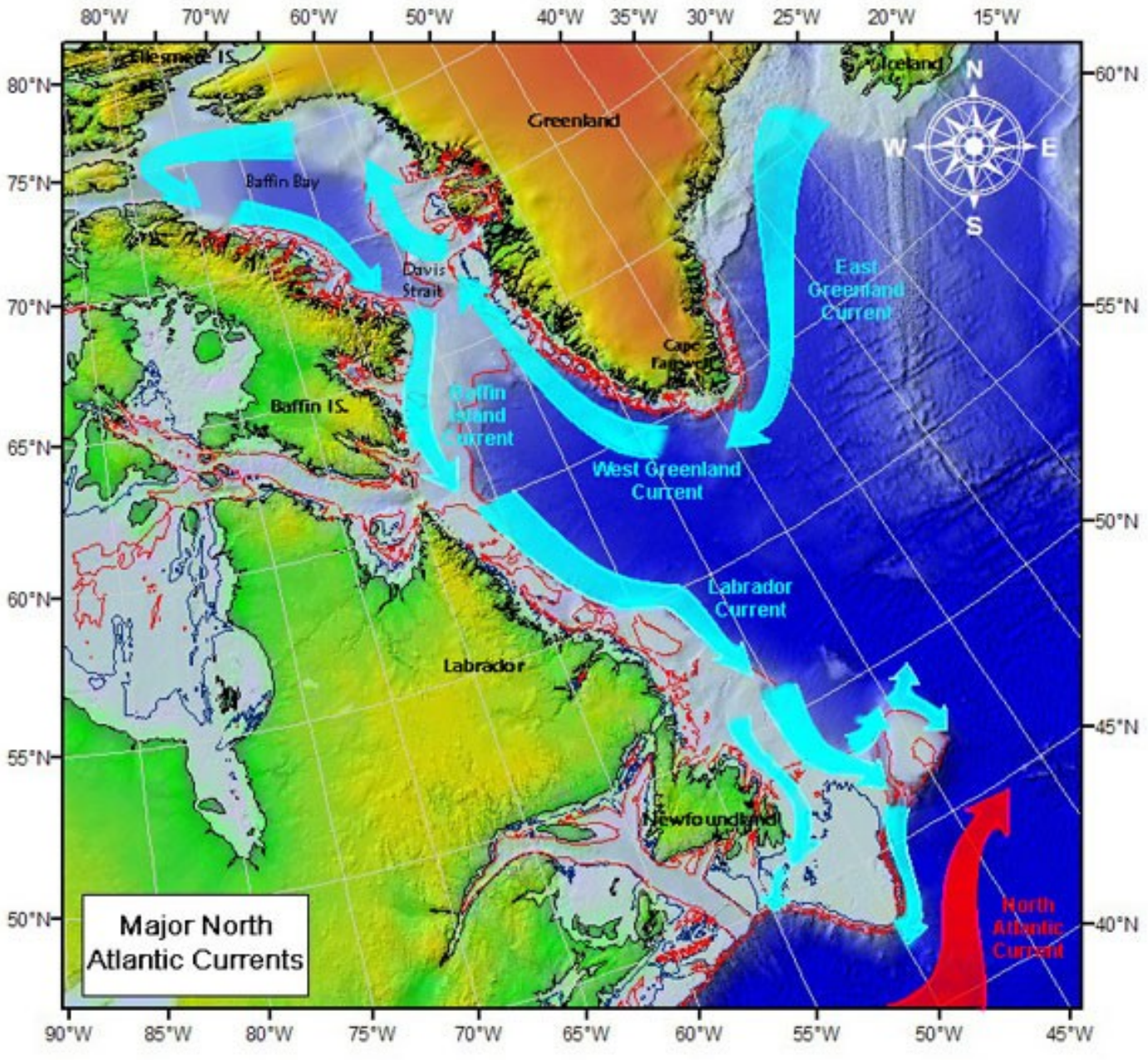
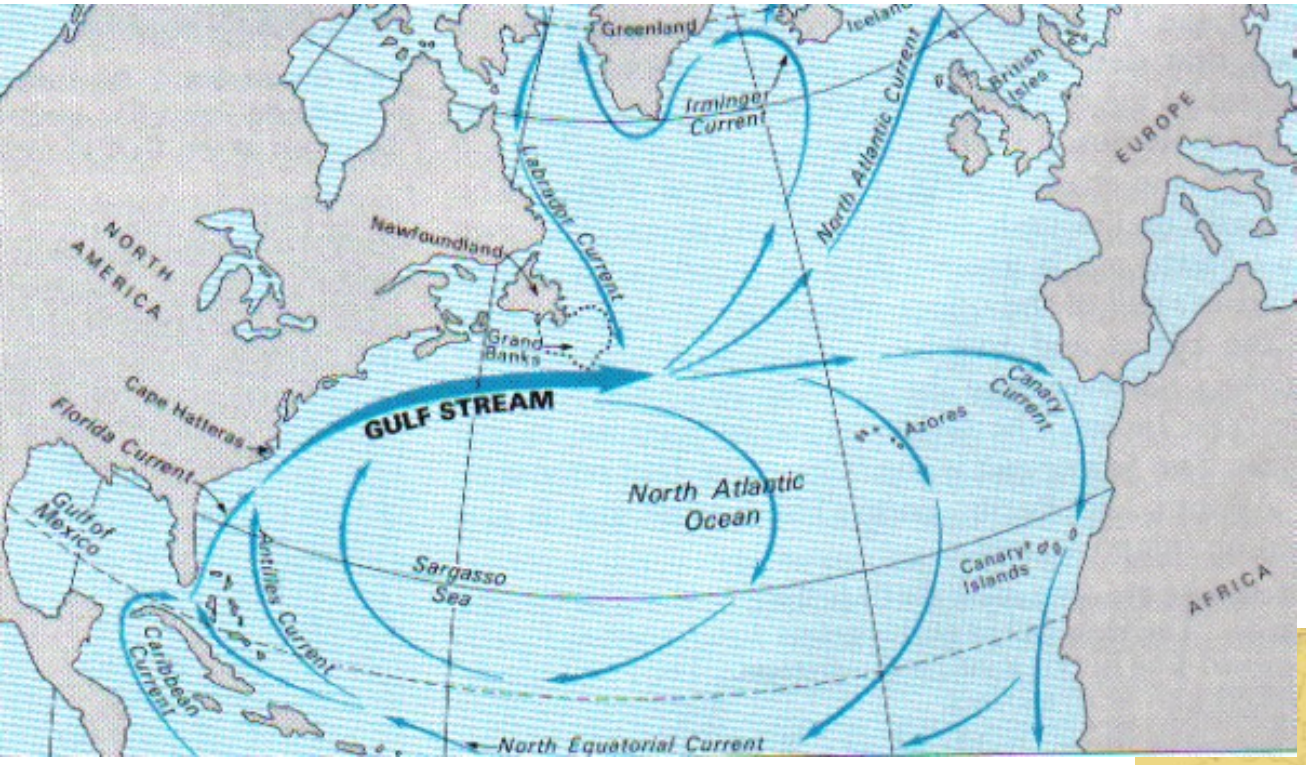



Figure 7. Palaeogeography of Atlantic Canada at 9000 $^{14}$C yr BP (from Shaw et al. 2002) (ka cal BP: calibrated $^{14}$C age, in thousands of years before present). At that time, islands ('subaerial banks') existed in the southern Gulf of Maine, on the Scotian Shelf and on the Grand Banks. Prince Edward Island (PEI) was part of the broad subaerial region extending from Nova Scotia and New Brunswick north into the Gulf of St. Lawrence.

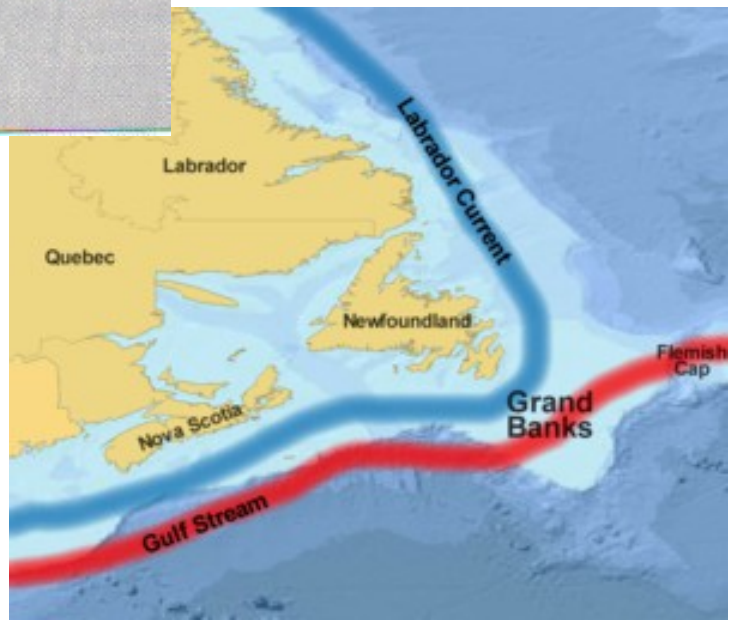
Labrador Current



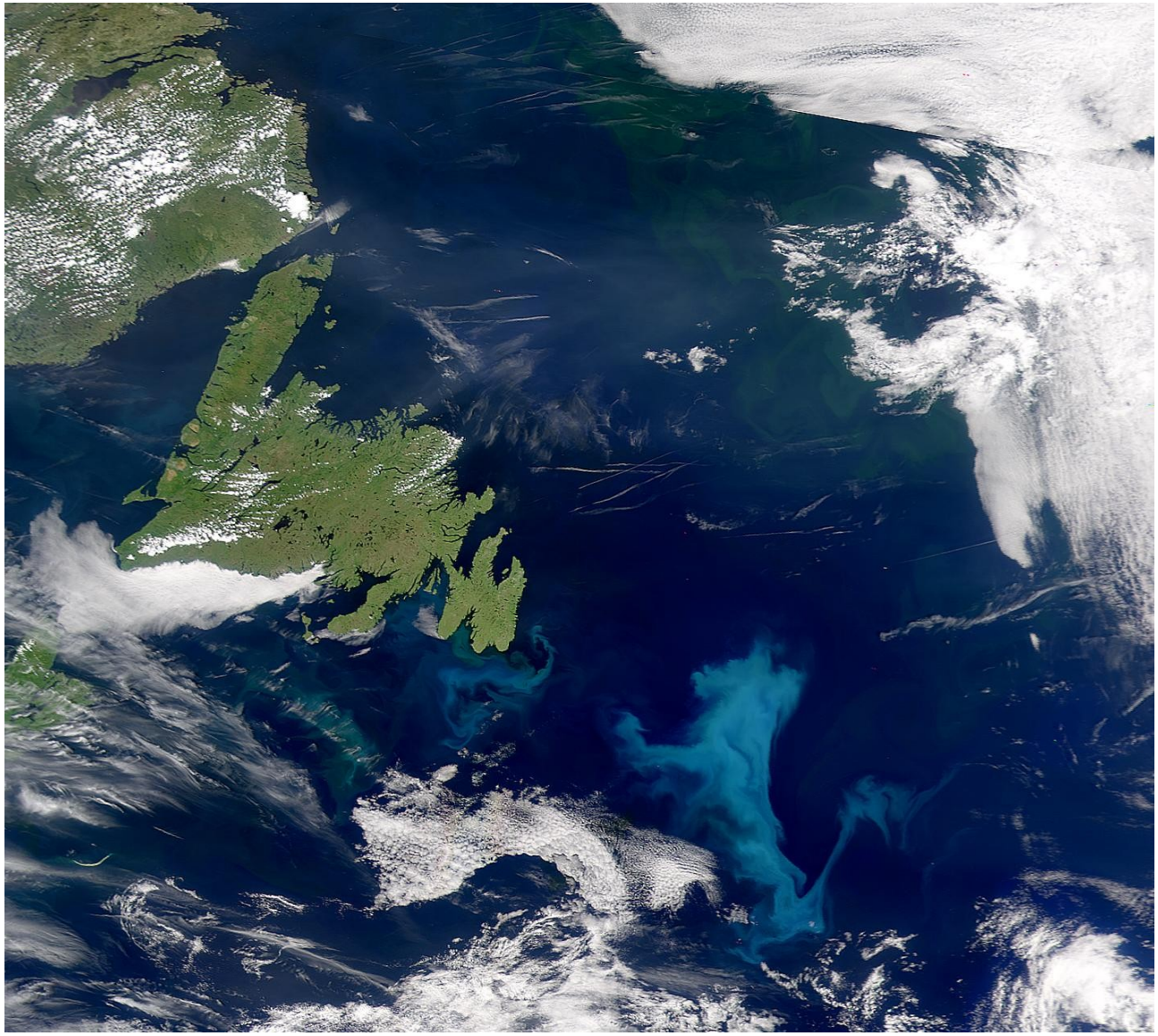

NOAA



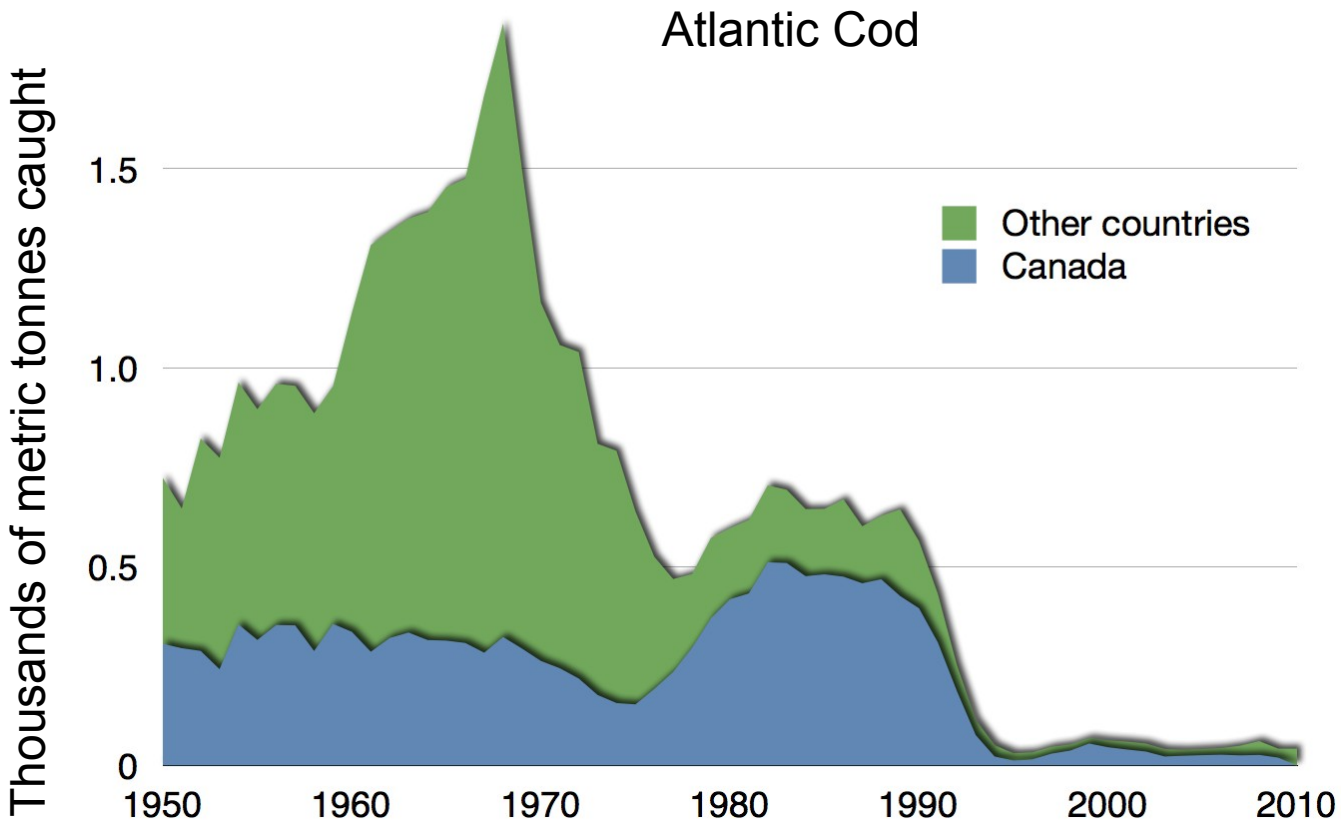

The meeting of the Labrador Current with the Gulf Stream at an area of shallow water: extremely biologically productive.



Phytoplankton Bloom: Grand Banks

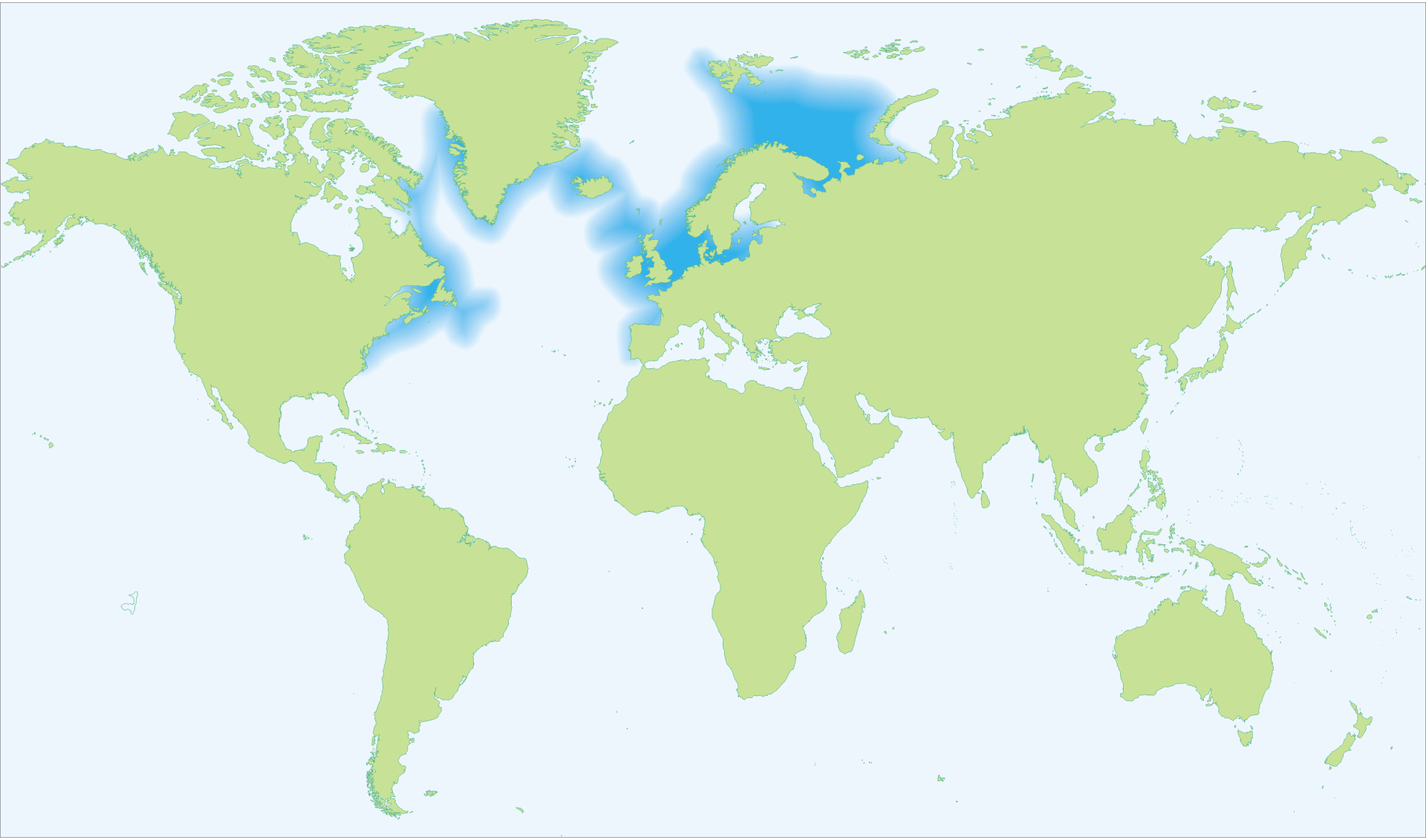


MODIS



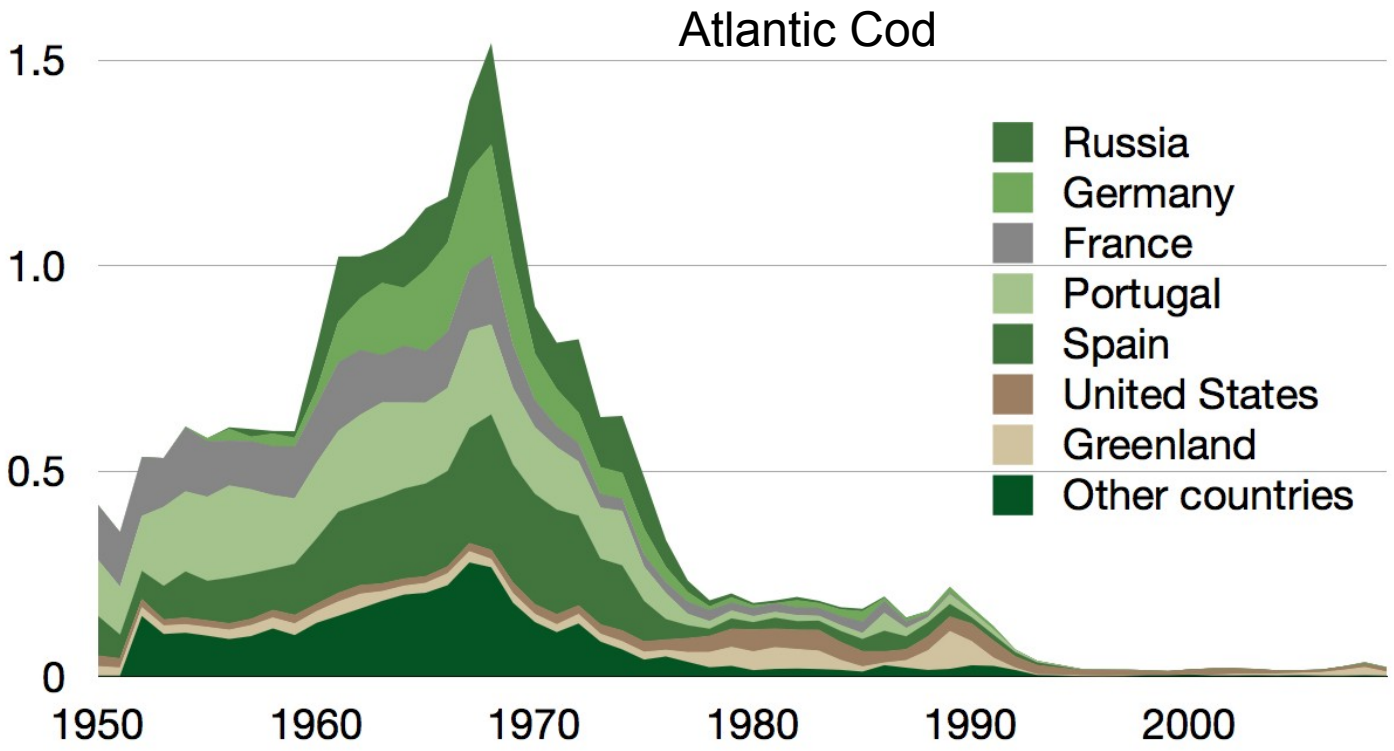

Fishing was going along fine… until the collapse

Atlantic Cod Range



Separated into Northeast and Northwest populations

Cod catch, exclusive of Canada




This lead to expansion and enforcement of Canada's 200 nautical mile exclusive economic zone.

## Regime shift:

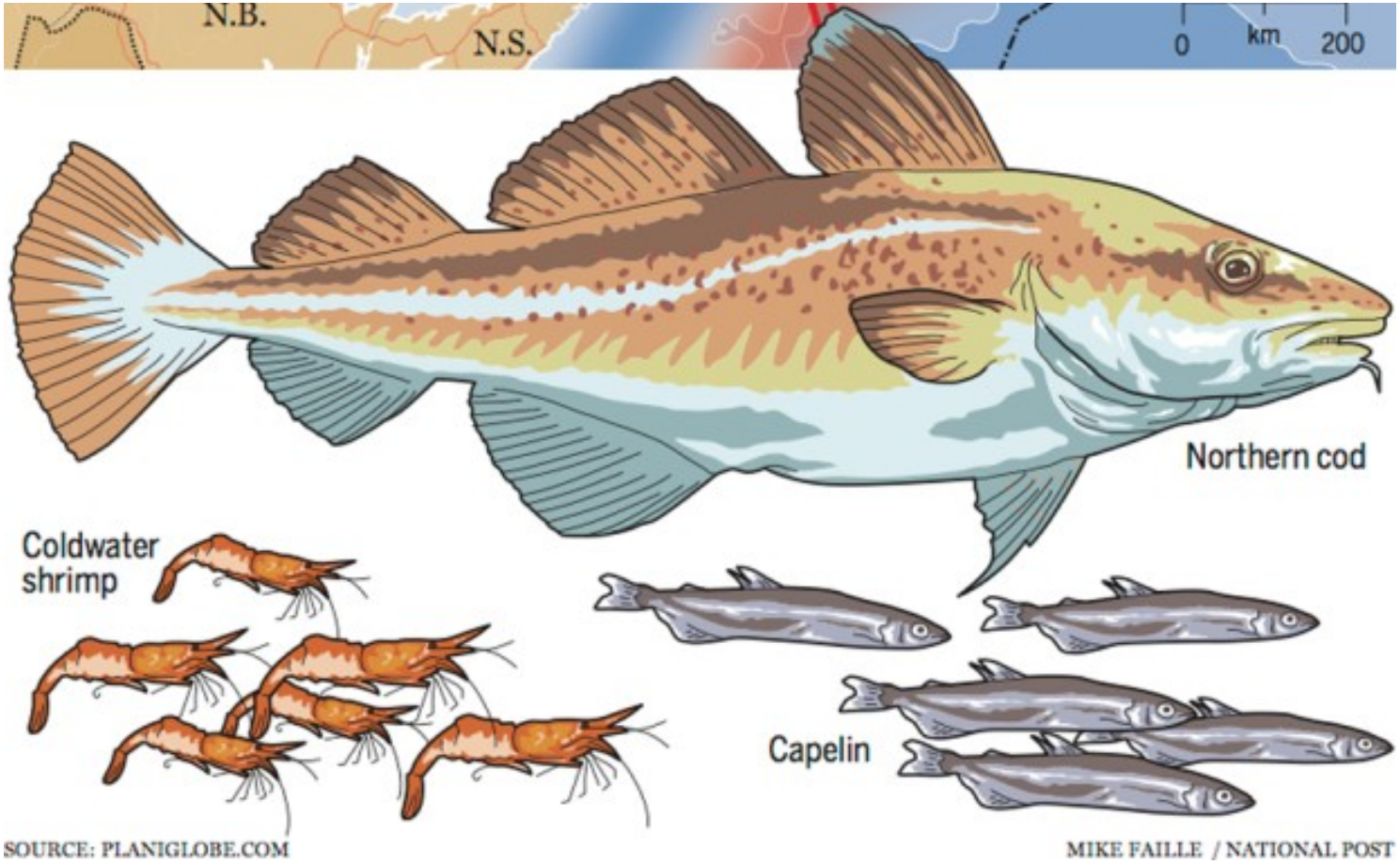
Invertebrates have taken over the cod's (and the food fish for the cod) role in the food chain.

Warming waters reduce cod's resilience

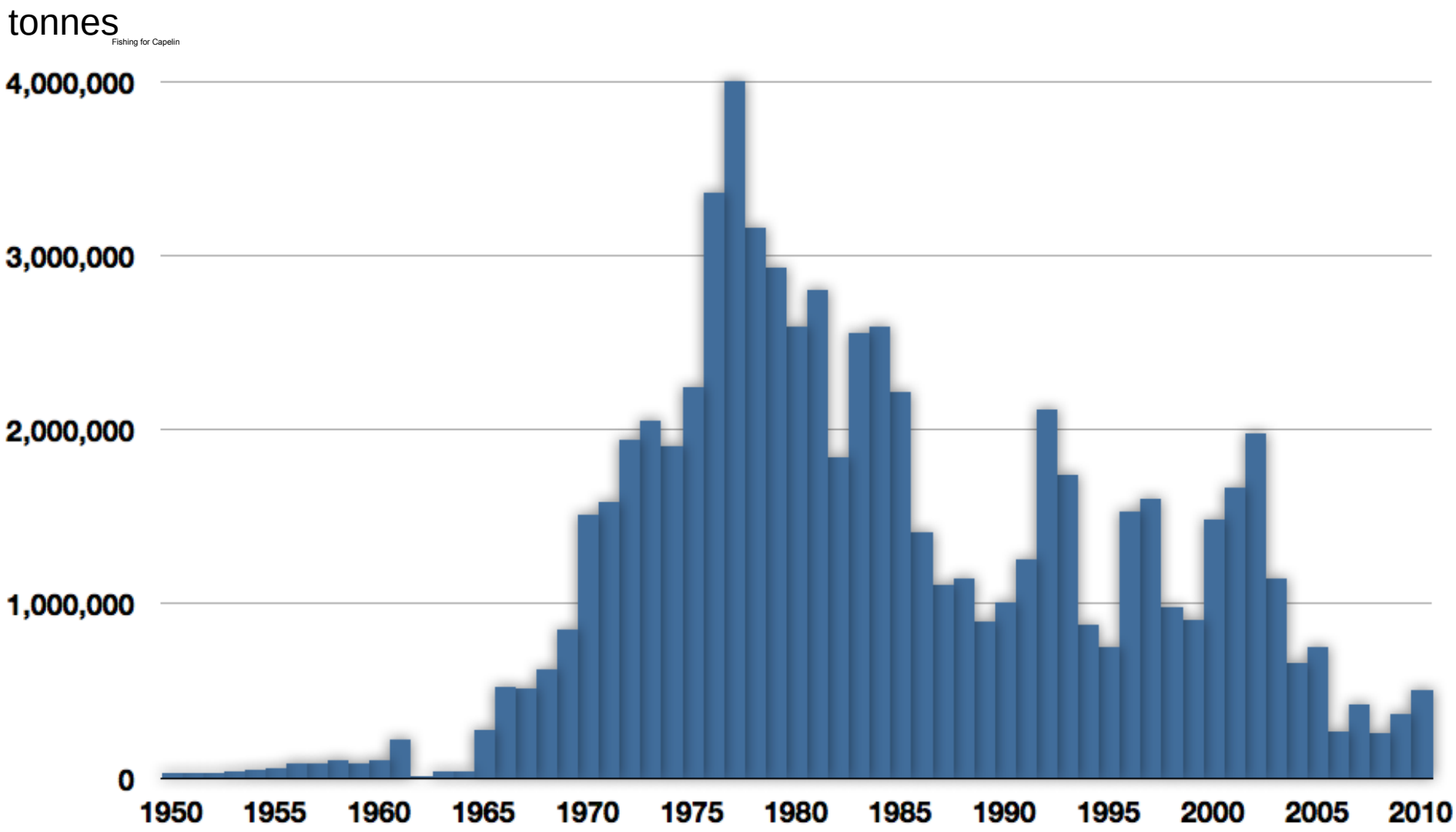
Continued fishing for big cod and cod taken due to bycatch

Continued fishing down the food chain (Capelin)

# Food chain: Cod and below



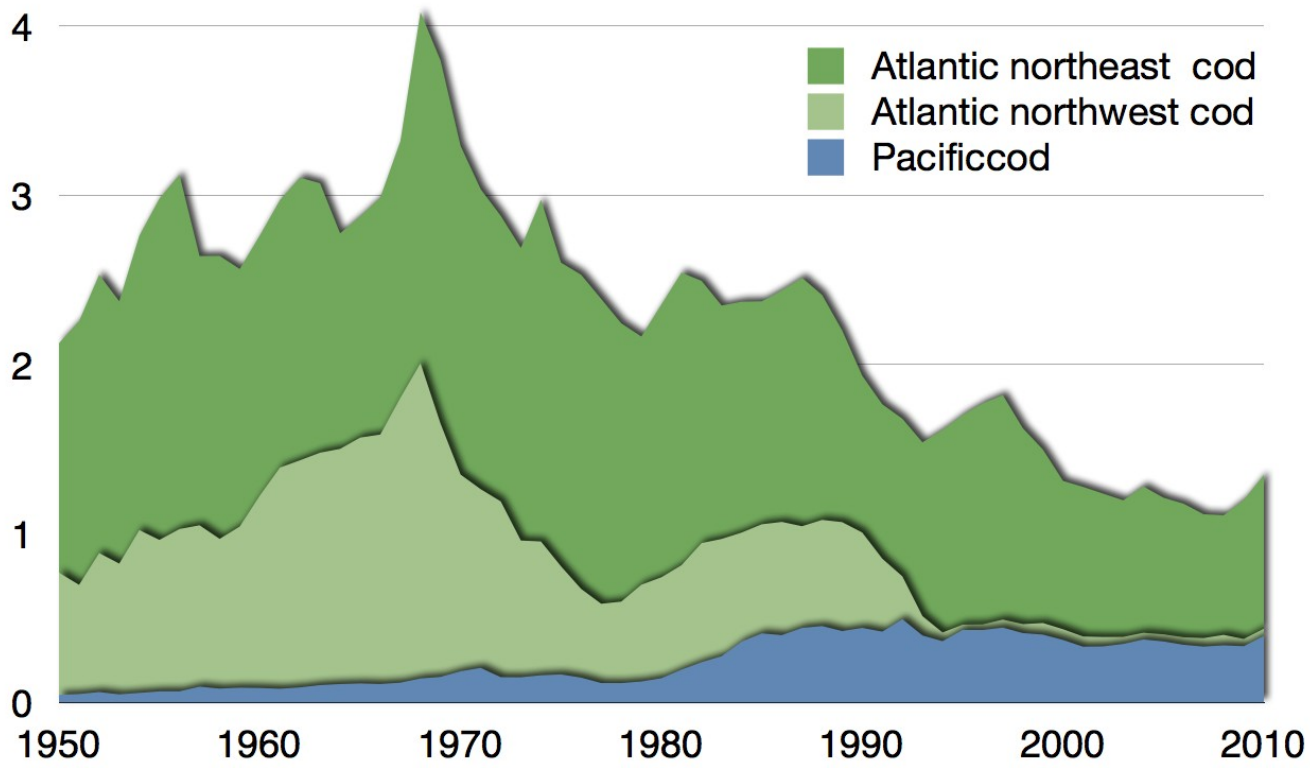

Grand Banks is now an invertebrate-dominated ecosystem




Capelin is a key food source for Northern cod and other species, and plays a significant role in sustaining the marine ecosystem

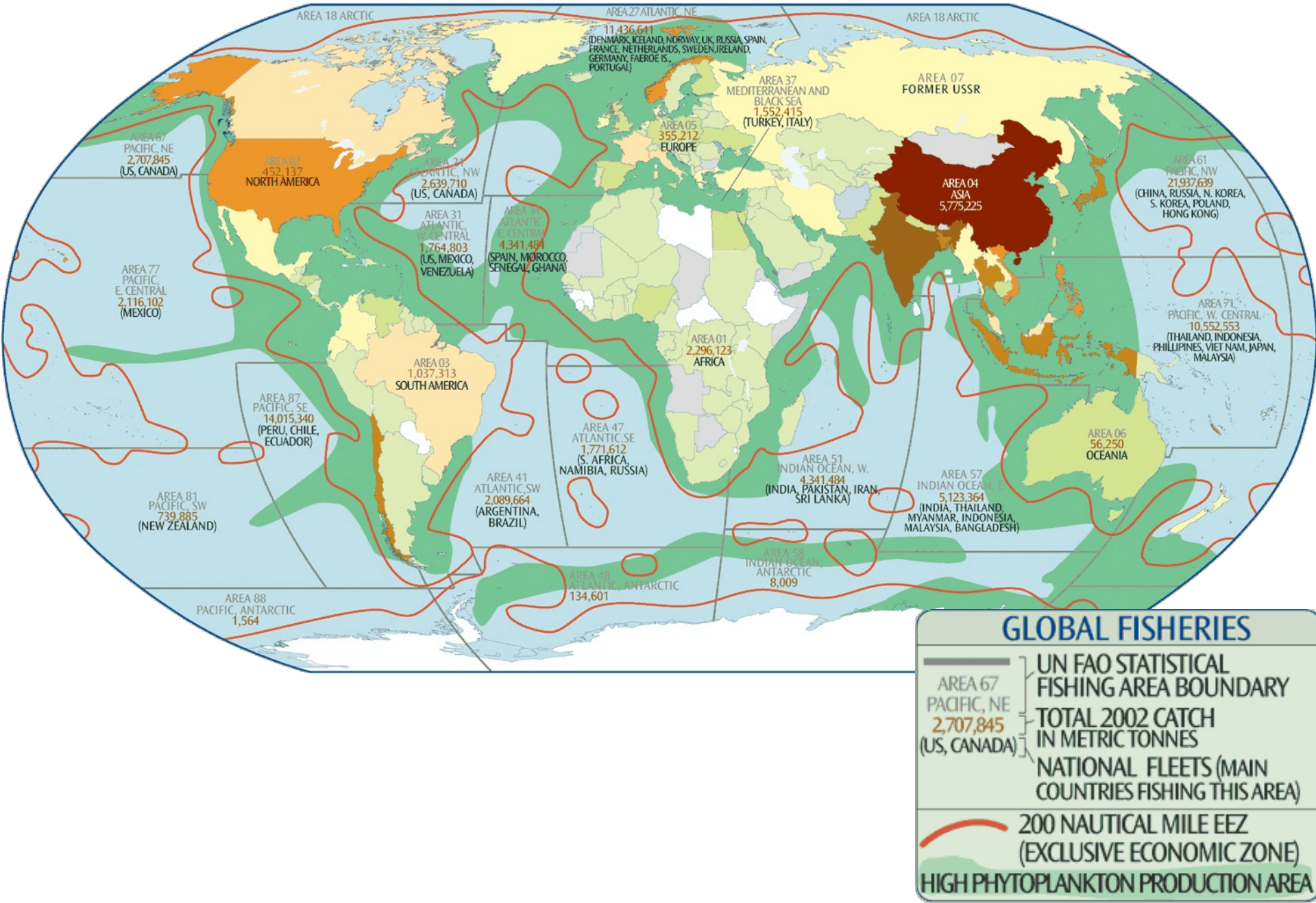



Canada now manages the Grand Banks Fisheries

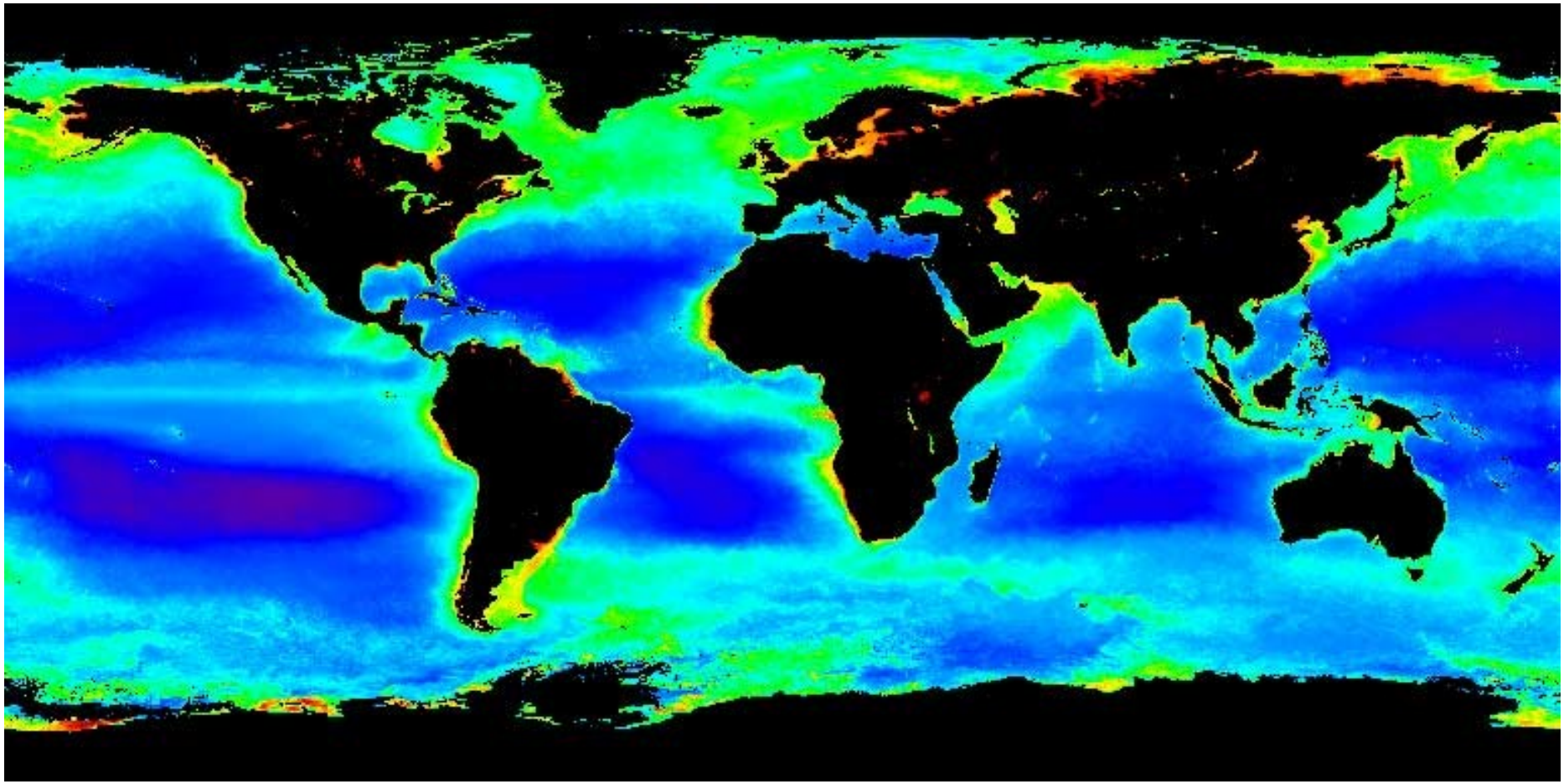



The northeast Atlantic Cod population held on longer (more habitat)

Fisheries, EEZ

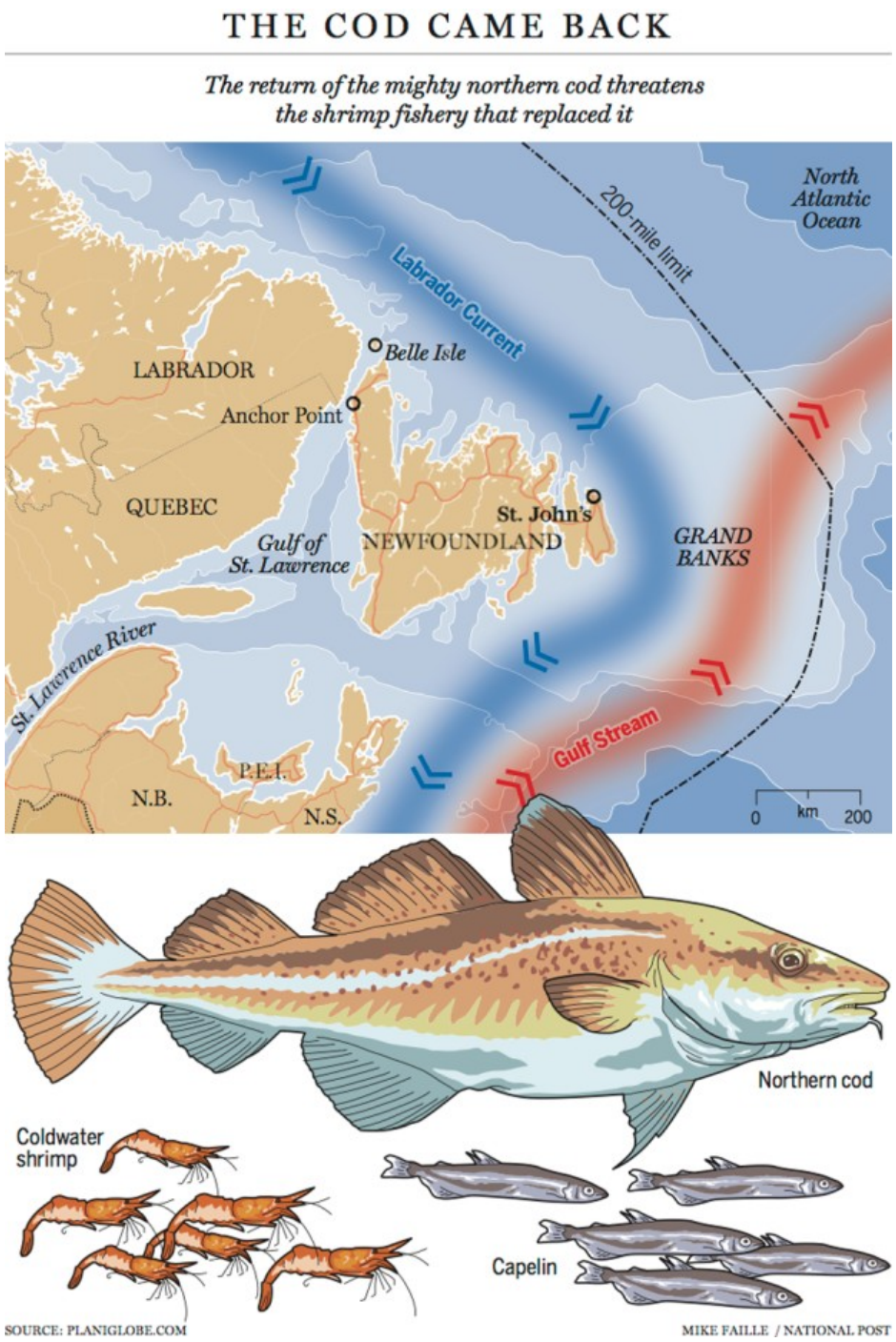


Global Chlorophyll Distribution



NASA

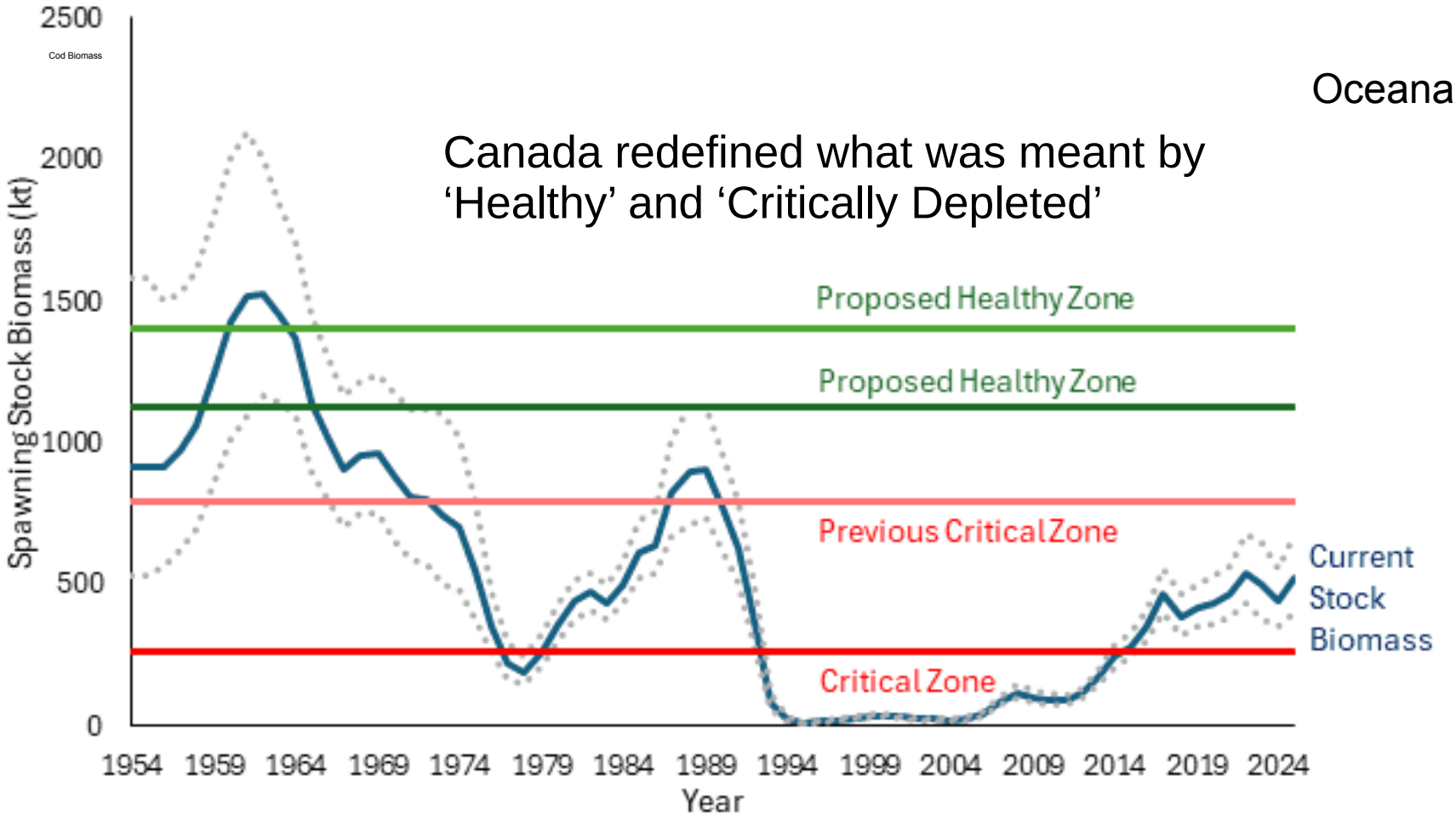
Return of Cod?




Cod have been learning to eat shrimp (also heavily fished)

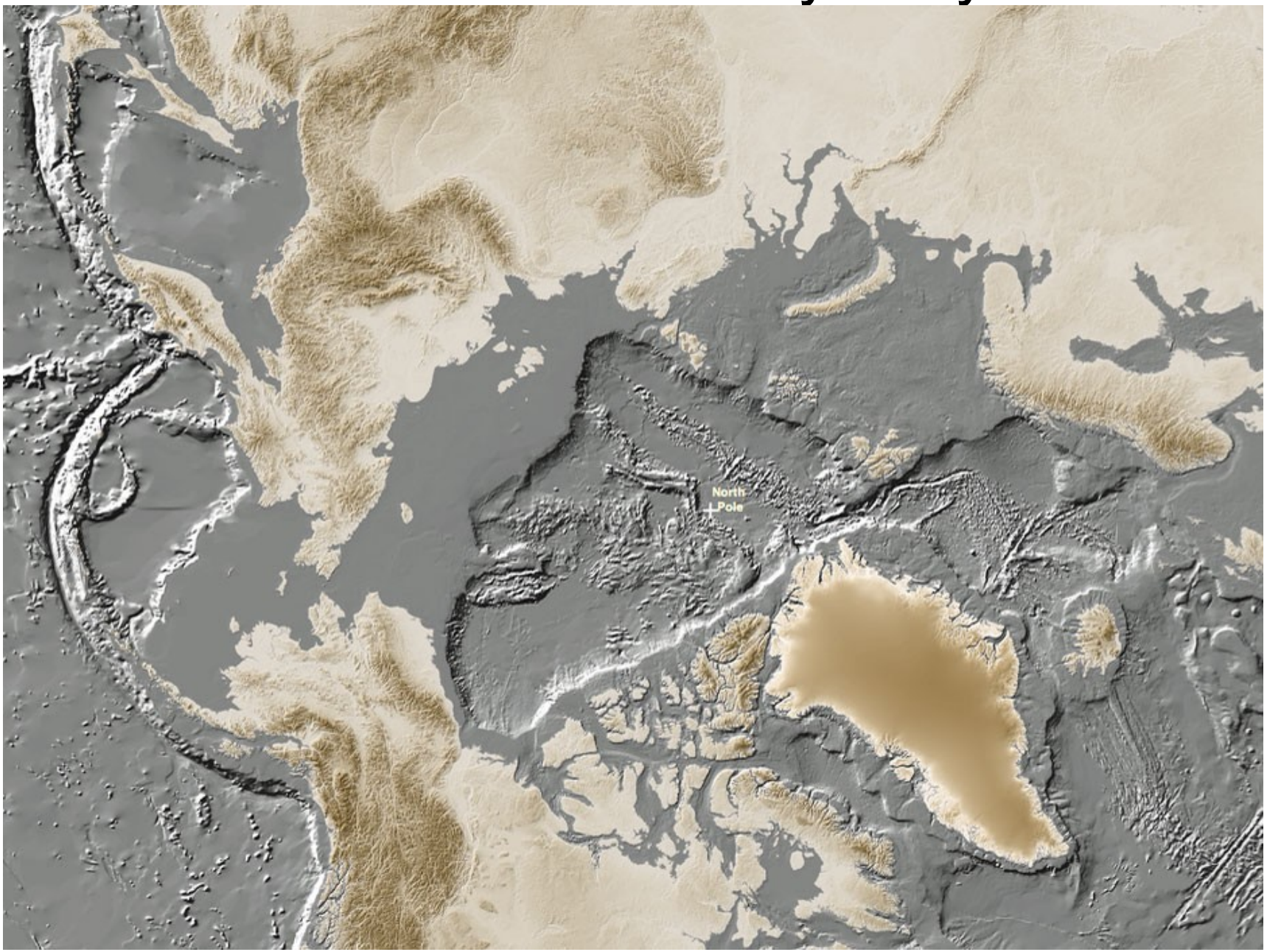
Canada reopened the commercial cod fishery in 2024—18,000 tonnes after a 32-year moratorium, and 2025 allowed for a “responsible” total allowable catch (TAC) increase to 38,000 tonnes. For 2025 the TAC for Capelin will remain at 14,533 tonnes.

National Post




Fisheries and Oceans Canada’s March 2025 assessment: northern cod remains deeply depleted, with little evidence of meaningful rebuilding underway. Spawning biomass, the total weight of fish that have reached reproductive maturity, sits near 524,000 tonnes. About half of what it was during the productive years from 1954 to 1970, and just a quarter of its pre-fishing abundance. Worse, there is a 56 to 71% chance of further decline.

# Arctic Ocean Bathymetry



# Arctic, EEZ (Energy and Minerals)!

